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1 Introduction

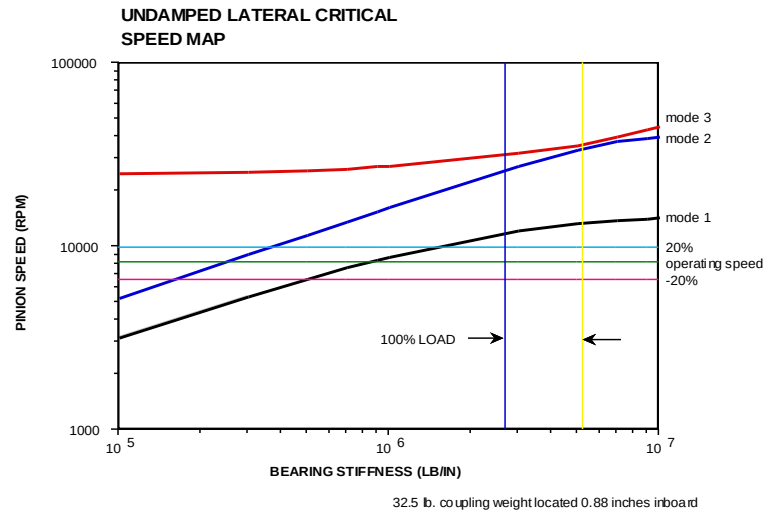
This is a presentation of the in-house rotordynamic analysis performed on a LUFKIN NF1819E Speed increaser, order number 01864H1. Lufkin is supplying this unit to  Nuovo Pignone to comply with the following specifications:

Input speeds	: 1953 rpm
Output speeds	: 12225 rpm
Ratio	: 6.2593:1
Transmitted power	: 2532 kW
AGMA Service factor	: 2.99
API Service factor	: 1.63
Maximum continuous input speed	: 2051 rpm
Maximum continuous output speed	: 12836 rpm

The purpose of this analysis is to determine the undamped lateral critical speeds and damped response of the rotor bearing system. This is accomplished by modeling the bearings, rotor geometry, and added weights and inertia's with computer programs. These programs calculate the bearing stiffness and damping coefficients, rotor bearing system undamped critical speeds, and damped rotor response. This report presents all calculated information in tabular or graphic form.

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Below is an *example* of an Undamped Lateral Critical Speed Map that shows the undamped critical speeds versus the bearing stiffness of the example rotor bearing system. The undamped critical speeds are the lines marked mode 1, 2, and 3.

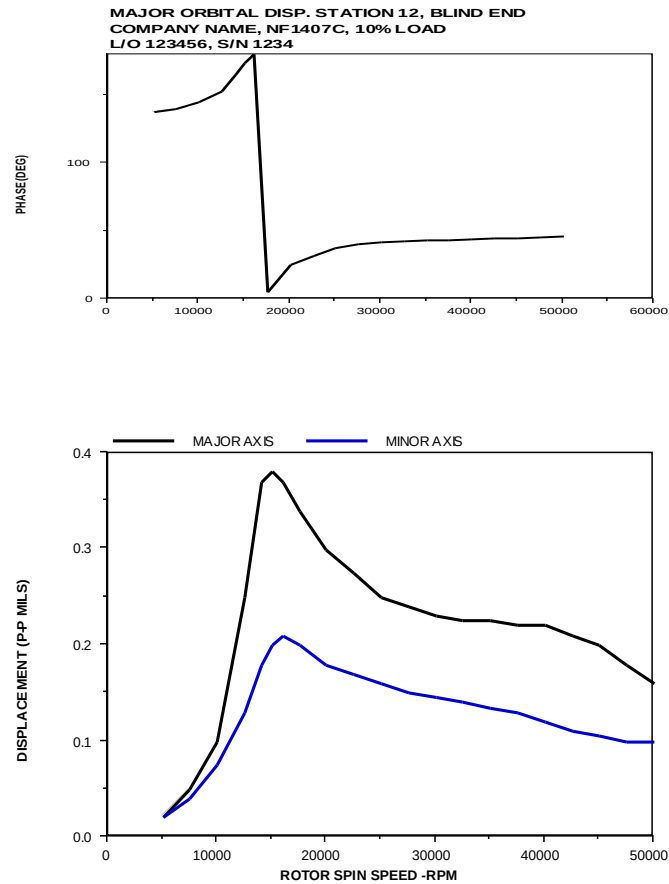


Superimposed vertically on the speed map are the bearing stiffness coefficients. The stiffness lines represent the minimum and maximum stiffness values at the 100% load condition.

Stiffness values at other load conditions of interest may also be plotted if desired. The map shows the maximum continuous operating speed of the rotor as a horizontal line with additional lines representing $\pm 20\%$ maximum continuous operating speed. The box formed by the minimum and maximum bearing stiffness values and the $\pm 20\%$ maximum continuous operating speed lines represents an area of interest. Should any of the undamped modes pass through this box, a critical speed resonance could occur. If an intersection does occur, a response plot will reveal if a resonance condition exists.

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When generating a response plot, unbalance is placed on the rotor to excite any possible resonance. The unbalance is located at positions that have been analytically determined to affect the mode of interest (i.e. at the midspan for translational, at the coupling for conical).



Each plot shows the diametral size of the orbit ellipse versus speed. The phase portion of the plot shows the angular orientation of the major axis with respect to the x axis. This angle is always between 0 and 180 degrees. Looking at the *example* Displacement and Phase Plot above, a peak in the response displacement is seen near 16000 rpm, indicating a possible critical speed. The smaller plot above the response shows an abrupt phase change at the same location as the peak, verifying a definite critical speed (an abrupt phase change usually occurs when passing through a critical speed).

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2 High Speed Pinion

Lufkin performed a computer analysis on the high speed bearing and pinion models for order number 01864H1. The analysis produced tables showing the bearing stiffness and damping coefficients, and graphs showing the undamped natural frequencies, and the rotor response to unbalance as described in the introduction. Undamped mode shapes of the rotor were also generated.

2.1 Bearing and Gear Mesh Stiffness and Damping Coefficients

The high speed journal bearings are Pressure-Dam bearings. The bearings are designed to operate with a diametral clearance between 0.178 mm and 0.216 mm. The bearing and gear mesh stiffness (K_{xx} , K_{xy} , K_{yx} , K_{yy}) and damping coefficients (C_{xx} , C_{xy} , C_{yx} , C_{yy}) for eight speeds, including maximum continuous, are presented in tabular form in **Appendix A**. Based on past experience, the bearing support stiffness (gear housing) was assumed to be much greater than the bearing stiffness and was not considered in the analysis.

2.2 Rotor-Bearing Model

LUFKIN mass elastic drawing YOH000020305 defines the geometry used to model the pinion with a 7.1 kg coupling weight located 11.3 mm inboard from the shaft end added to the model. Figure 1 shows the computer rotor-bearing model used for the analysis. Springs denote the bearing locations. The figure also shows station numbering for the model with vibration probe locations at stations 5 and 21.

2.3 Undamped Critical Speed Map and Mode Shapes

Figure 2 shows the first three undamped lateral critical speeds versus the bearing support stiffness for the rotor bearing system. The undamped speed map does not include the effects of damping or cross-coupled stiffness values. The map shows bearing stiffness values at 10%, 50%, and 100% load conditions (three boxes) for both minimum and maximum bearing clearances.

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The first undamped mode pass through boxes formed by the bearing stiffnesses values at 10%, 50% load and +/-20% of the operating speed margin.

Figures 3, 4, and 5 show the first three undamped mode shapes for the rotor in a three dimensional view. An assumed bearing stiffness of 6.77E+08 Newtons per Meter was used when generating the mode shapes. Calculation of the mode shapes included the added weight of the coupling at the shaft end.

2.4 Unbalance Response Plots

The response plots were generated using a rotor unbalance of 81.48 g-mm (4 x 6350 x rotor weight / rpm) located at the midspan (center of the gear mesh), and a coupling unbalance of 14.05 g-mm (4 x 6350 x Coupling weight / rpm) located at the coupling center of gravity. These values are equal or greater than 4 times maximum allowable.

Figures 6 through 29 show the peak to peak response levels of the pinion with the unbalance placed at the midspan. **Figures 30 through 53** show the response levels with the unbalance placed at the coupling. All of the plots show the pinion response to unbalance from standstill to a speed of 20538 rpm. Response levels are taken at the blind end vibration probe, midspan, extension end vibration probe and the coupling locations. The response plots are shown for load conditions of 10%, 50%, and 100% of rated horsepower for both minimum and maximum bearing clearances.

The amplitudes of the responses are the results of the unbalances only and do not include any effects due to misalignment or rotor imperfections (runout), because it is not possible to predict either. Therefore, the amplitudes shown on the response plots should not be expected to match those obtained during the testing of the gear. However, the response plots are still valuable in predicting the rotor's sensitivity to unbalance.

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On each response curve are presented, in case of a peak, the speed at the peak **Nc** with amplification factor **AF** calculated as defined in API specifications. The "Separation Margin" is also calculated for the peak: the actual separation margin **S_{Ma}** and the "required" separation margin **S_{Mr}**, which depends on the amplification factor value (see **API 684** Rotordynamic Tutorial).

All of the plots show only a steady rise in amplitude with speed as the result of unbalance. No plots show peak responses within the $\pm 20\%$ range of maximum continuous speed. The maximum predicted response at the vibration probes for the maximum continuous speed of 12836 rpm is 3.554 micrometers peak-to-peak. The maximum response occurs at the extension end vibration probe for the 10% load, maximum bearing clearance condition with the unbalance placed at the midspan (**Figure 10**).

3 Low Speed Gear

Lufkin performed a computer analysis on the low speed bearing and gear models for order number 01864H1. The analysis produced tables showing the bearing stiffness and damping coefficients, and graphs showing the undamped natural frequencies, and the rotor response to unbalance as described in the introduction. Undamped mode shapes of the rotor were also generated.

3.1 Bearing and Gear Mesh Stiffness and Damping Coefficients

The low speed journal bearings are pressure DAM bearings. The bearings are designed to operate with a diametral clearance between 0.191 mm and 0.249 mm. The bearing and gear mesh stiffness (**K_{xx}**, **K_{xy}**, **K_{yx}**, **K_{yy}**) and damping coefficients (**C_{xx}**, **C_{xy}**, **C_{yx}**, **C_{yy}**) for eight speeds, including maximum continuous, are presented in tabular form following the introduction. Based on past experience, the bearing support stiffness (gear housing) was assumed to be much greater than the bearing stiffness and was not considered in the analysis.

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3.2 Rotor-Bearing Model

LUFKIN mass elastic drawing YOH000020305 defines the geometry used to model the gear with a 53.1 kg coupling weight located 90.2 mm outboard from the shaft end added to the model. **Figure 54** shows the computer rotor-bearing model used for the analysis. Springs denote the bearing locations. The figure also shows station numbering for the model with vibration probe locations at stations 2 and 18.

3.3 Undamped Critical Speed Map and Mode Shapes

Figure 55 shows the first three undamped lateral critical speeds versus the bearing support stiffness for the rotor bearing system. The undamped speed map does not include the effects of damping or cross-coupled stiffness values. The map shows bearing stiffness values at 10%, 50%, and 100% load conditions (three boxes) for both minimum and maximum bearing clearances.

None undamped mode pass through boxes formed by the bearing stiffnesses values and +/-20% of the operating speed margin.

Figures 56, 57, and 58 show the first three undamped mode shapes for the rotor in a three dimensional view. An assumed bearing stiffness of 1.04E+09 Newtons per Meter was used when generating the mode shapes. Calculation of the mode shapes included the added weight of the coupling at the shaft end.

3.4 Unbalance Response Plots

The response plots were generated using a rotor unbalance of 11491.25 g-mm (4x 6350 x rotor weight / rpm) located at the midspan (center of the gear mesh), and a coupling unbalance of 657.60 g-mm (4 x 6350 x Coupling weight / rpm) located at the coupling center of gravity. These values are equal or greater than 4 times maximum allowable.

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Figures 59 through 82 show the peak to peak response levels of the gear with the unbalance placed at the midspan. **Figures 83 through 106** show the response levels with the unbalance placed at the coupling. All of the plots show the gear response to unbalance from standstill to a speed of 3282 rpm. Response levels are taken at the blind end vibration probe, midspan, extension end vibration probe and the coupling locations. The response plots are shown for load conditions of 10%, 50%, and 100% of rated horsepower for both minimum and maximum bearing clearances.

The amplitudes of the responses are the results of the unbalances only and do not include any effects due to misalignment or rotor imperfections (runout), because it is not possible to predict either. Therefore, the amplitudes shown on the response plots should not be expected to match those obtained during the testing of the gear. However, the response plots are still valuable in predicting the rotor's sensitivity to unbalance.

On each response curve are presented, in case of a peak, the speed at the peak **Nc** with amplification factor **AF** calculated as defined in API specifications. The "Separation Margin" is also calculated for the peak: the actual separation margin **S_{Ma}** and the "required" separation margin **S_{Mr}**, which depends on the amplification factor value (see **API 684** Rotordynamic Tutorial).

All of the plots show only a steady rise in amplitude with speed as the result of unbalance. No plots show peak responses within the $\pm 20\%$ range of maximum continuous speed. The maximum predicted response at the vibration probes for the maximum continuous speed of 2051 rpm is 5.169 micrometers peak-to-peak. The maximum response occurs at the blind end vibration probe for the 10% load, maximum bearing clearance condition with the unbalance placed at the midspan (Figure 63).

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4 Conclusions

4.1 High Speed Pinion

The high speed rotor-bearing system will be free of any critical speeds within a 20% margin of maximum continuous operating speed. This deduction is due to the absence of a sharp response (amplification factor > 2.5) and abrupt phase change at the same frequency within the separation margin specified by API 613 Fifth Edition.

The maximum predicted vibration level at maximum continuous operating speed for the high speed pinion due to unbalance is 3.554 micrometers peak-to-peak. This value was obtained using 4 times the allowable unbalance for the rotor and coupling.

4.2 Low Speed Gear

The low speed rotor-bearing system will be free of any critical speeds within a 20% margin of maximum continuous operating speed. This deduction is due to the absence of a sharp response (amplification factor > 2.5) and abrupt phase change at the same frequency within the separation margin specified by API 613 Fifth Edition.

The maximum predicted vibration level at maximum continuous operating speed for the low speed gear due to unbalance is 5.169 micrometers peak-to-peak. This value was obtained using 4 times the allowable unbalance for the rotor and coupling.

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Appendix A - Rotordynamics Coefficients

Rotordynamic Coefficients for Blind End Bearing, 10% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	7.16E+07	1.75E+07	-9.16E+07	2.24E+07	3.49E+05	-2.76E+05	-2.74E+05	5.05E+05
5134	9.94E+07	2.51E+07	-1.35E+08	2.39E+07	2.35E+05	-1.88E+05	-1.86E+05	3.77E+05
7702	1.28E+08	3.16E+07	-1.79E+08	2.75E+07	2.00E+05	-1.59E+05	-1.57E+05	3.30E+05
10269	1.57E+08	3.72E+07	-2.20E+08	3.21E+07	1.81E+05	-1.43E+05	-1.42E+05	3.03E+05
12836	1.85E+08	4.19E+07	-2.60E+08	3.73E+07	1.68E+05	-1.32E+05	-1.31E+05	2.82E+05
15403	2.12E+08	4.59E+07	-2.98E+08	4.29E+07	1.59E+05	-1.25E+05	-1.24E+05	2.66E+05
17970	2.39E+08	4.94E+07	-3.35E+08	4.86E+07	1.51E+05	-1.18E+05	-1.17E+05	2.52E+05
20538	2.66E+08	5.26E+07	-3.70E+08	5.43E+07	1.45E+05	-1.13E+05	-1.12E+05	2.42E+05

Rotordynamic Coefficients for Blind End Bearing, 10% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	5.39E+07	1.21E+07	-6.65E+07	1.89E+07	2.65E+05	-2.08E+05	-2.07E+05	3.57E+05
5134	7.11E+07	1.64E+07	-9.36E+07	1.89E+07	1.65E+05	-1.32E+05	-1.31E+05	2.55E+05
7702	9.03E+07	2.03E+07	-1.22E+08	2.08E+07	1.36E+05	-1.10E+05	-1.09E+05	2.21E+05
10269	1.10E+08	2.38E+07	-1.51E+08	2.33E+07	1.22E+05	-9.84E+04	-9.73E+04	2.02E+05
12836	1.29E+08	2.70E+07	-1.78E+08	2.61E+07	1.13E+05	-9.10E+04	-9.00E+04	1.89E+05
15403	1.47E+08	2.99E+07	-2.05E+08	2.90E+07	1.07E+05	-8.56E+04	-8.46E+04	1.80E+05
17970	1.66E+08	3.25E+07	-2.31E+08	3.20E+07	1.02E+05	-8.14E+04	-8.04E+04	1.72E+05
20538	1.83E+08	3.48E+07	-2.56E+08	3.47E+07	9.73E+04	-7.77E+04	-7.68E+04	1.65E+05

Rotordynamic Coefficients for Blind End Bearing, 50% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	3.15E+08	-4.43E+06	-3.96E+08	2.09E+08	1.66E+06	-1.38E+06	-1.54E+06	1.92E+06
5134	3.06E+08	5.80E+07	-3.68E+08	1.50E+08	8.52E+05	-6.58E+05	-6.70E+05	9.45E+05
7702	3.17E+08	7.05E+07	-3.83E+08	1.37E+08	5.64E+05	-4.32E+05	-4.31E+05	6.67E+05
10269	3.35E+08	7.45E+07	-4.08E+08	1.31E+08	4.26E+05	-3.28E+05	-3.27E+05	5.36E+05
12836	3.55E+08	7.79E+07	-4.36E+08	1.29E+08	3.50E+05	-2.70E+05	-2.69E+05	4.58E+05
15403	3.76E+08	8.11E+07	-4.66E+08	1.30E+08	3.02E+05	-2.33E+05	-2.32E+05	4.06E+05
17970	3.98E+08	8.40E+07	-4.95E+08	1.31E+08	2.68E+05	-2.07E+05	-2.06E+05	3.67E+05
20538	4.21E+08	8.67E+07	-5.26E+08	1.33E+08	2.44E+05	-1.88E+05	-1.86E+05	3.39E+05



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Rotordynamic Coefficients for Blind End Bearing, 50% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	2.82E+08	-4.35E+07	-3.65E+08	2.21E+08	1.36E+06	-1.21E+06	-1.39E+06	1.69E+06
5134	2.52E+08	2.69E+07	-3.08E+08	1.40E+08	6.90E+05	-5.49E+05	-5.83E+05	7.66E+05
7702	2.53E+08	4.98E+07	-3.04E+08	1.19E+08	4.65E+05	-3.56E+05	-3.60E+05	5.17E+05
10269	2.61E+08	5.59E+07	-3.14E+08	1.10E+08	3.44E+05	-2.63E+05	-2.62E+05	4.04E+05
12836	2.71E+08	5.81E+07	-3.29E+08	1.06E+08	2.74E+05	-2.10E+05	-2.09E+05	3.39E+05
15403	2.83E+08	6.00E+07	-3.46E+08	1.03E+08	2.31E+05	-1.78E+05	-1.77E+05	2.97E+05
17970	2.95E+08	6.19E+07	-3.63E+08	1.01E+08	2.02E+05	-1.55E+05	-1.54E+05	2.67E+05
20538	3.08E+08	6.37E+07	-3.82E+08	1.00E+08	1.81E+05	-1.39E+05	-1.38E+05	2.45E+05

Rotordynamic Coefficients for Blind End Bearing, 100% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	8.16E+08	-2.75E+08	-1.10E+09	7.92E+08	3.52E+06	-3.37E+06	-3.95E+06	4.77E+06
5134	6.55E+08	-5.44E+07	-8.37E+08	4.72E+08	1.65E+06	-1.41E+06	-1.61E+06	1.98E+06
7702	6.24E+08	4.11E+07	-7.70E+08	3.70E+08	1.12E+06	-9.05E+05	-9.80E+05	1.26E+06
10269	6.22E+08	8.72E+07	-7.52E+08	3.27E+08	8.53E+05	-6.65E+05	-6.94E+05	9.35E+05
12836	6.29E+08	1.15E+08	-7.53E+08	3.05E+08	6.94E+05	-5.27E+05	-5.35E+05	7.54E+05
15403	6.40E+08	1.27E+08	-7.64E+08	2.92E+08	5.77E+05	-4.35E+05	-4.36E+05	6.39E+05
17970	6.54E+08	1.32E+08	-7.79E+08	2.83E+08	4.92E+05	-3.70E+05	-3.69E+05	5.58E+05
20538	6.68E+08	1.35E+08	-7.96E+08	2.77E+08	4.29E+05	-3.22E+05	-3.21E+05	4.99E+05

Rotordynamic Coefficients for Blind End Bearing, 100% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	7.82E+08	-4.18E+08	-1.11E+09	9.26E+08	2.87E+06	-2.95E+06	-3.52E+06	4.37E+06
5134	5.93E+08	-1.37E+08	-7.81E+08	5.05E+08	1.36E+06	-1.23E+06	-1.45E+06	1.75E+06
7702	5.37E+08	-2.42E+07	-6.79E+08	3.67E+08	9.25E+05	-7.78E+05	-8.72E+05	1.07E+06
10269	5.16E+08	2.53E+07	-6.37E+08	3.07E+08	6.89E+05	-5.58E+05	-6.03E+05	7.70E+05
12836	5.11E+08	5.92E+07	-6.20E+08	2.73E+08	5.56E+05	-4.35E+05	-4.57E+05	6.06E+05
15403	5.12E+08	8.24E+07	-6.15E+08	2.53E+08	4.67E+05	-3.57E+05	-3.66E+05	5.03E+05
17970	5.16E+08	9.36E+07	-6.16E+08	2.40E+08	3.98E+05	-3.00E+05	-3.04E+05	4.33E+05
20538	5.22E+08	1.01E+08	-6.21E+08	2.30E+08	3.47E+05	-2.60E+05	-2.60E+05	3.83E+05



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LANGUAGE

Rotordynamic Coefficients for Extension End Bearing, 10% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	6.32E+07	1.17E+07	-7.63E+07	1.66E+07	2.78E+05	-2.16E+05	-2.14E+05	4.01E+05
5134	9.16E+07	1.95E+07	-1.21E+08	1.91E+07	2.05E+05	-1.62E+05	-1.60E+05	3.28E+05
7702	1.21E+08	2.62E+07	-1.65E+08	2.30E+07	1.81E+05	-1.43E+05	-1.42E+05	3.00E+05
10269	1.49E+08	3.19E+07	-2.08E+08	2.77E+07	1.67E+05	-1.32E+05	-1.30E+05	2.80E+05
12836	1.78E+08	3.68E+07	-2.49E+08	3.30E+07	1.57E+05	-1.23E+05	-1.22E+05	2.66E+05
15403	2.05E+08	4.08E+07	-2.87E+08	3.86E+07	1.50E+05	-1.17E+05	-1.16E+05	2.53E+05
17970	2.32E+08	4.45E+07	-3.25E+08	4.44E+07	1.44E+05	-1.12E+05	-1.11E+05	2.42E+05
20538	2.59E+08	4.77E+07	-3.60E+08	5.02E+07	1.39E+05	-1.07E+05	-1.07E+05	2.33E+05

Rotordynamic Coefficients for Extension End Bearing, 10% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	4.71E+07	7.27E+06	-5.39E+07	1.36E+07	2.02E+05	-1.56E+05	-1.54E+05	2.71E+05
5134	6.51E+07	1.18E+07	-8.22E+07	1.48E+07	1.39E+05	-1.11E+05	-1.09E+05	2.14E+05
7702	8.47E+07	1.59E+07	-1.12E+08	1.70E+07	1.20E+05	-9.69E+04	-9.56E+04	1.96E+05
10269	1.04E+08	1.95E+07	-1.41E+08	1.98E+07	1.11E+05	-8.95E+04	-8.82E+04	1.84E+05
12836	1.24E+08	2.28E+07	-1.69E+08	2.26E+07	1.04E+05	-8.42E+04	-8.31E+04	1.76E+05
15403	1.43E+08	2.58E+07	-1.96E+08	2.56E+07	9.97E+04	-8.01E+04	-7.91E+04	1.69E+05
17970	1.61E+08	2.84E+07	-2.23E+08	2.85E+07	9.56E+04	-7.68E+04	-7.58E+04	1.63E+05
20538	1.79E+08	3.09E+07	-2.48E+08	3.13E+07	9.24E+04	-7.39E+04	-7.30E+04	1.58E+05

Rotordynamic Coefficients for Extension End Bearing, 50% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	3.07E+08	6.20E+06	-3.72E+08	1.87E+08	1.63E+06	-1.32E+06	-1.45E+06	1.78E+06
5134	2.98E+08	5.71E+07	-3.50E+08	1.40E+08	8.19E+05	-6.21E+05	-6.25E+05	8.86E+05
7702	3.09E+08	6.47E+07	-3.66E+08	1.28E+08	5.34E+05	-4.07E+05	-4.05E+05	6.31E+05
10269	3.27E+08	6.85E+07	-3.92E+08	1.24E+08	4.05E+05	-3.11E+05	-3.09E+05	5.10E+05
12836	3.47E+08	7.21E+07	-4.21E+08	1.23E+08	3.34E+05	-2.57E+05	-2.56E+05	4.38E+05
15403	3.68E+08	7.54E+07	-4.51E+08	1.23E+08	2.89E+05	-2.23E+05	-2.21E+05	3.89E+05
17970	3.91E+08	7.85E+07	-4.81E+08	1.25E+08	2.58E+05	-1.99E+05	-1.97E+05	3.54E+05
20538	4.13E+08	8.13E+07	-5.12E+08	1.27E+08	2.35E+05	-1.81E+05	-1.79E+05	3.27E+05



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LANGUAGE

Rotordynamic Coefficients for Extension End Bearing, 50% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	2.73E+08	-3.06E+07	-3.41E+08	1.95E+08	1.36E+06	-1.16E+06	-1.31E+06	1.56E+06
5134	2.46E+08	3.07E+07	-2.91E+08	1.28E+08	6.75E+05	-5.22E+05	-5.45E+05	7.13E+05
7702	2.47E+08	4.79E+07	-2.90E+08	1.11E+08	4.45E+05	-3.35E+05	-3.36E+05	4.86E+05
10269	2.54E+08	5.12E+07	-3.00E+08	1.03E+08	3.26E+05	-2.47E+05	-2.46E+05	3.82E+05
12836	2.64E+08	5.32E+07	-3.16E+08	9.92E+07	2.60E+05	-1.99E+05	-1.98E+05	3.22E+05
15403	2.76E+08	5.53E+07	-3.33E+08	9.69E+07	2.20E+05	-1.69E+05	-1.68E+05	2.84E+05
17970	2.89E+08	5.73E+07	-3.52E+08	9.56E+07	1.93E+05	-1.48E+05	-1.47E+05	2.56E+05
20538	3.02E+08	5.93E+07	-3.70E+08	9.49E+07	1.73E+05	-1.33E+05	-1.32E+05	2.35E+05

Rotordynamic Coefficients for Extension End Bearing, 100% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	7.97E+08	-2.72E+08	-1.06E+09	7.57E+08	3.39E+06	-3.20E+06	-3.78E+06	4.53E+06
5134	6.49E+08	-3.35E+07	-8.12E+08	4.43E+08	1.67E+06	-1.41E+06	-1.58E+06	1.92E+06
7702	6.17E+08	4.72E+07	-7.49E+08	3.53E+08	1.11E+06	-8.83E+05	-9.49E+05	1.21E+06
10269	6.14E+08	8.99E+07	-7.33E+08	3.14E+08	8.42E+05	-6.49E+05	-6.71E+05	9.05E+05
12836	6.21E+08	1.13E+08	-7.35E+08	2.94E+08	6.76E+05	-5.11E+05	-5.18E+05	7.31E+05
15403	6.33E+08	1.23E+08	-7.46E+08	2.82E+08	5.62E+05	-4.22E+05	-4.22E+05	6.21E+05
17970	6.46E+08	1.26E+08	-7.62E+08	2.74E+08	4.78E+05	-3.59E+05	-3.58E+05	5.43E+05
20538	6.61E+08	1.29E+08	-7.81E+08	2.69E+08	4.18E+05	-3.13E+05	-3.12E+05	4.86E+05

Rotordynamic Coefficients for Extension End Bearing, 100% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
2567	7.86E+08	-3.61E+08	-1.08E+09	8.46E+08	3.04E+06	-3.06E+06	-3.57E+06	4.32E+06
5134	5.86E+08	-1.11E+08	-7.56E+08	4.69E+08	1.40E+06	-1.24E+06	-1.43E+06	1.70E+06
7702	5.29E+08	-1.82E+07	-6.57E+08	3.48E+08	9.09E+05	-7.54E+05	-8.41E+05	1.03E+06
10269	5.09E+08	3.25E+07	-6.19E+08	2.92E+08	6.82E+05	-5.42E+05	-5.84E+05	7.42E+05
12836	5.04E+08	6.33E+07	-6.03E+08	2.62E+08	5.49E+05	-4.23E+05	-4.41E+05	5.85E+05
15403	5.05E+08	8.02E+07	-5.99E+08	2.44E+08	4.54E+05	-3.45E+05	-3.53E+05	4.87E+05
17970	5.10E+08	9.32E+07	-6.01E+08	2.31E+08	3.90E+05	-2.92E+05	-2.93E+05	4.20E+05
20538	5.15E+08	9.74E+07	-6.07E+08	2.22E+08	3.37E+05	-2.51E+05	-2.51E+05	3.72E+05

Rotordynamic Coefficients for Gear Mesh

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
1	1.62E+08	-4.44E+08	-4.44E+08	1.22E+09	1.89E+03	0.00E+00	0.00E+00	5.18E+03
50000	1.62E+08	-4.44E+08	-4.44E+08	1.22E+09	1.89E+03	0.00E+00	0.00E+00	5.18E+03



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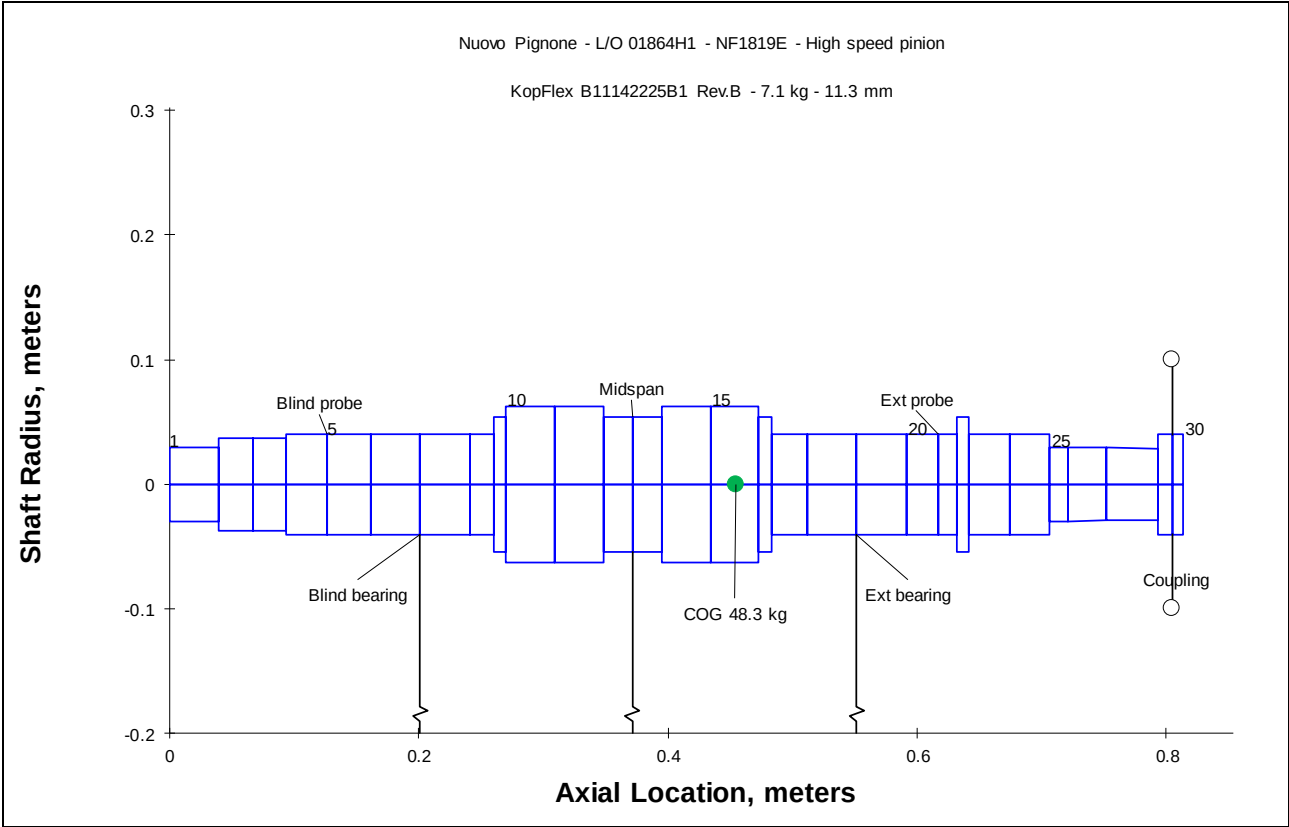
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Appendix B - Rotor-Bearing Model



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Appendix C - Undamped Critical Speed Map

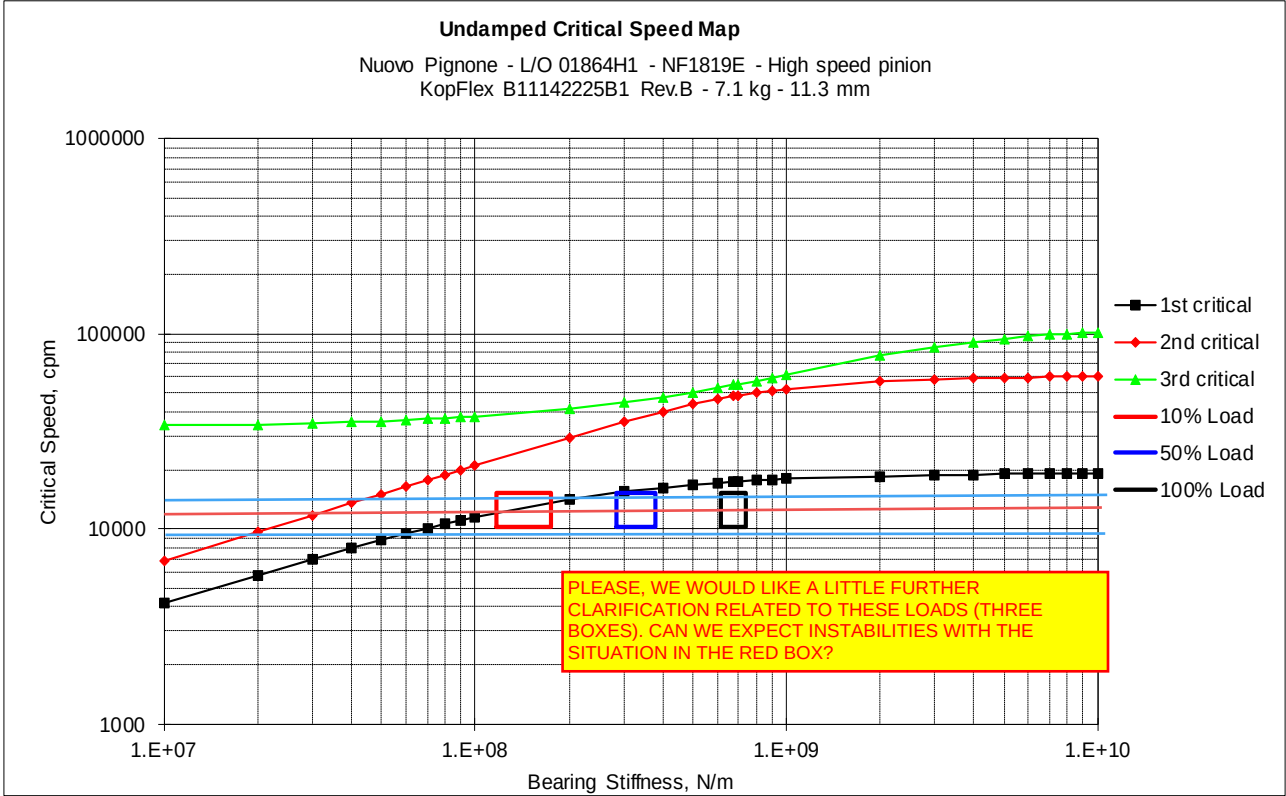


Figure 2

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Appendix D - Undamped Mode Shape Plots

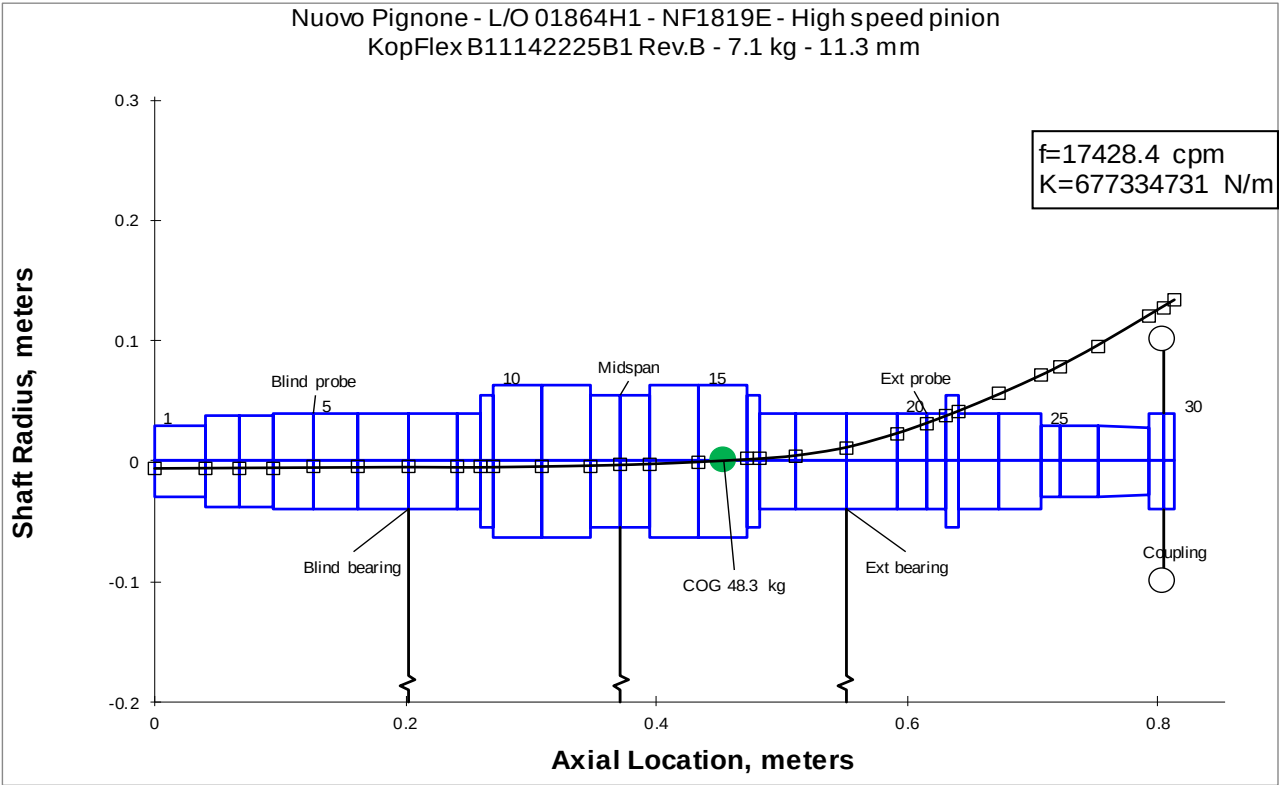


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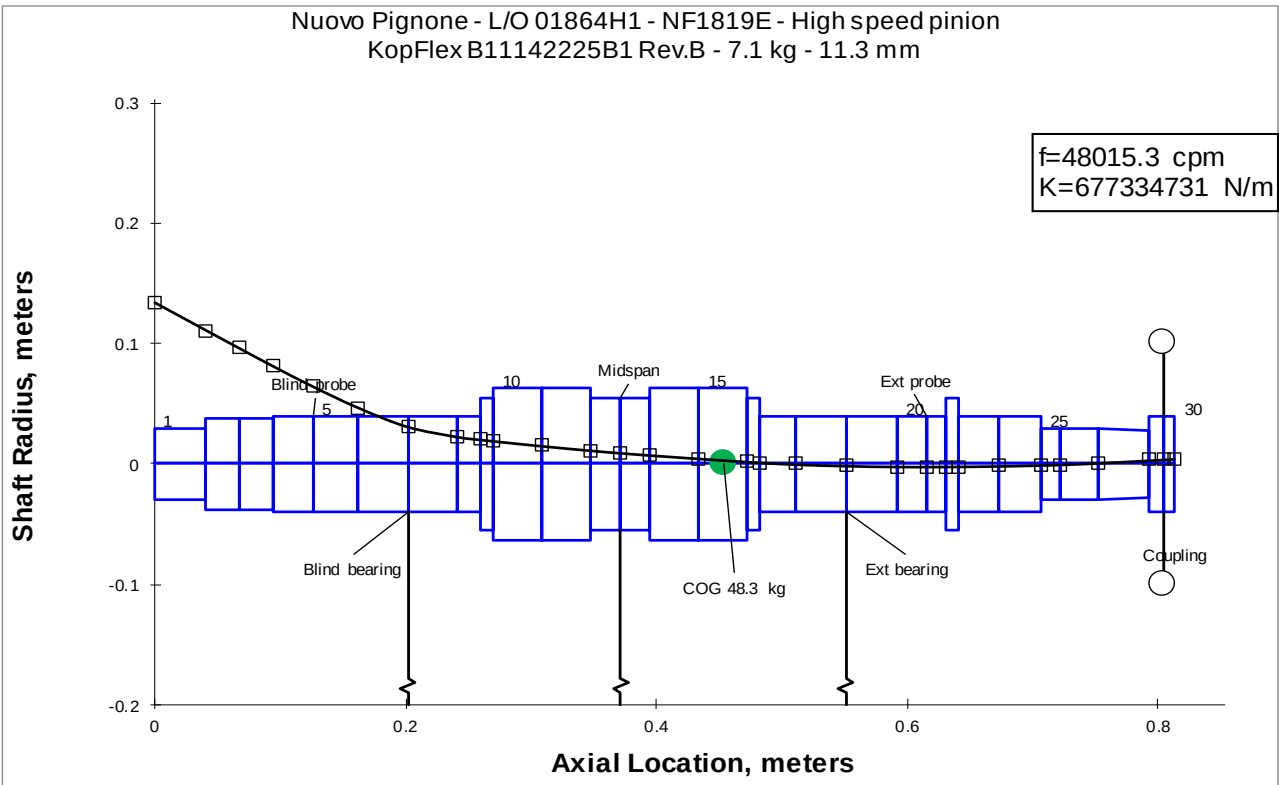
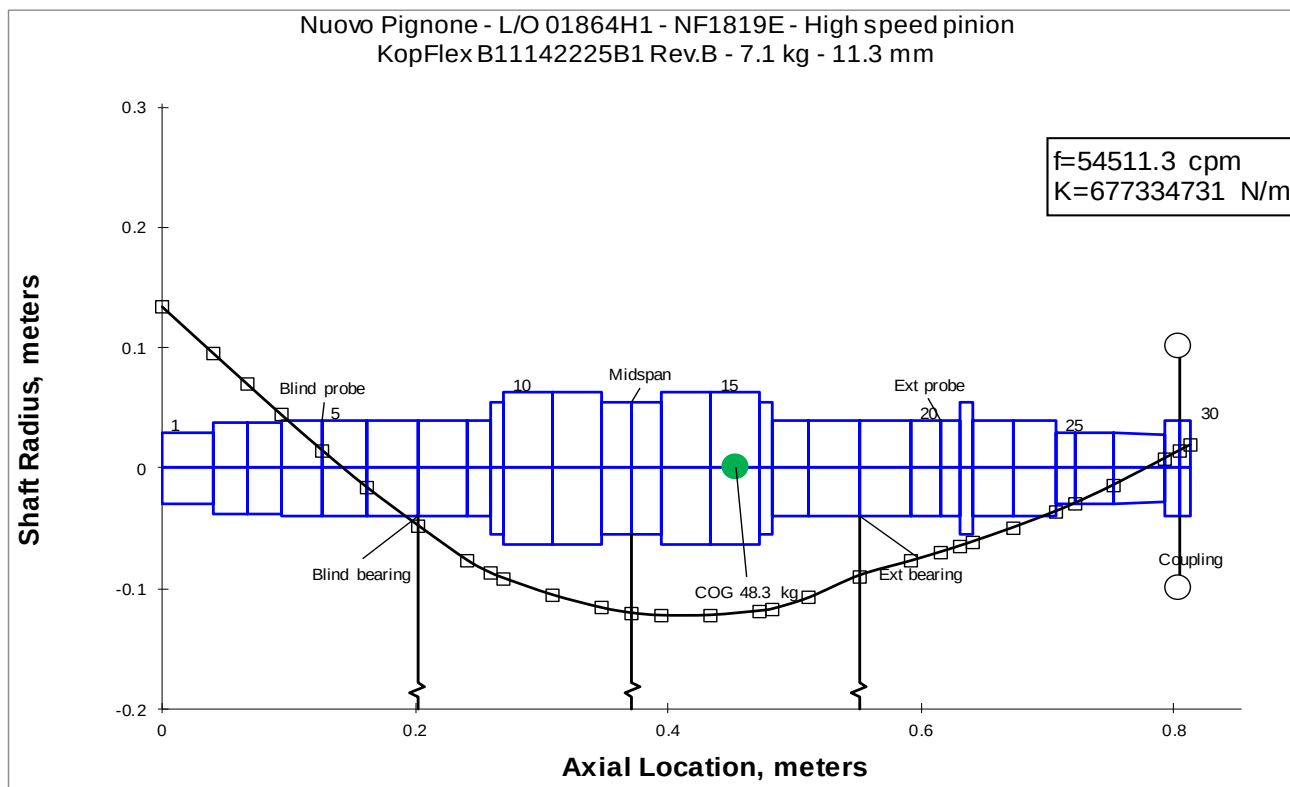


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Appendix E - Midspan Unbalance Response Plots

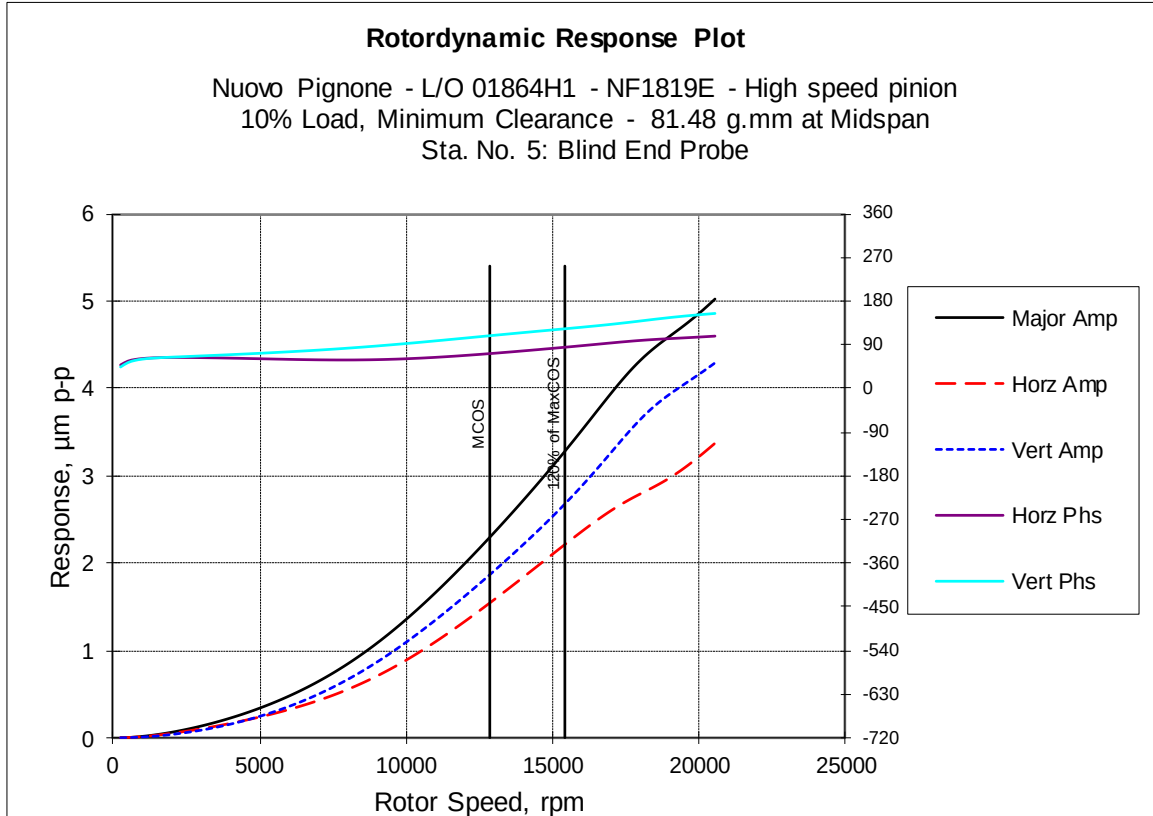


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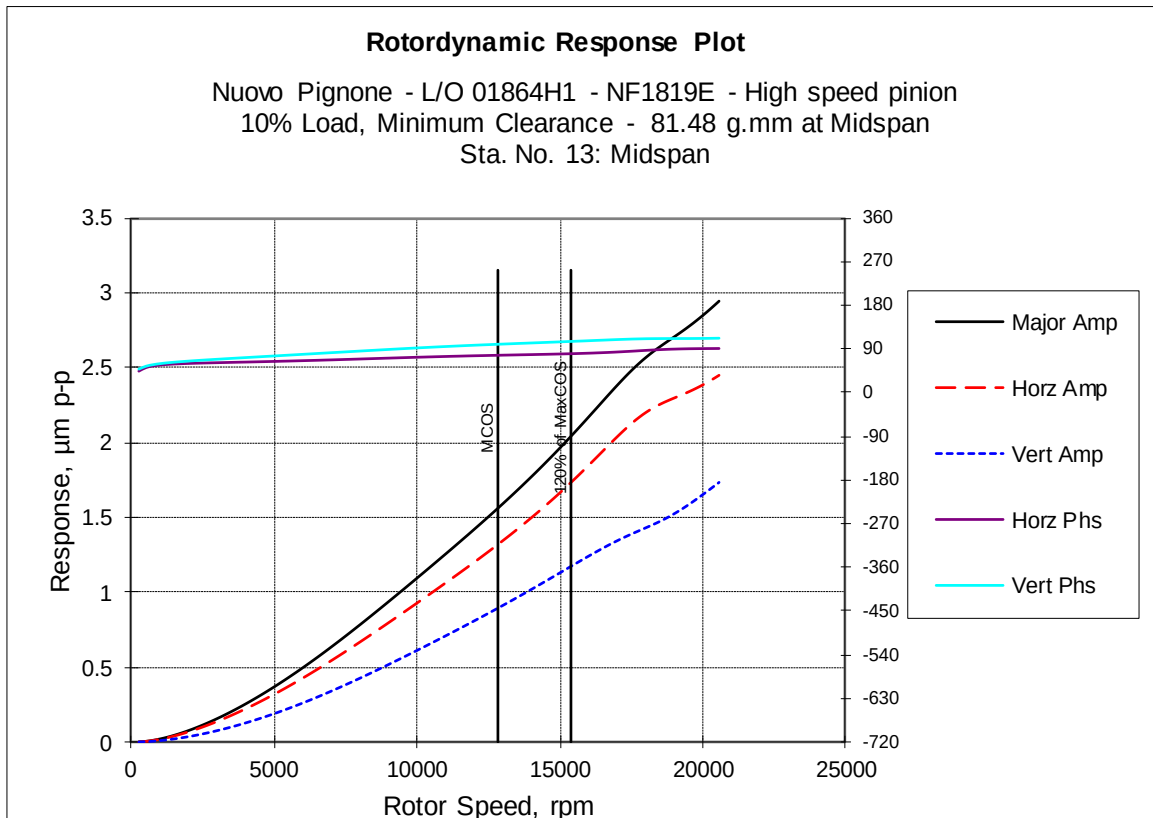


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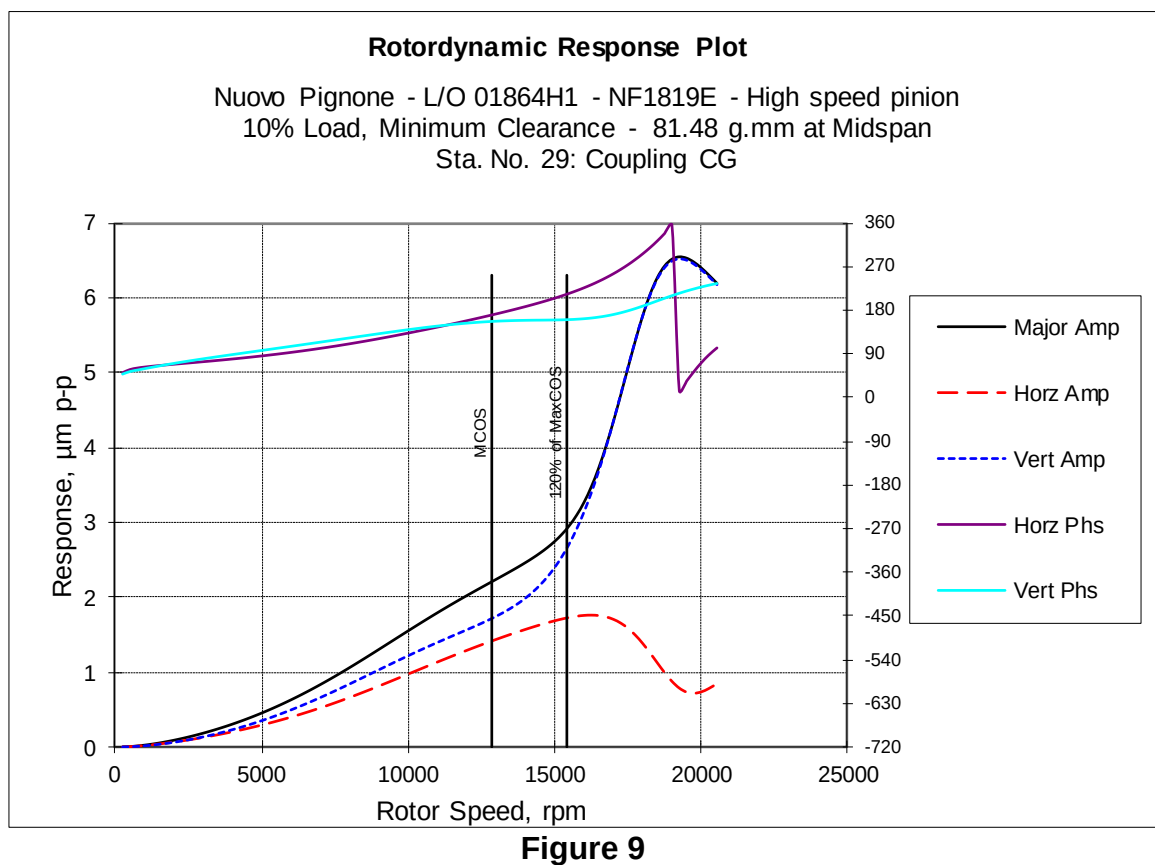
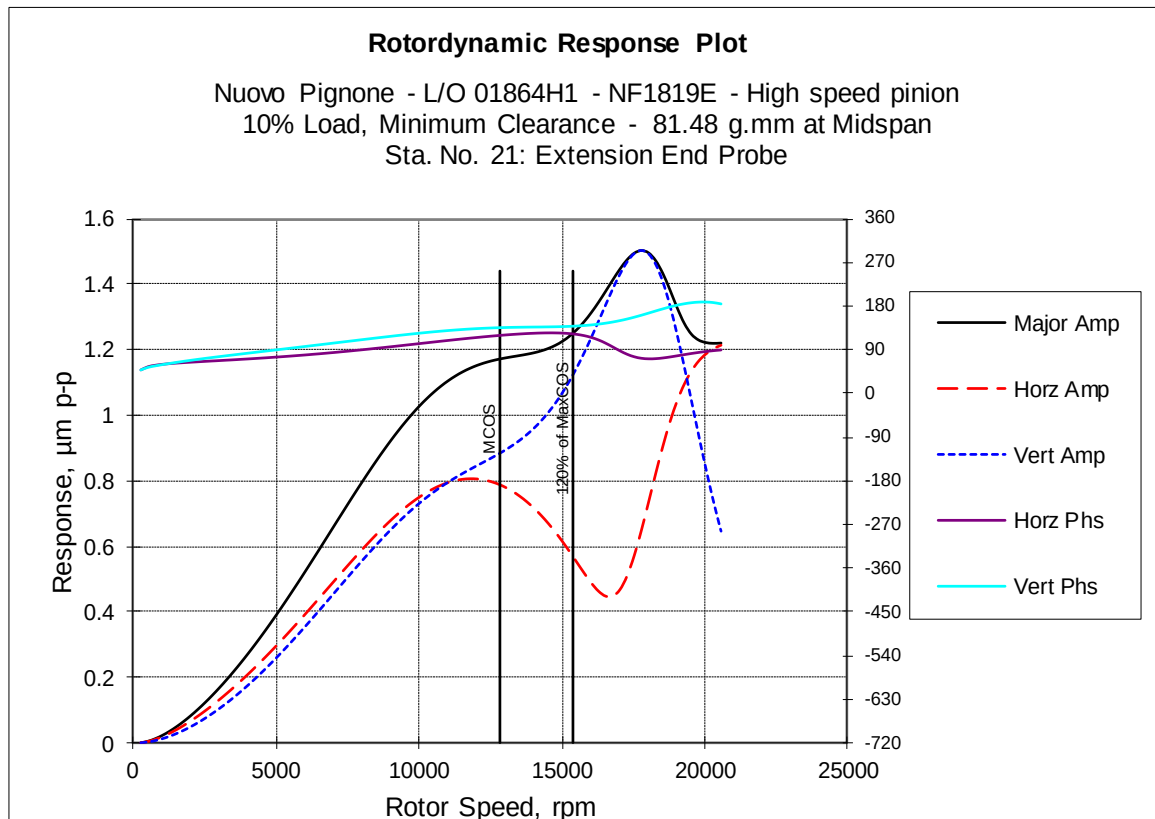
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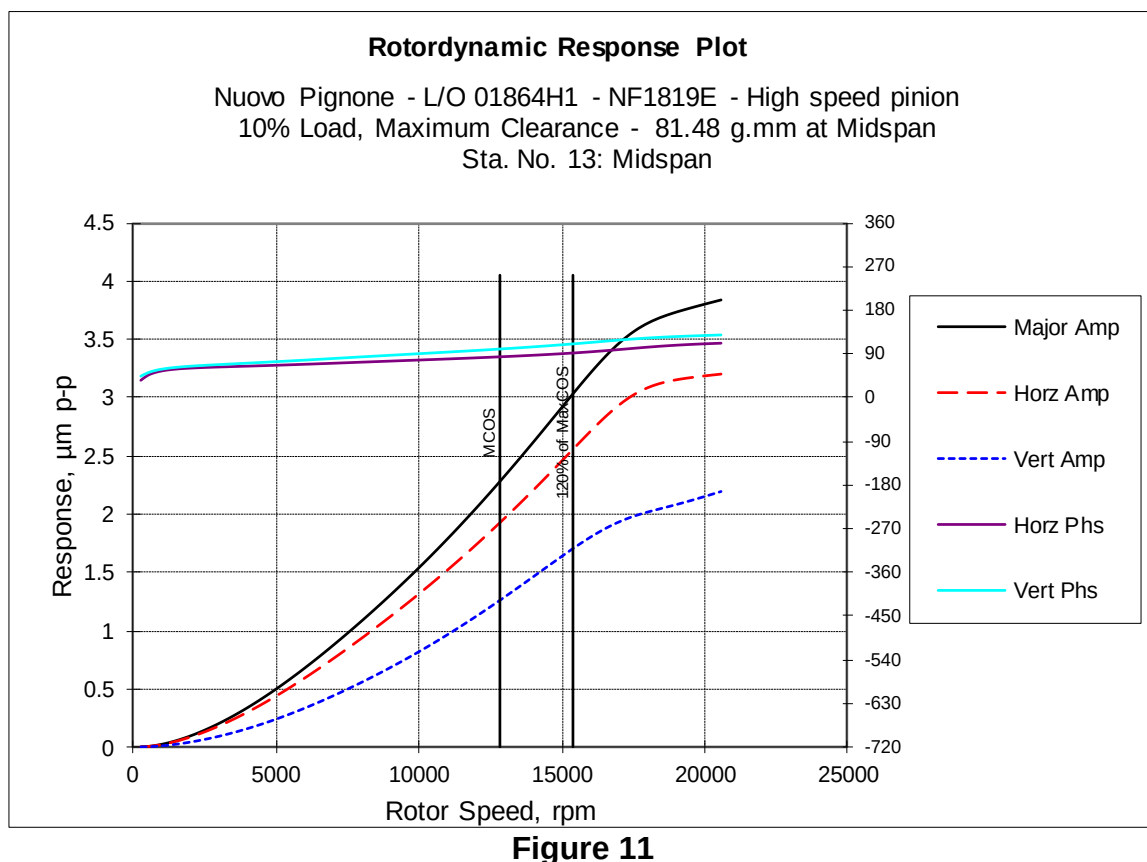
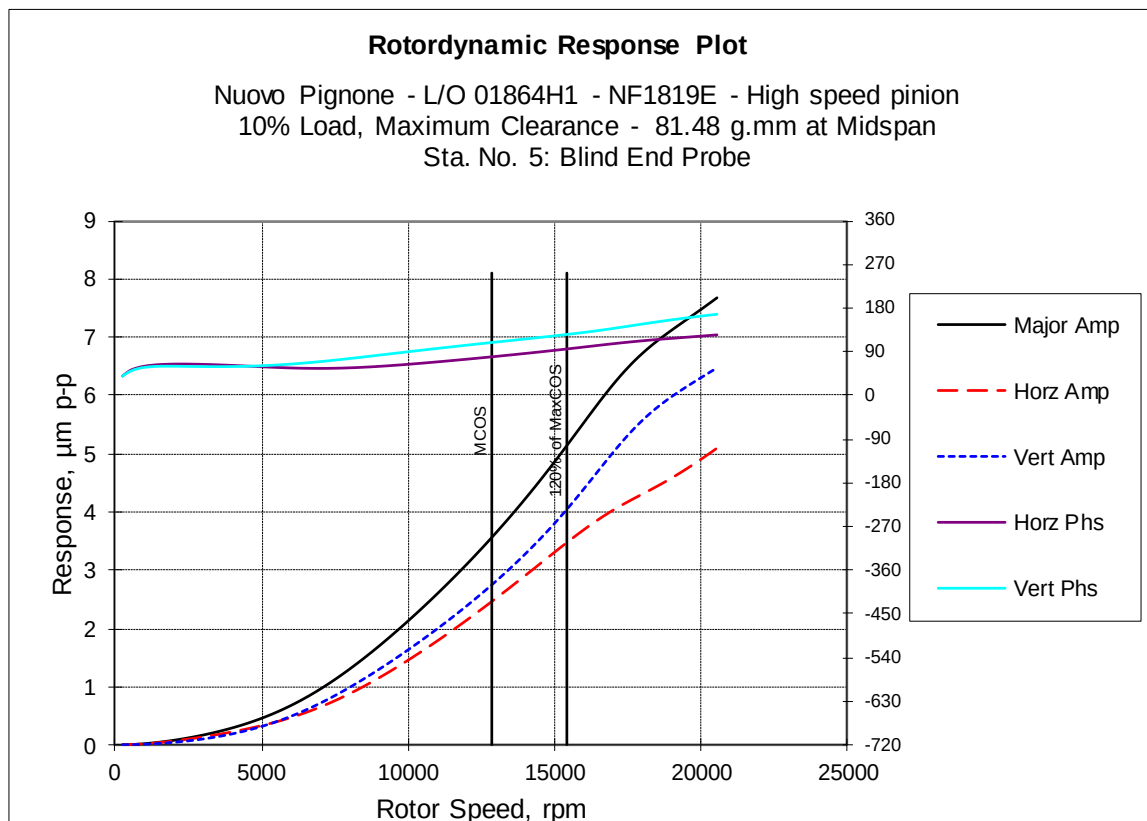
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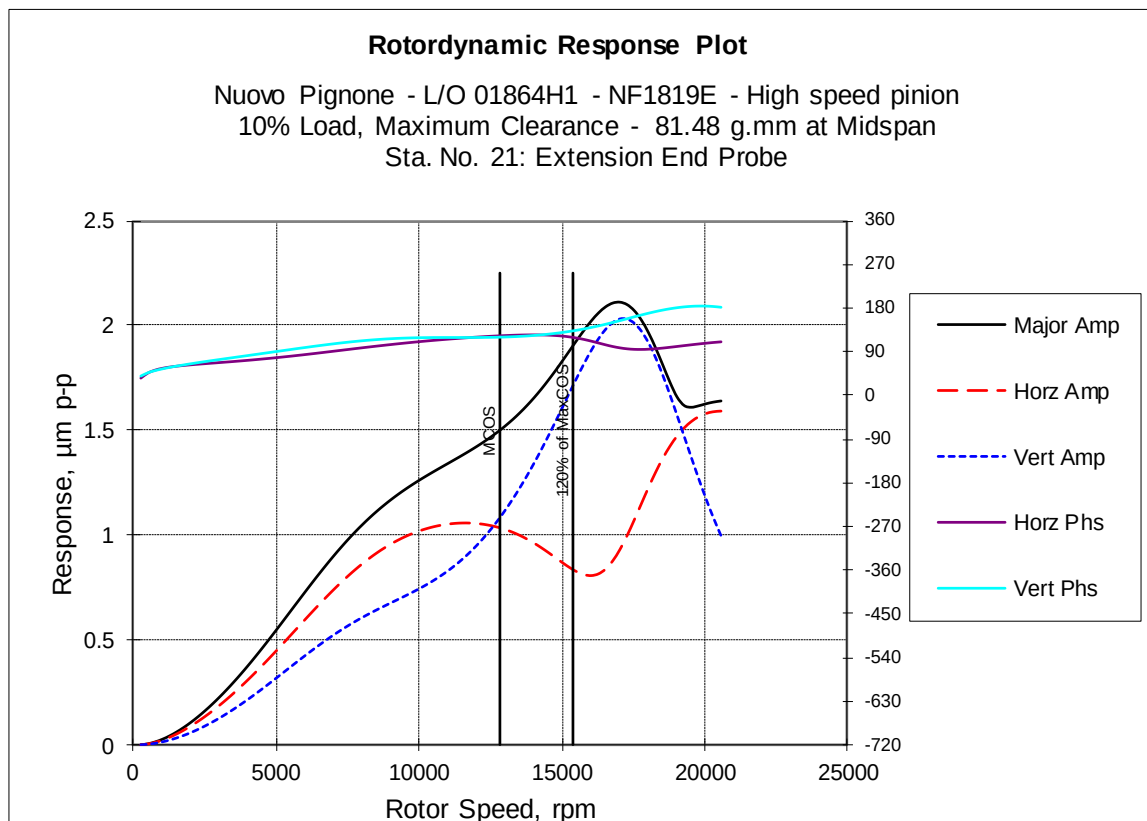


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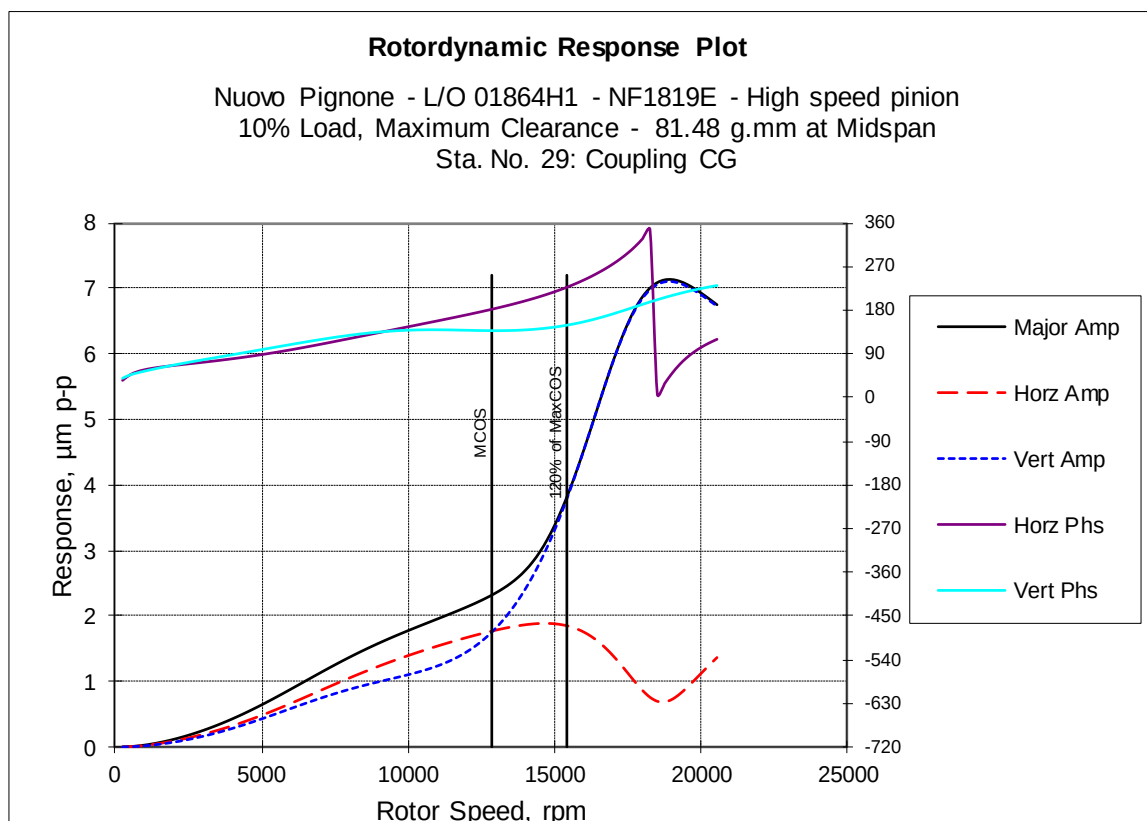
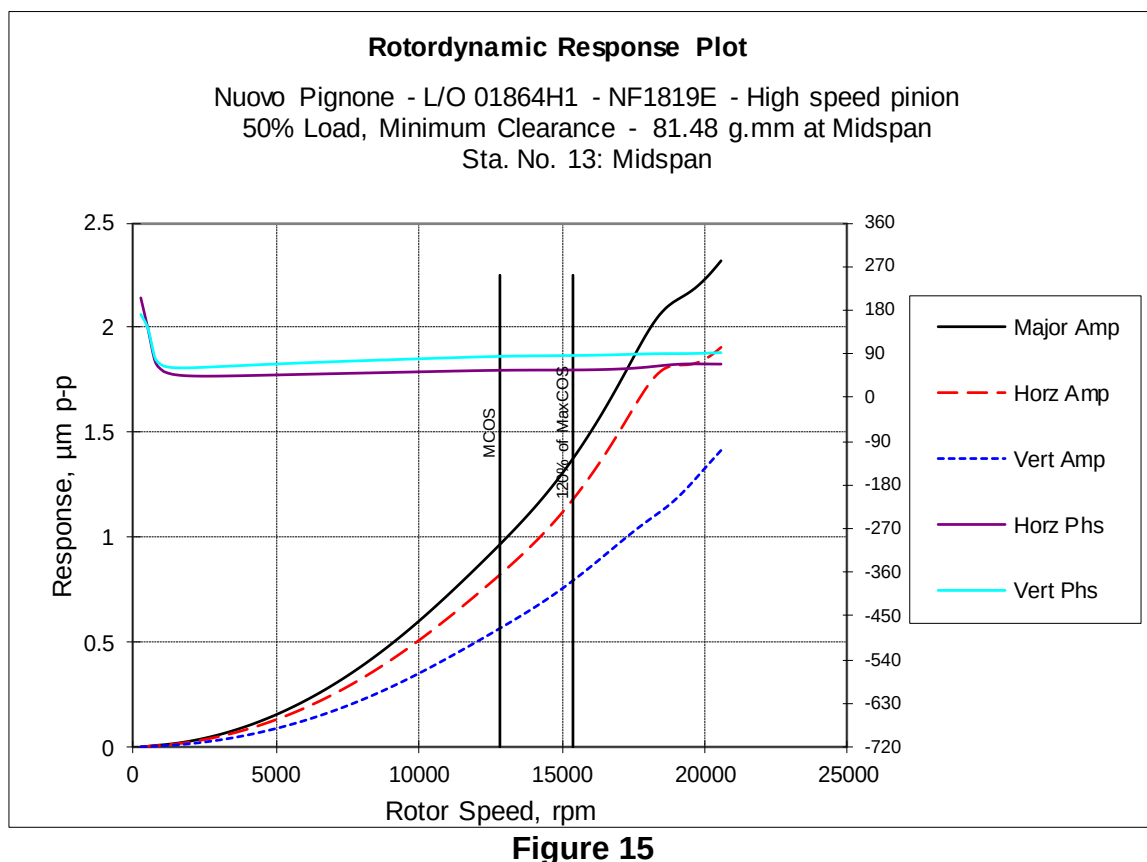
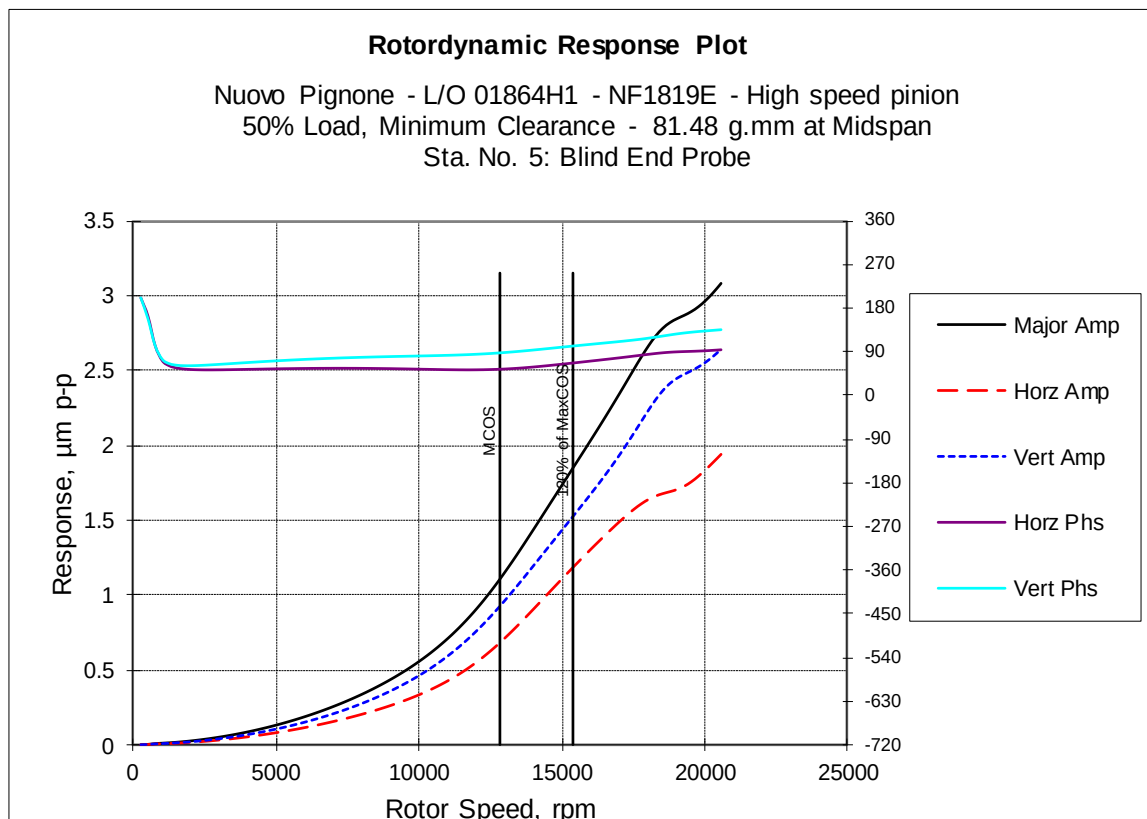


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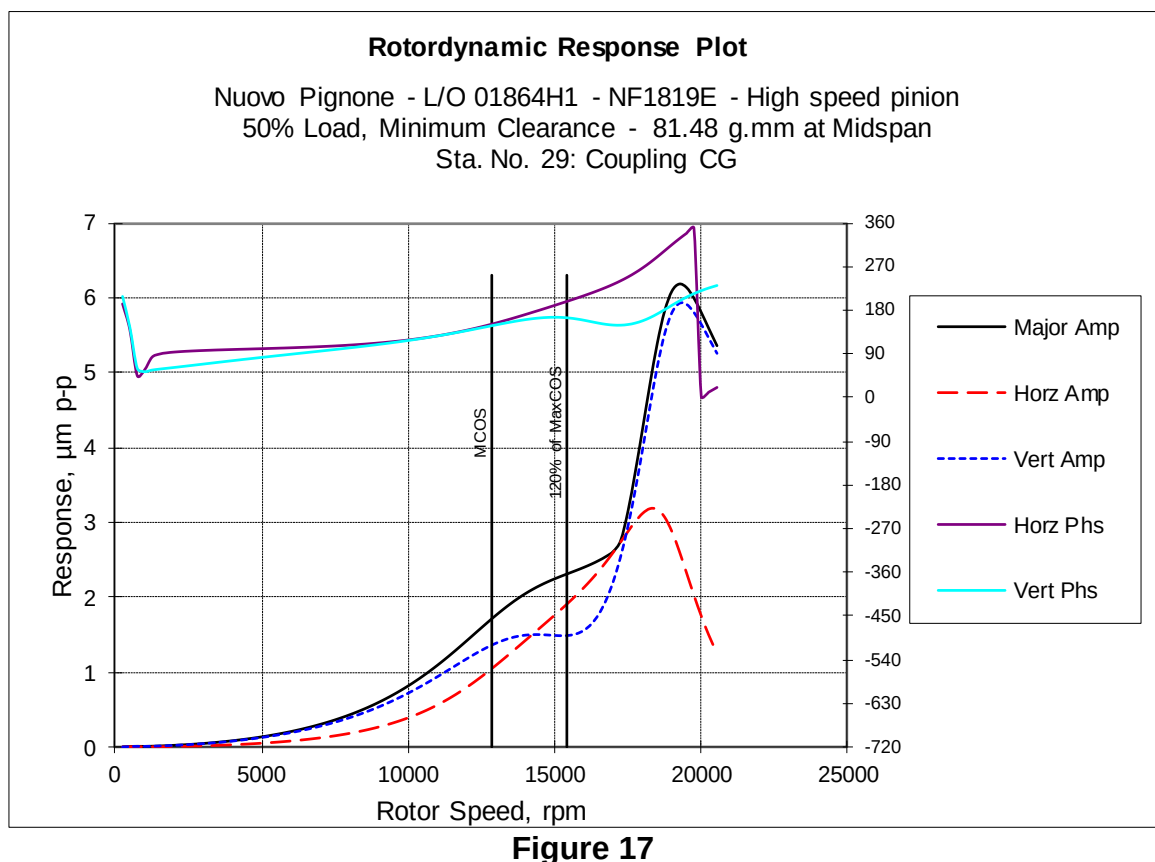
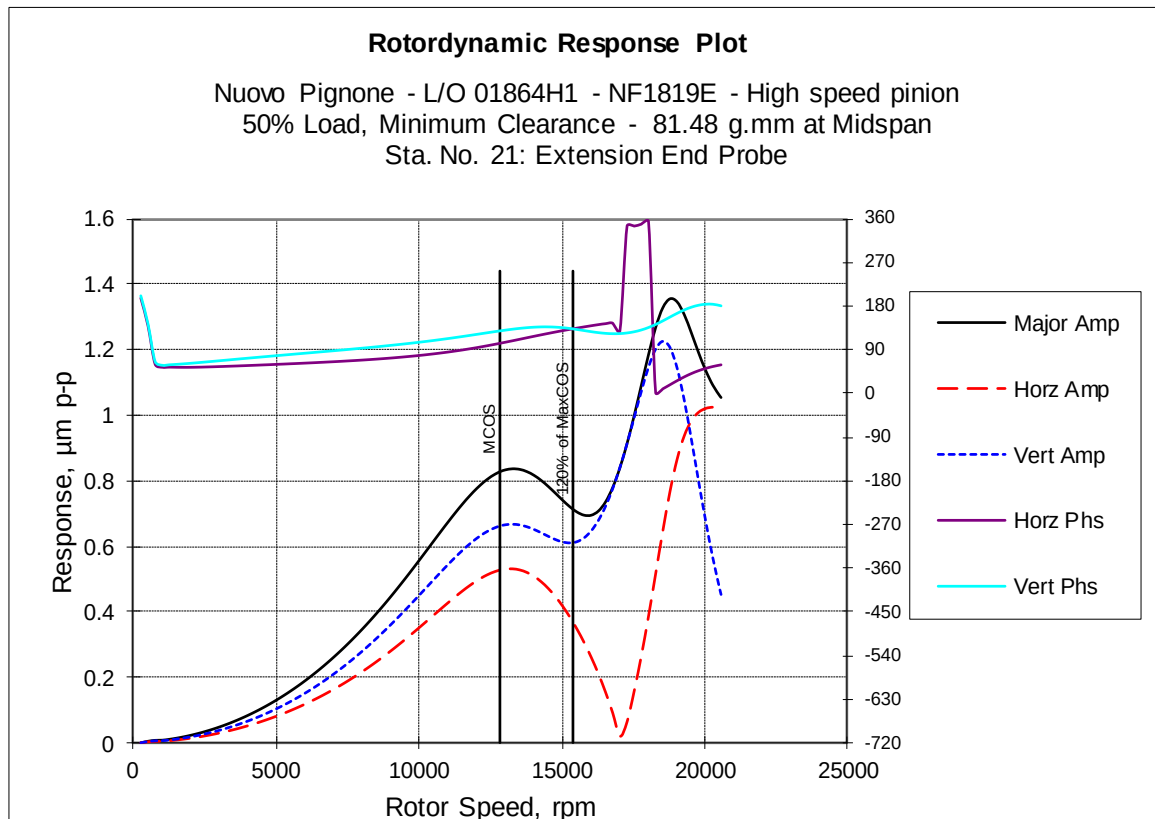
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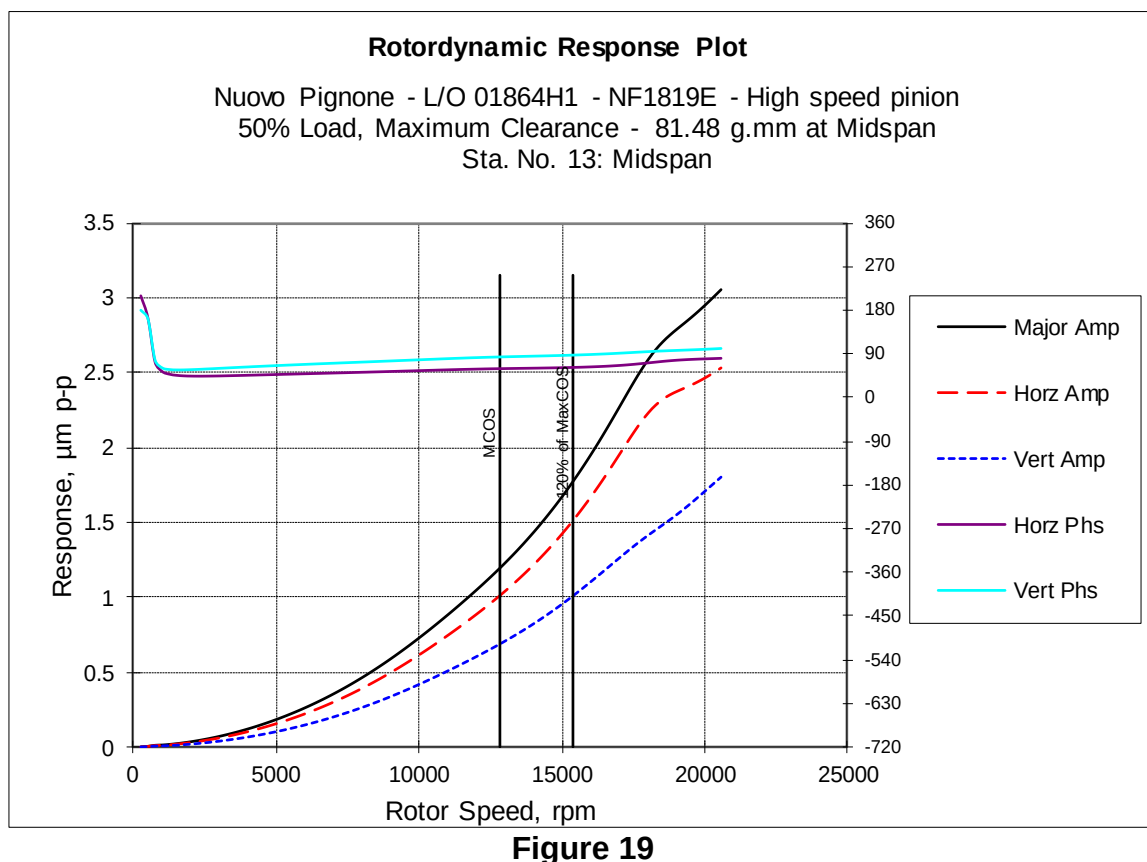
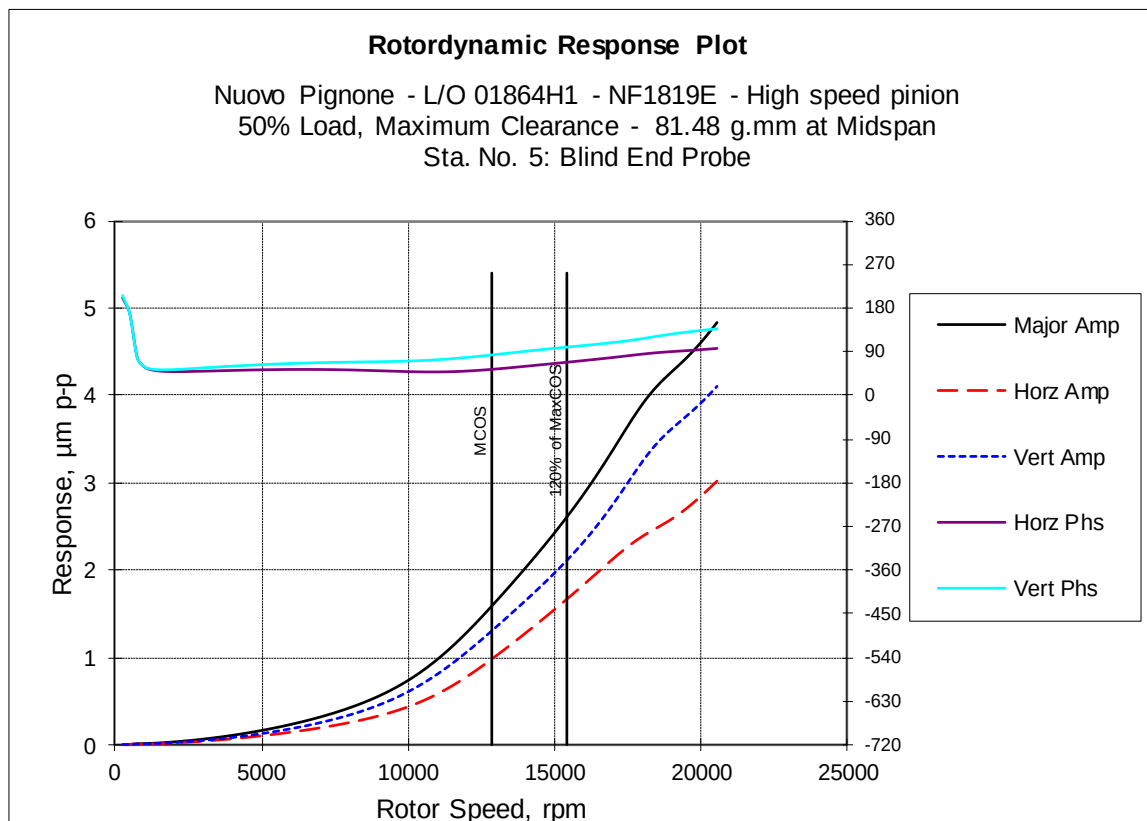
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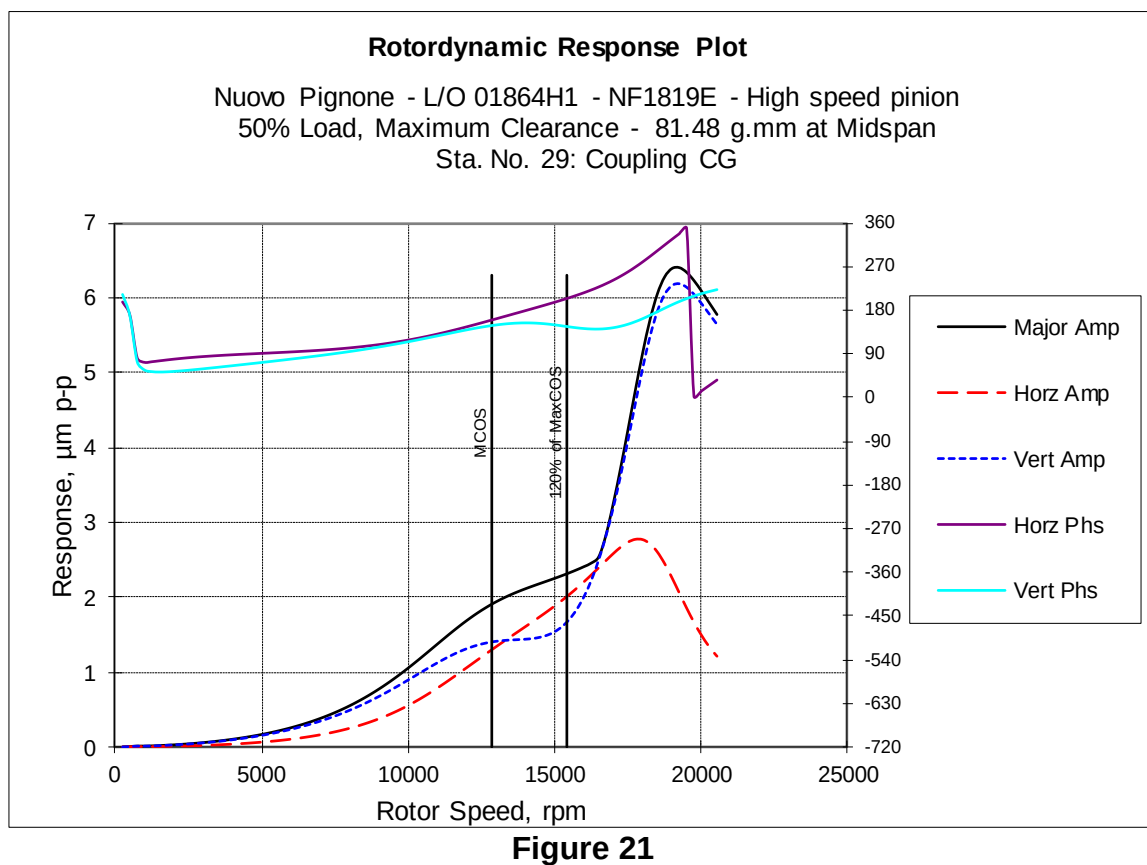
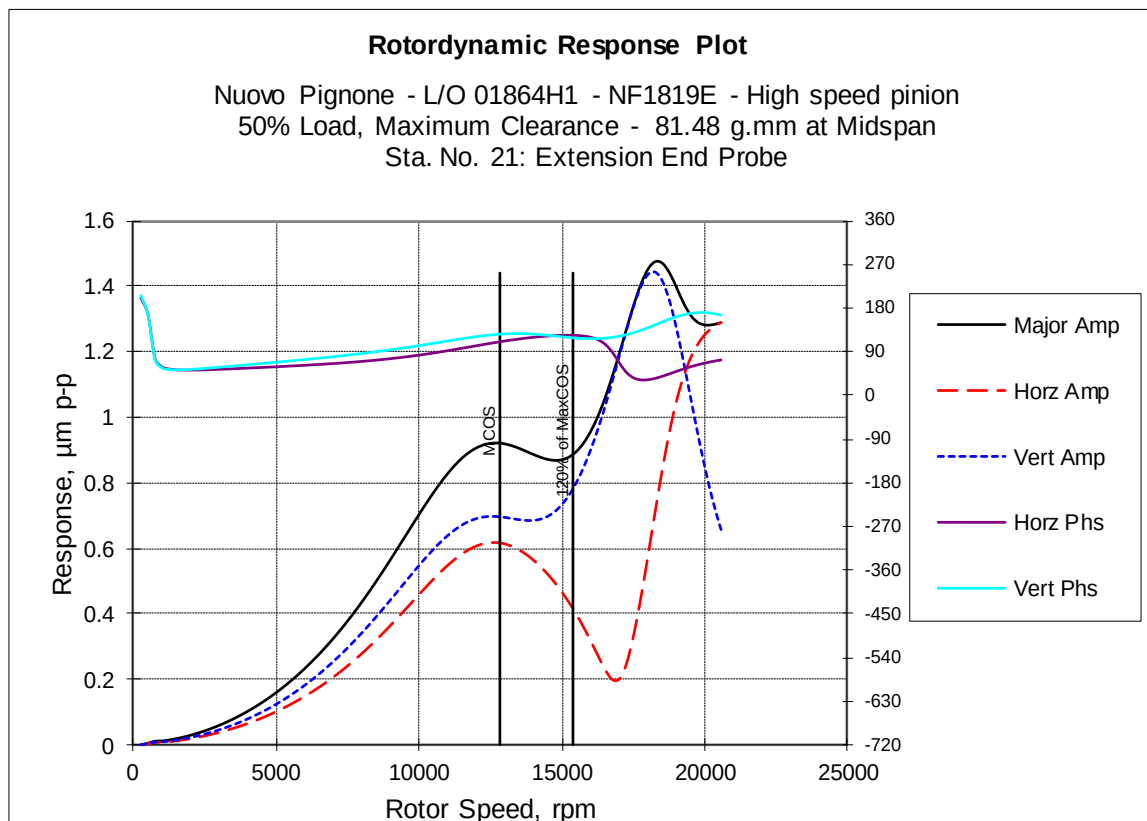
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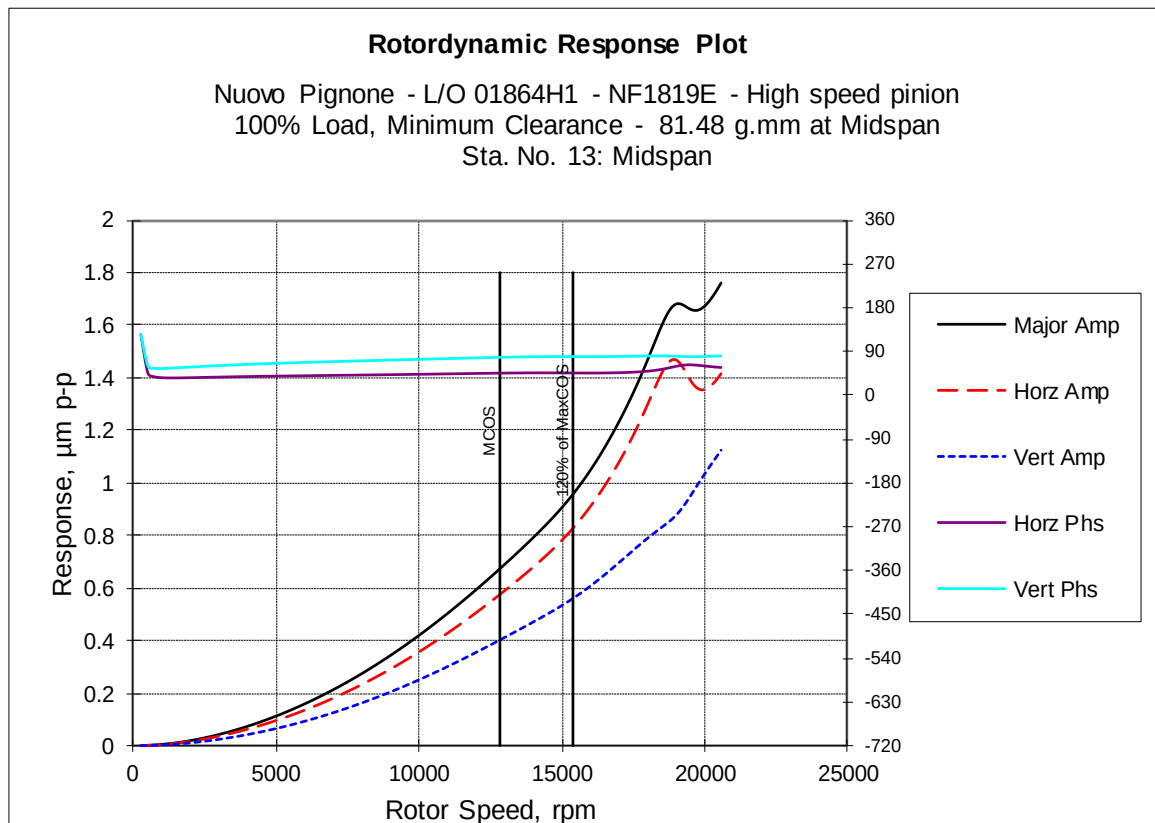
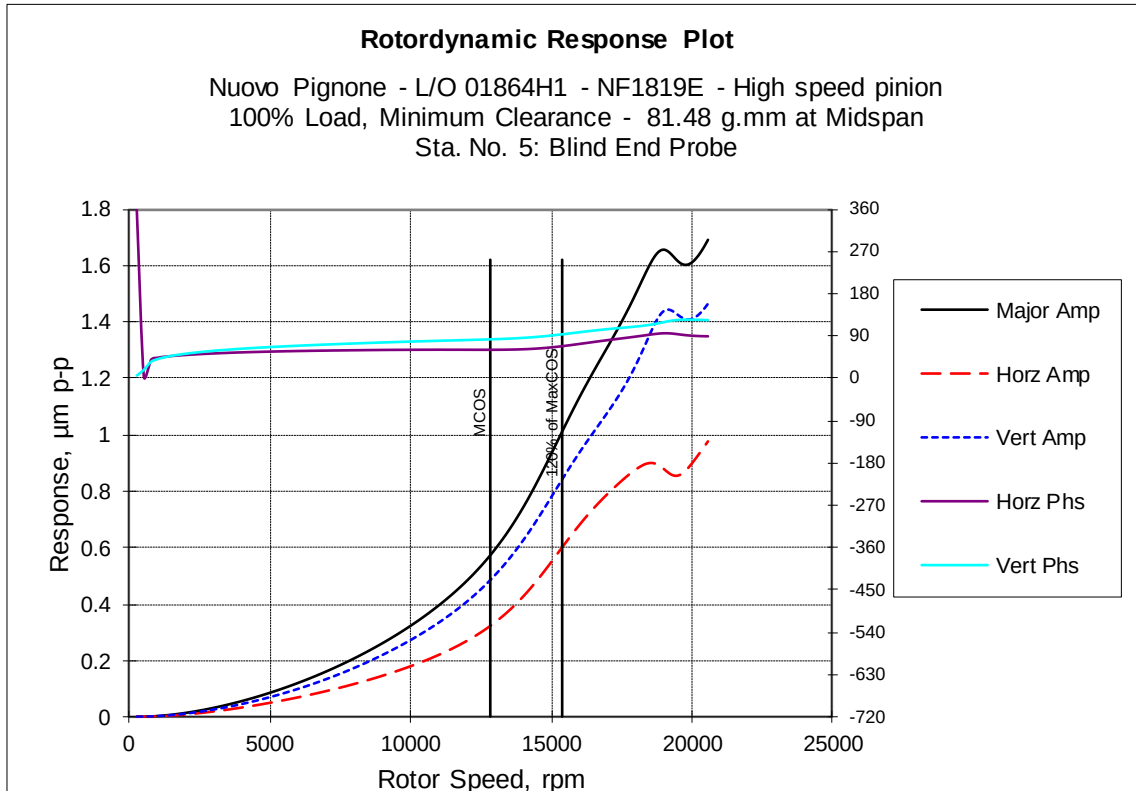
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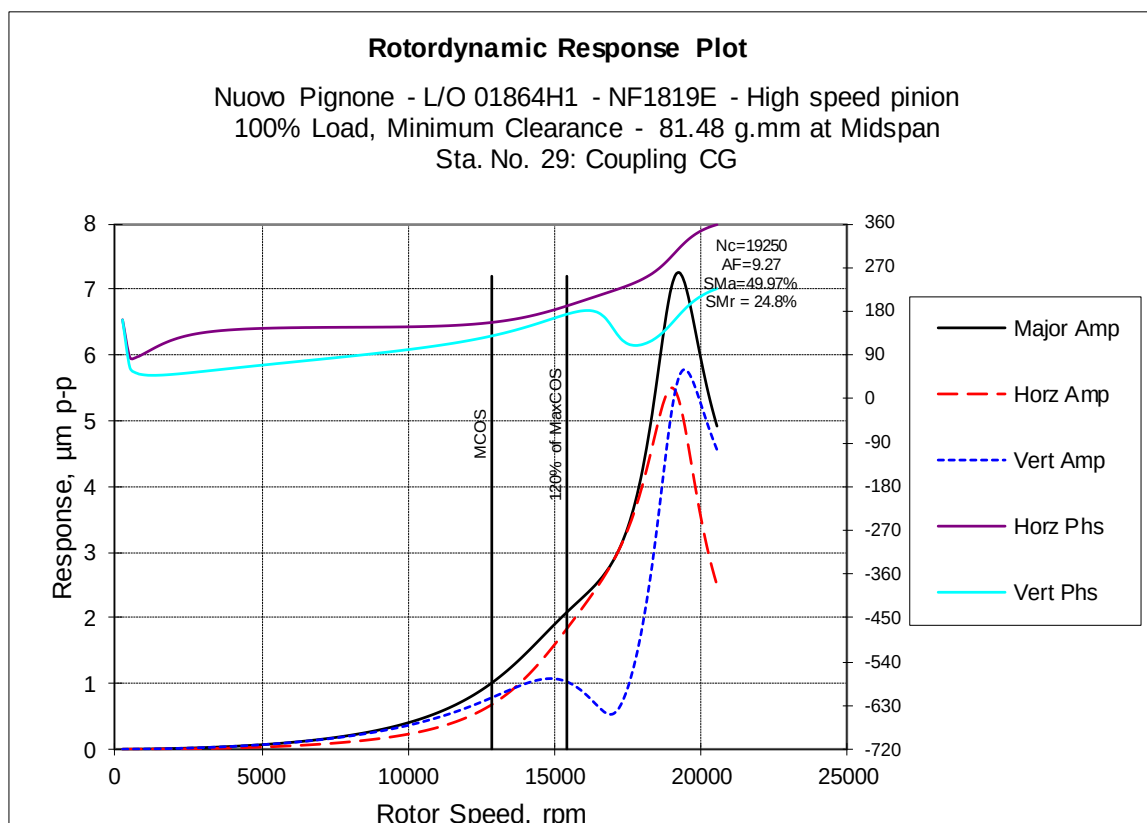
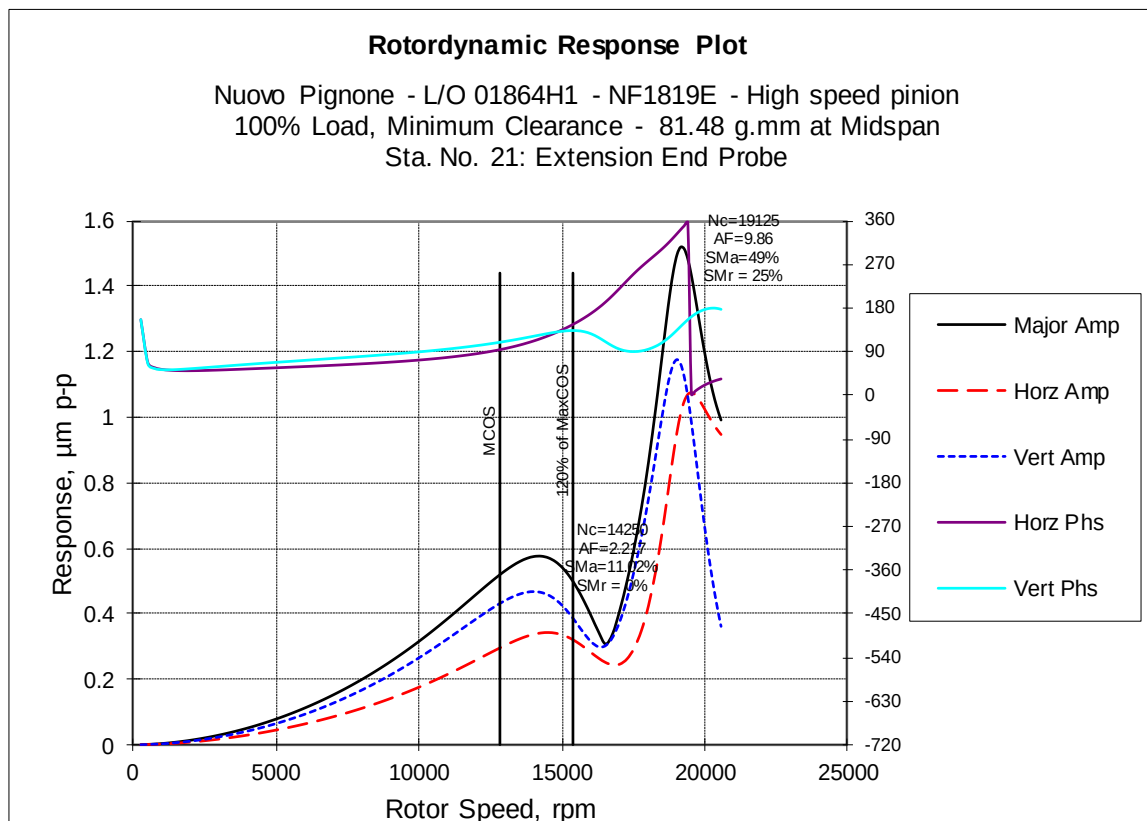
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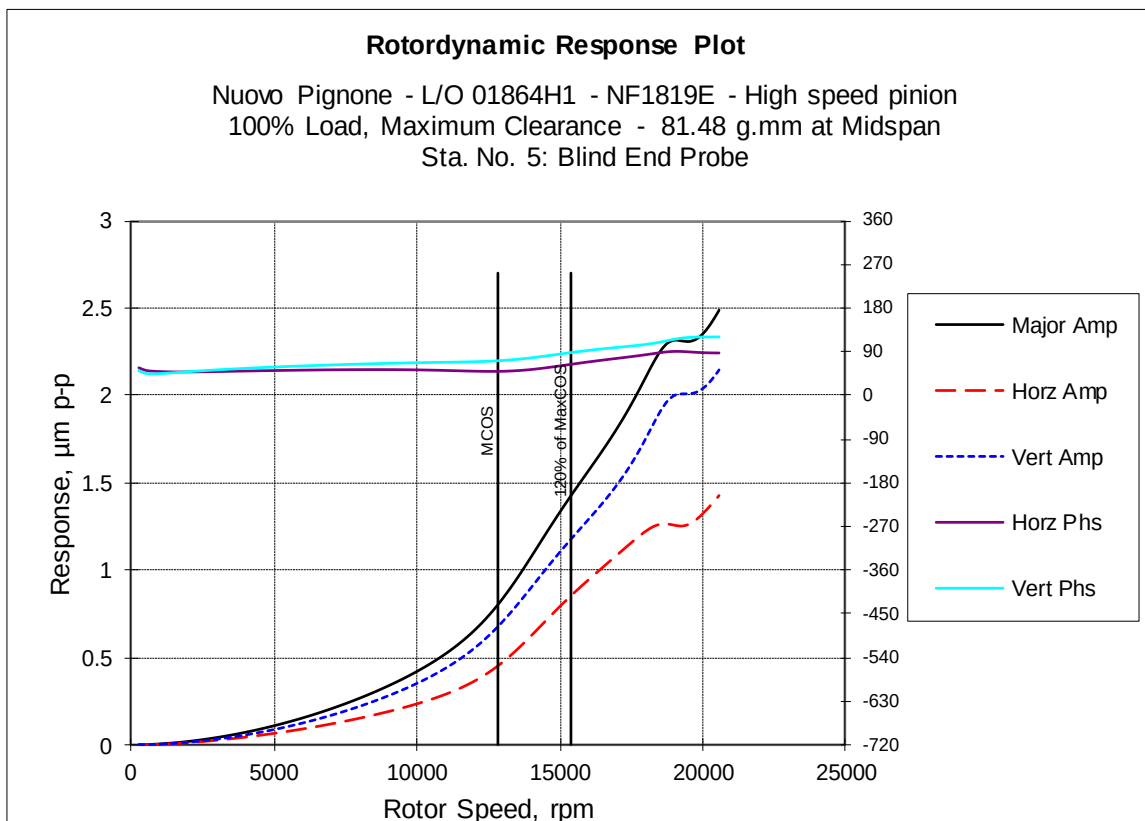


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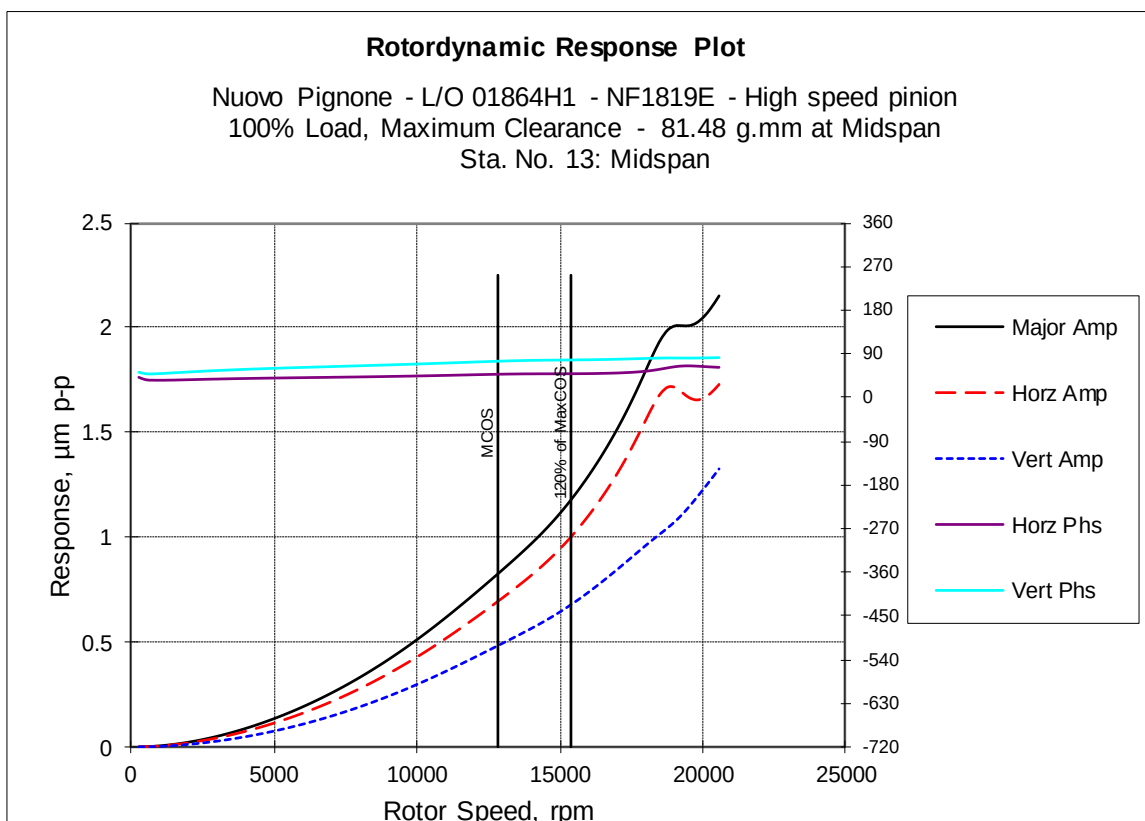


Figure 27



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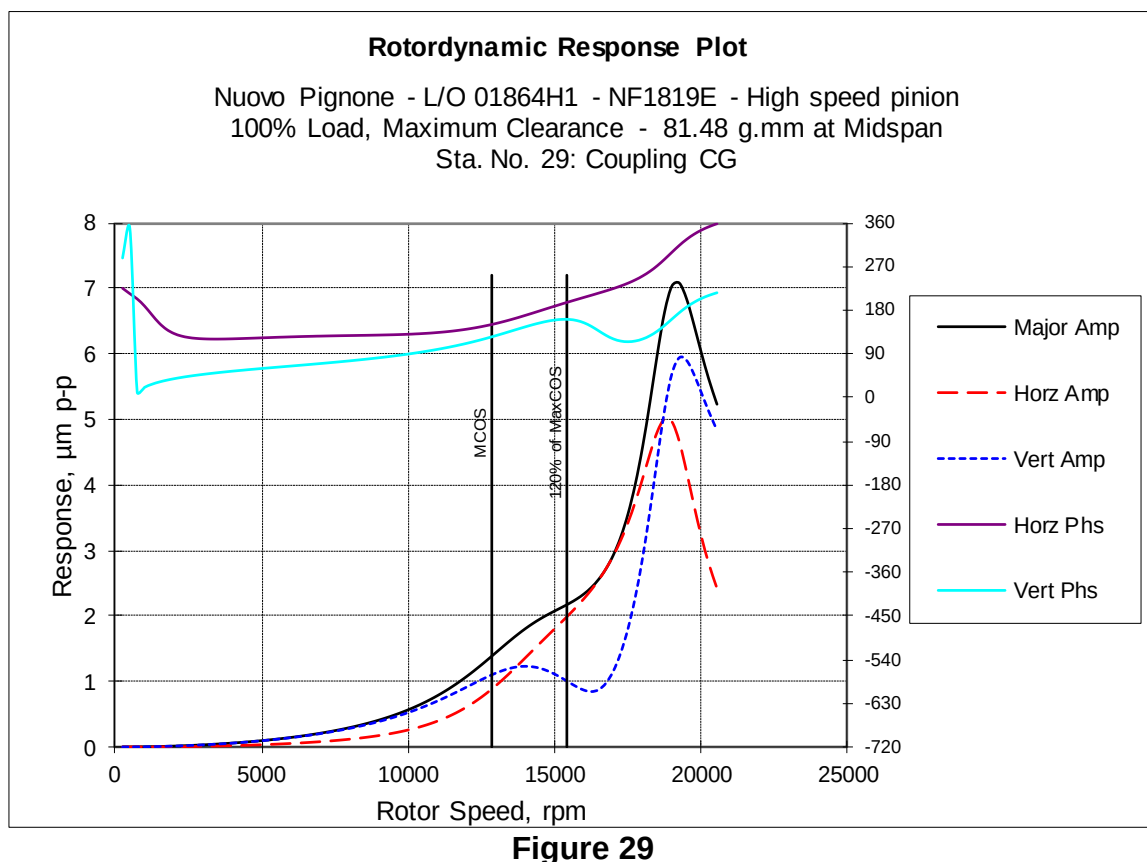
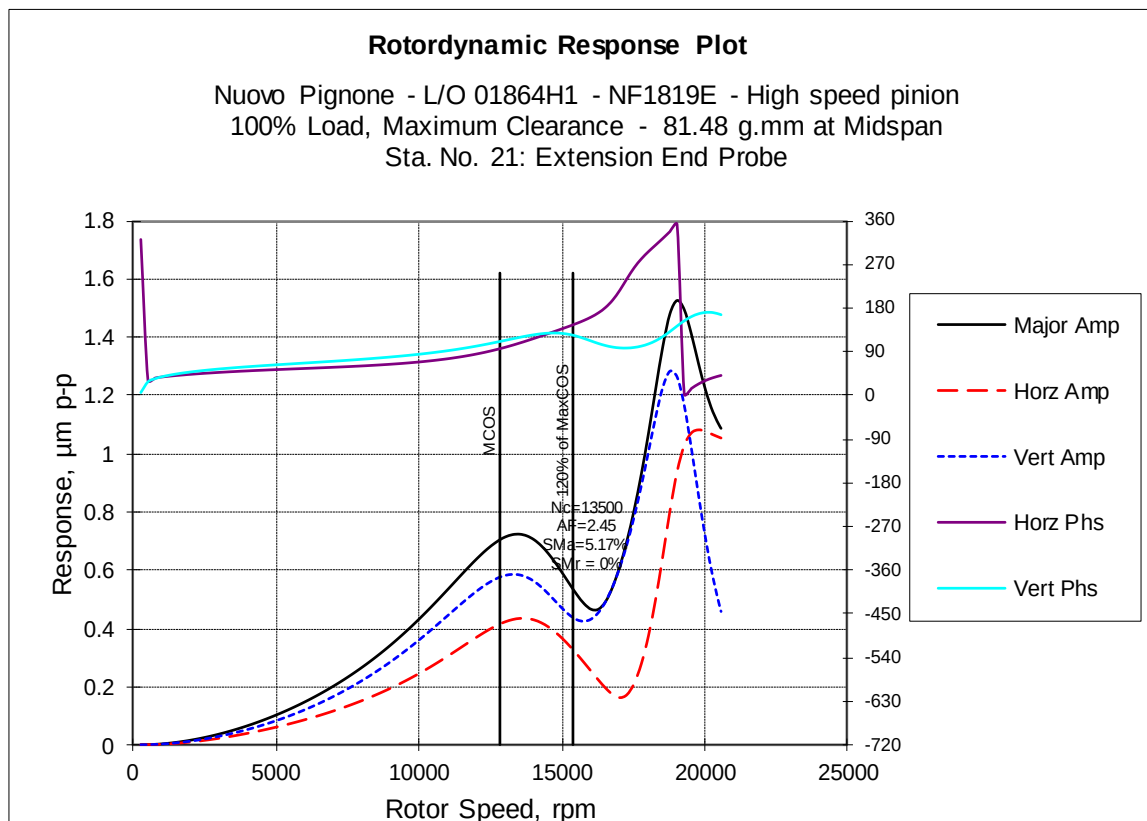
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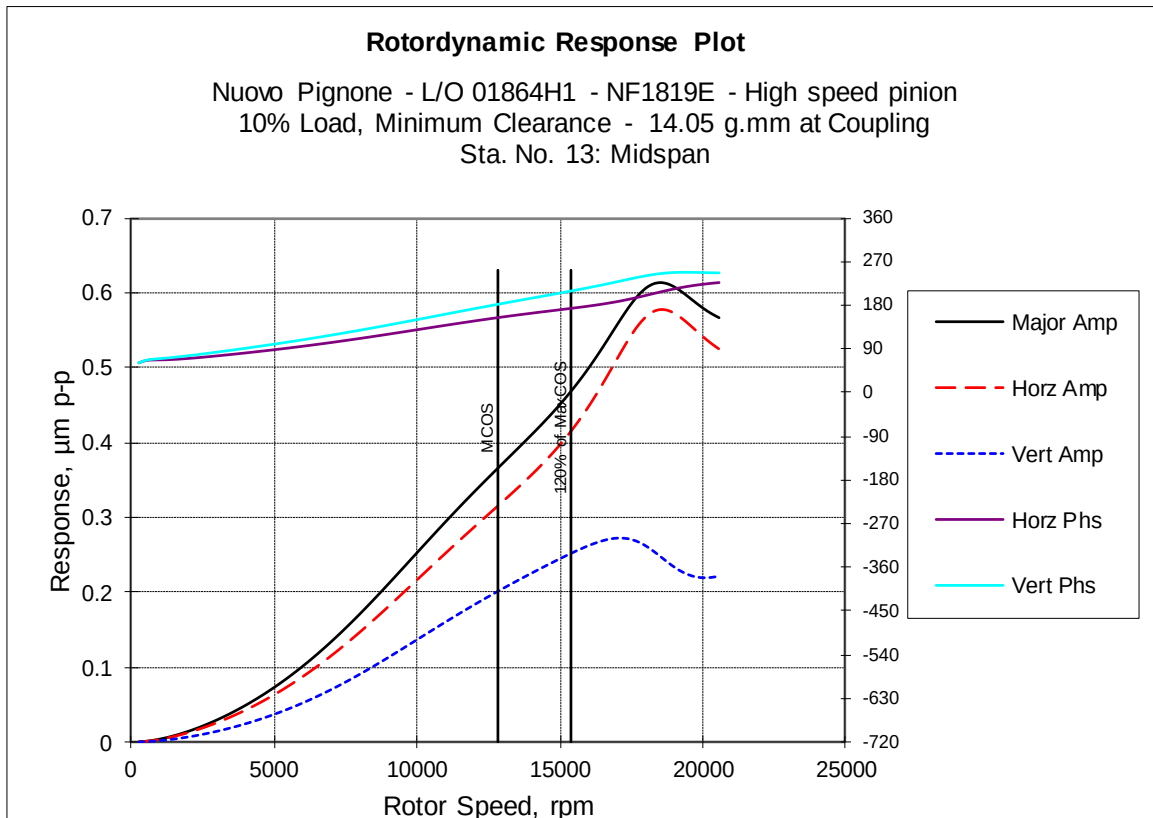
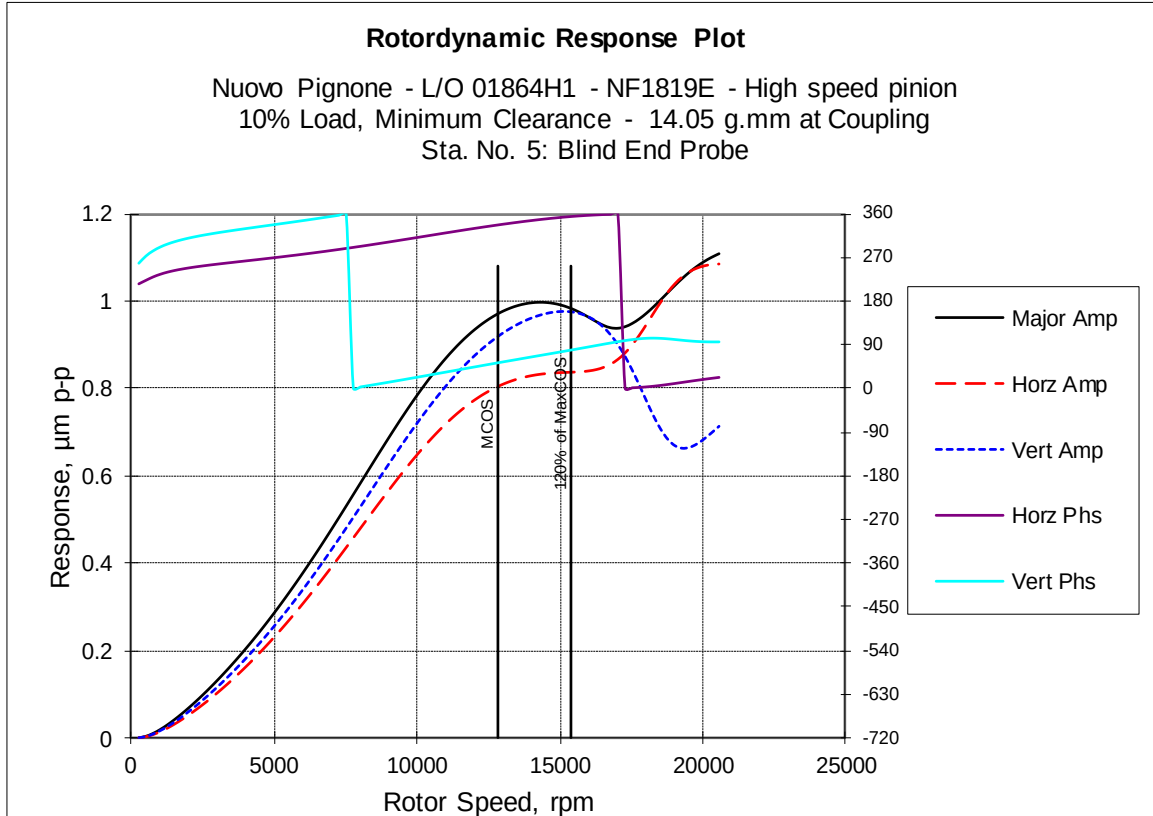
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Appendix F - Coupling Unbalance Response Plots



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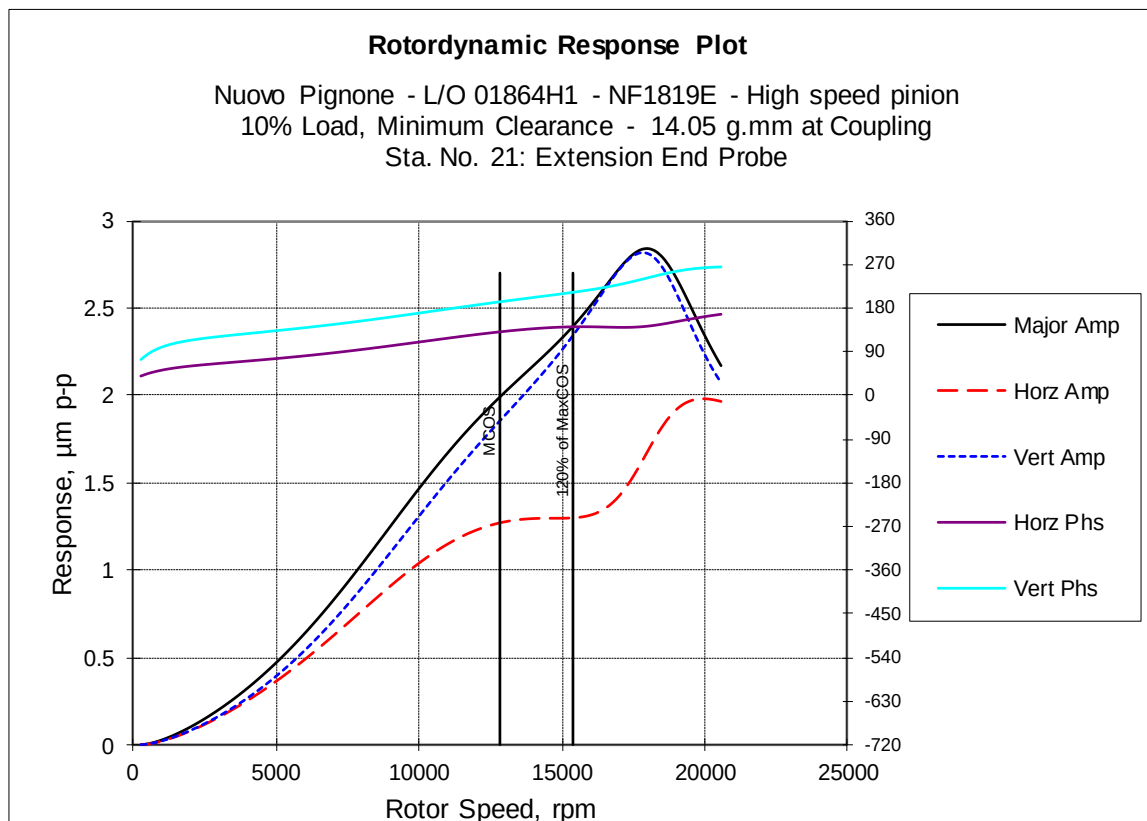


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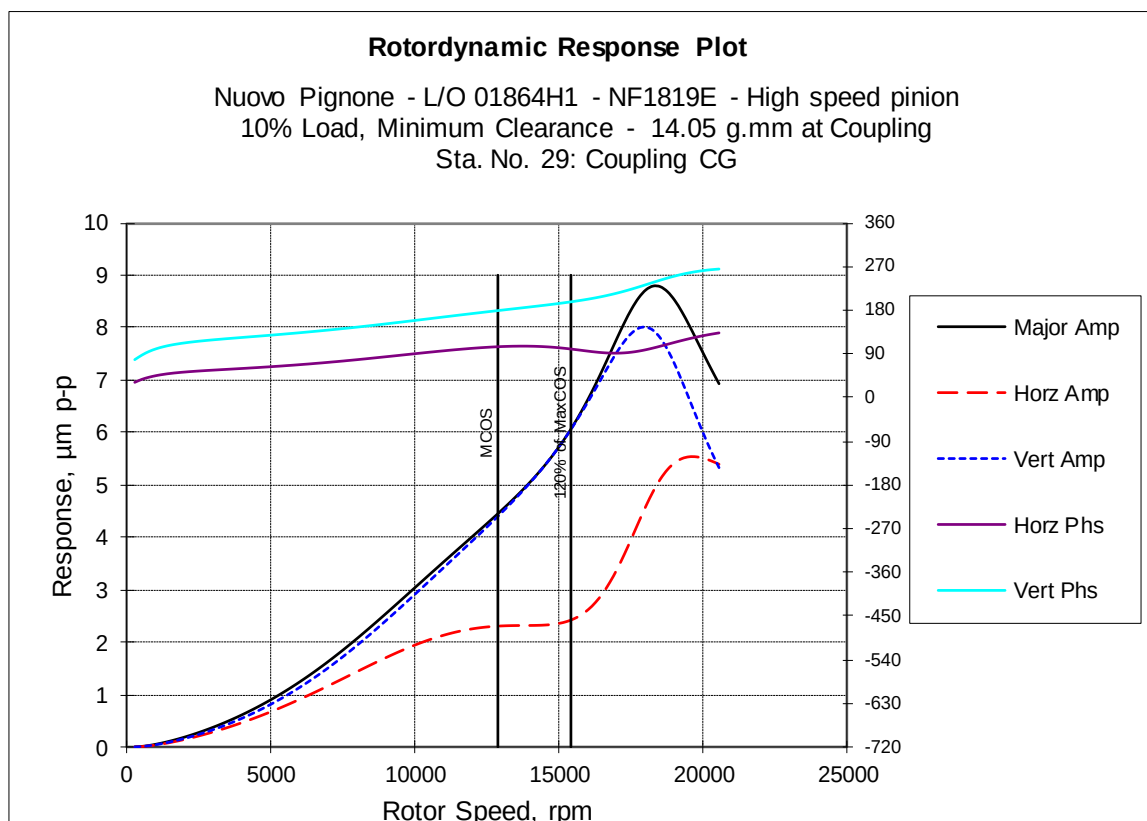


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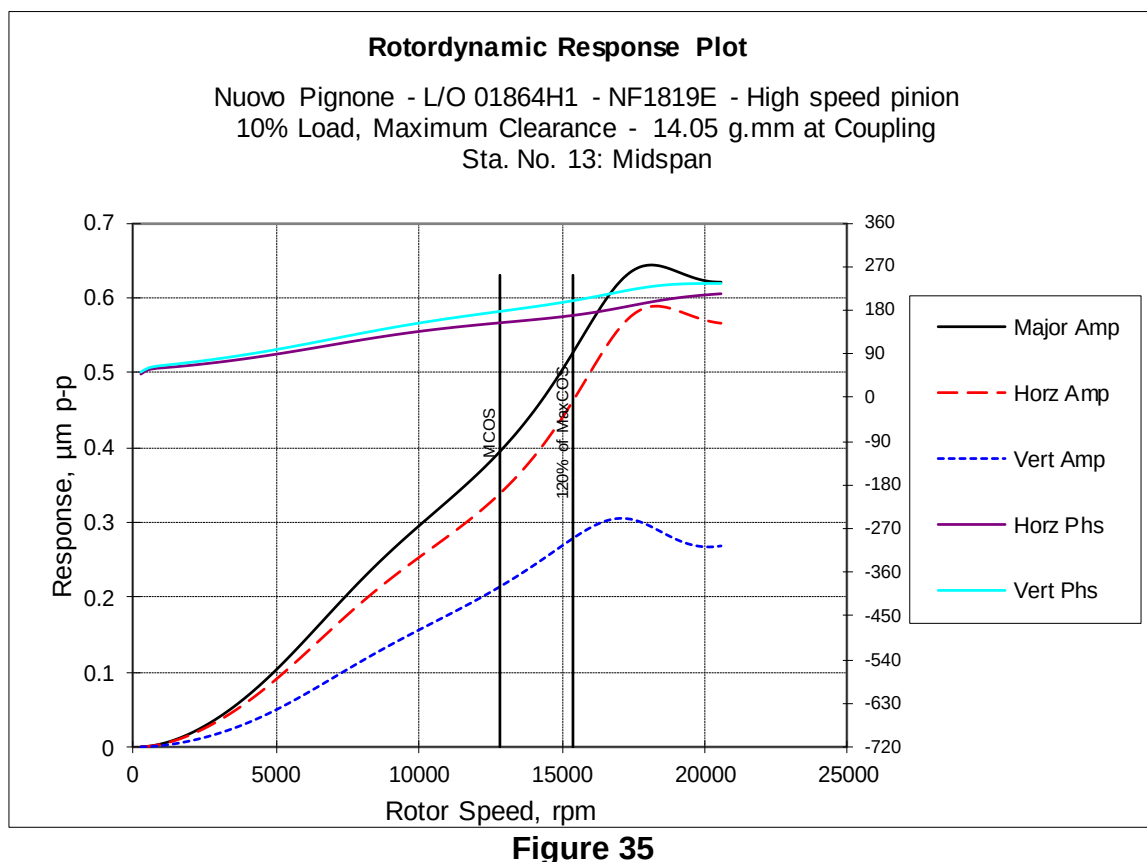
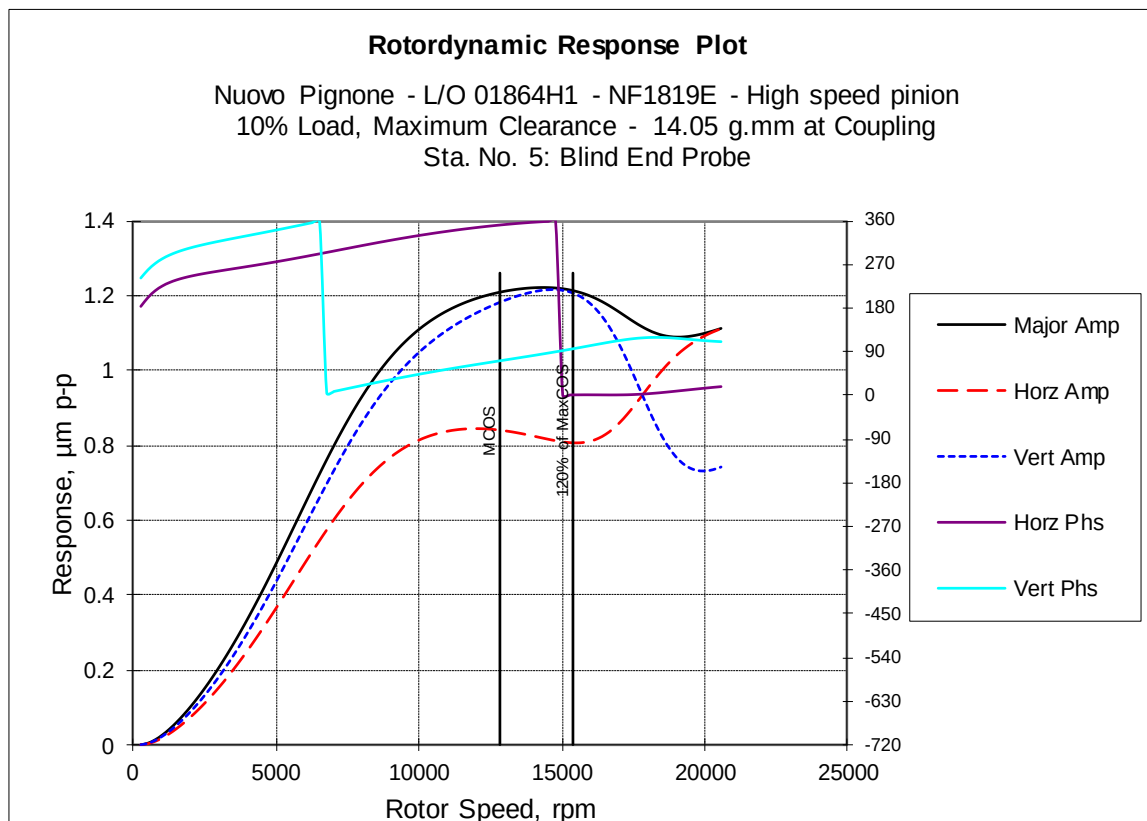
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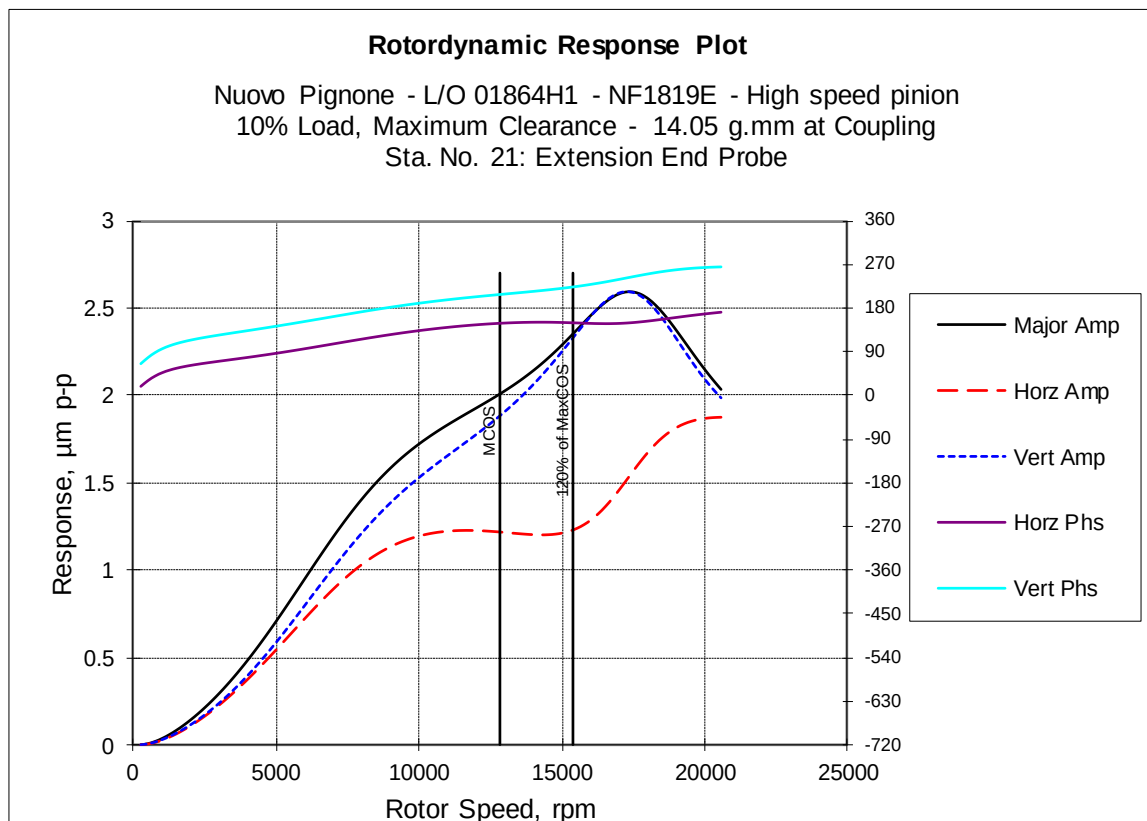


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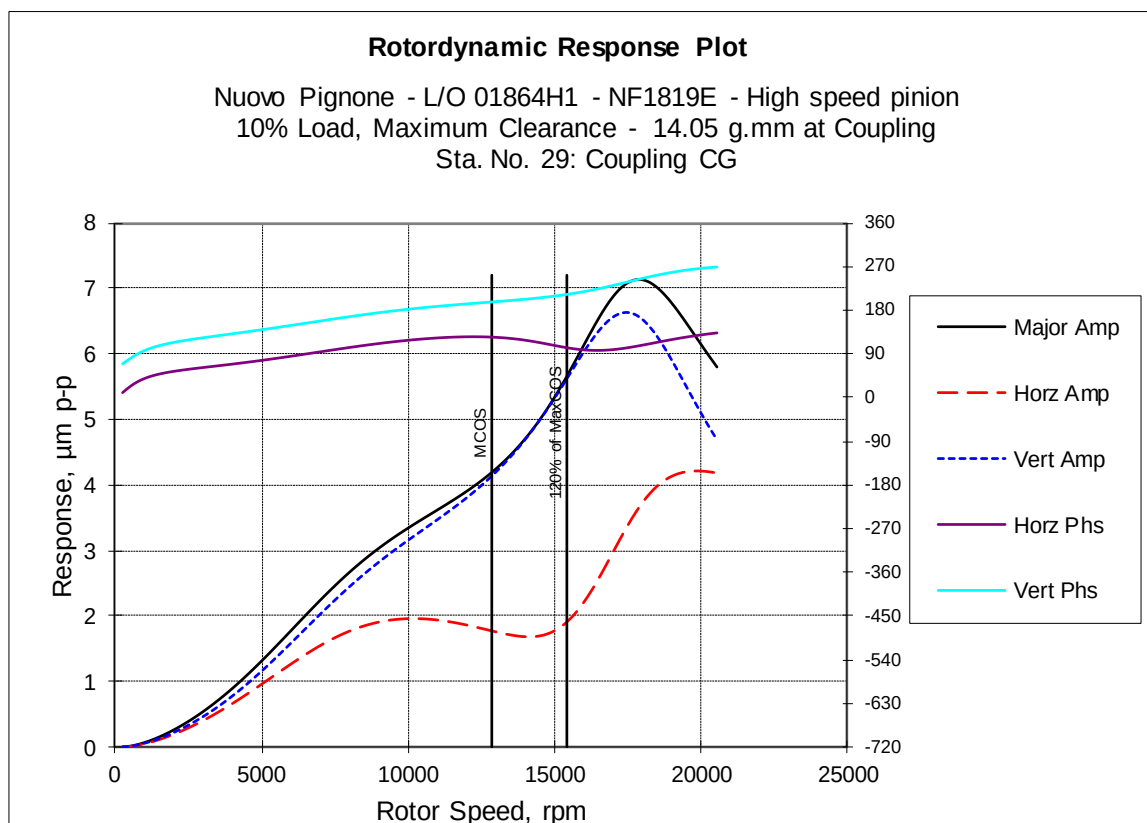


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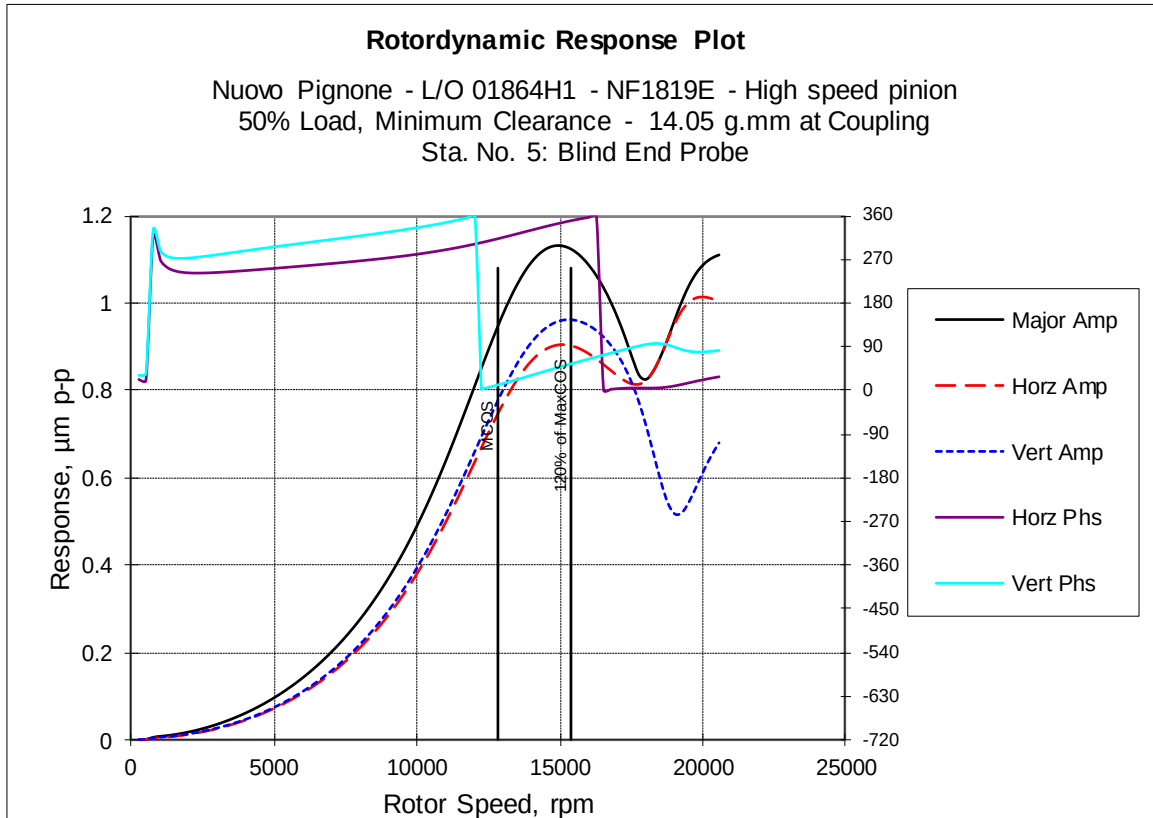


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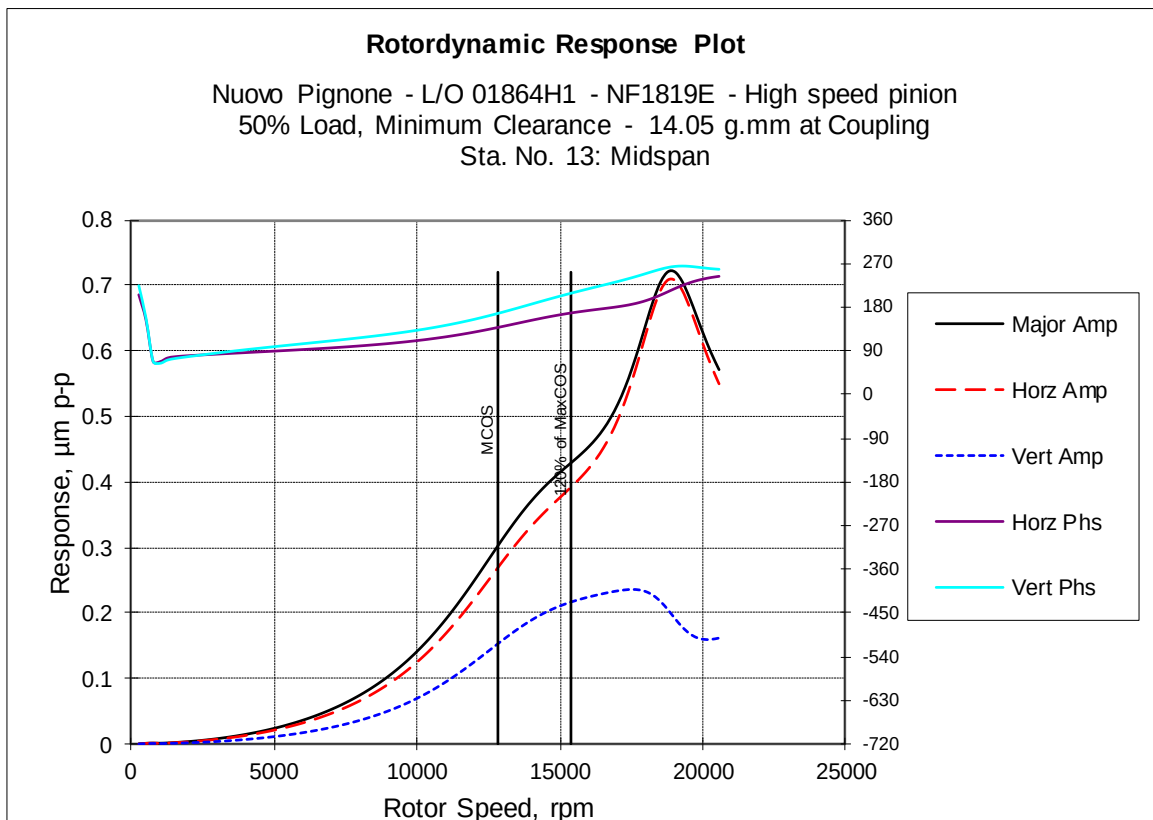


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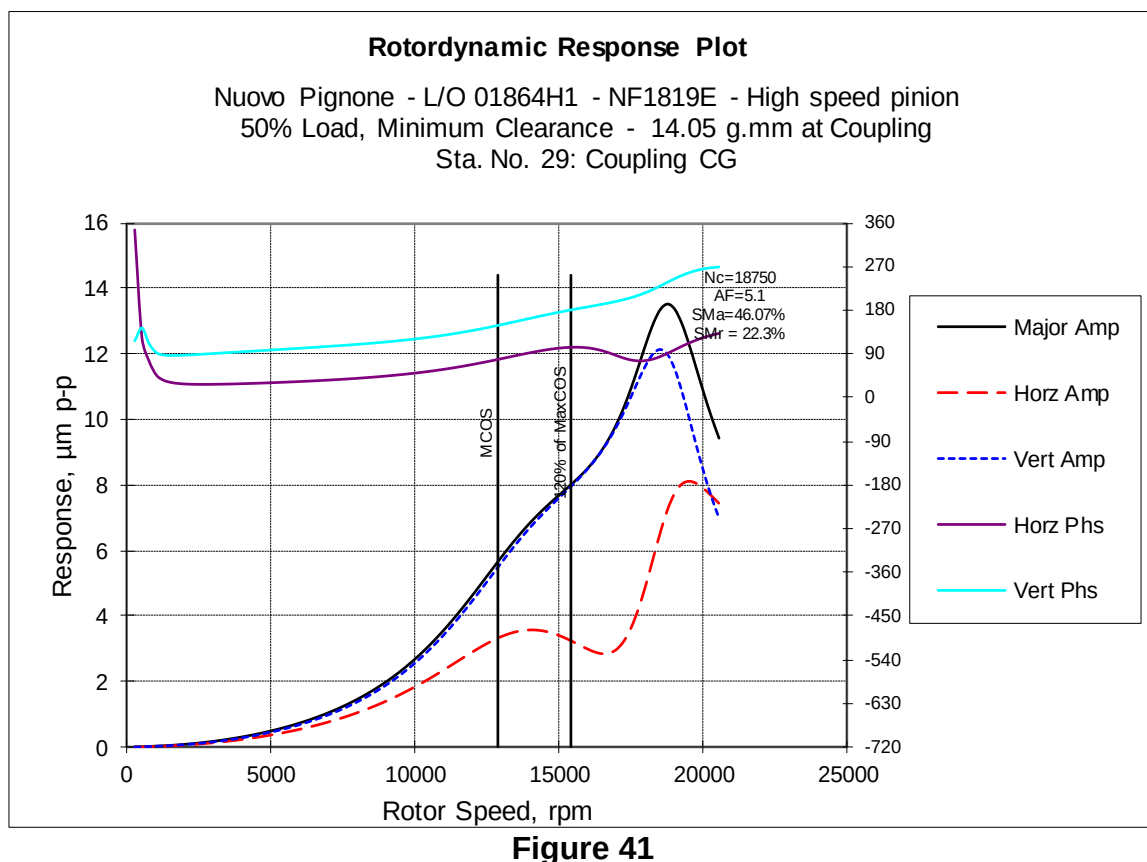
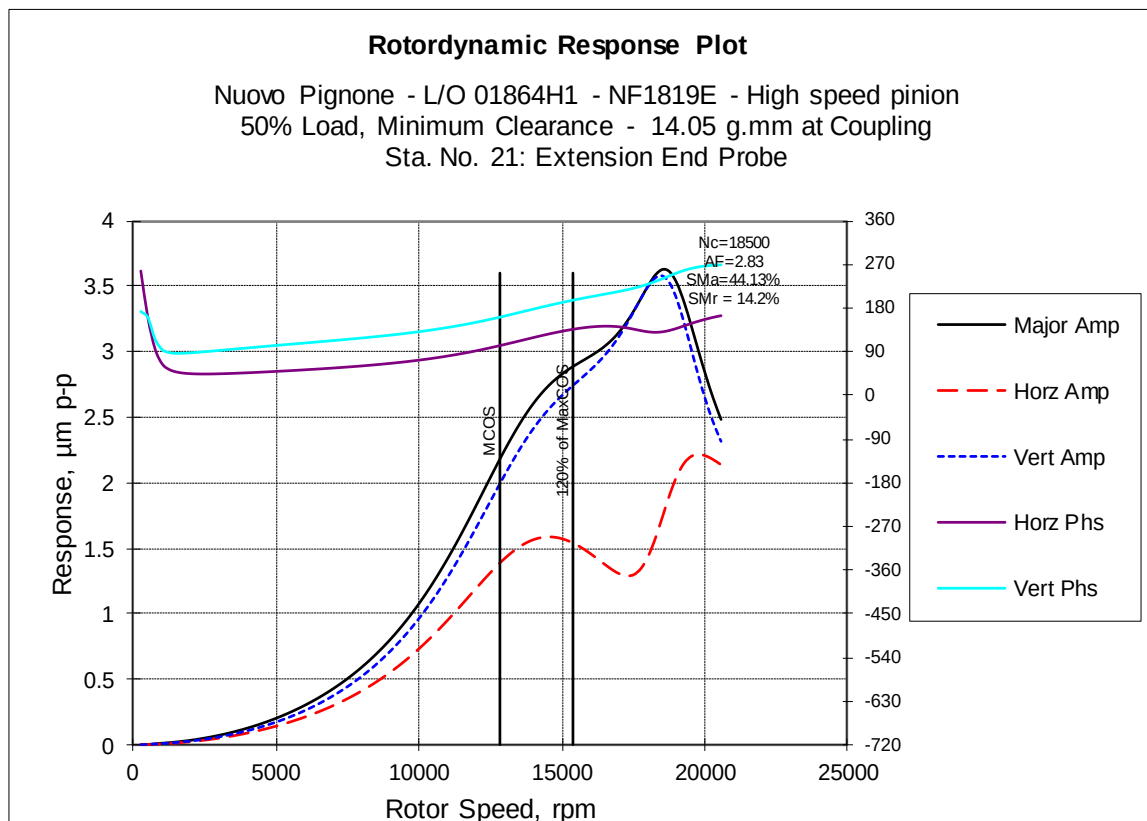
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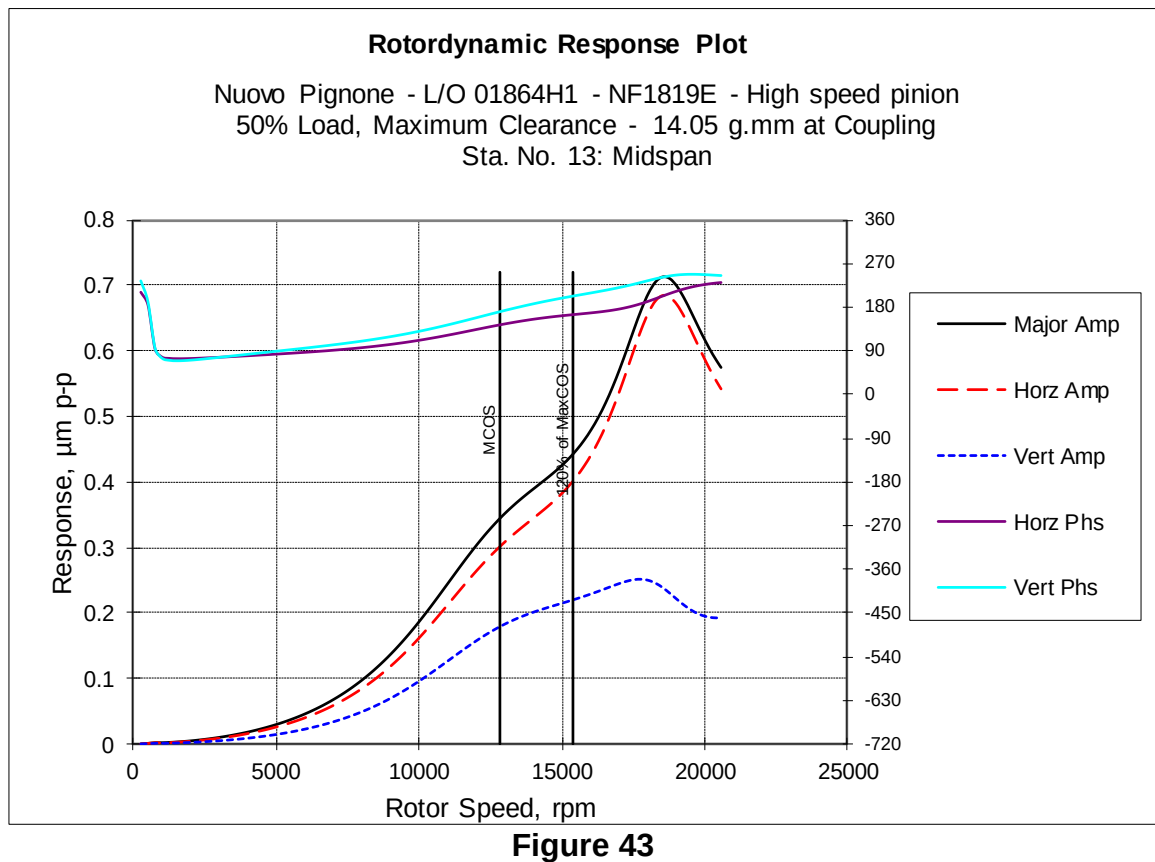
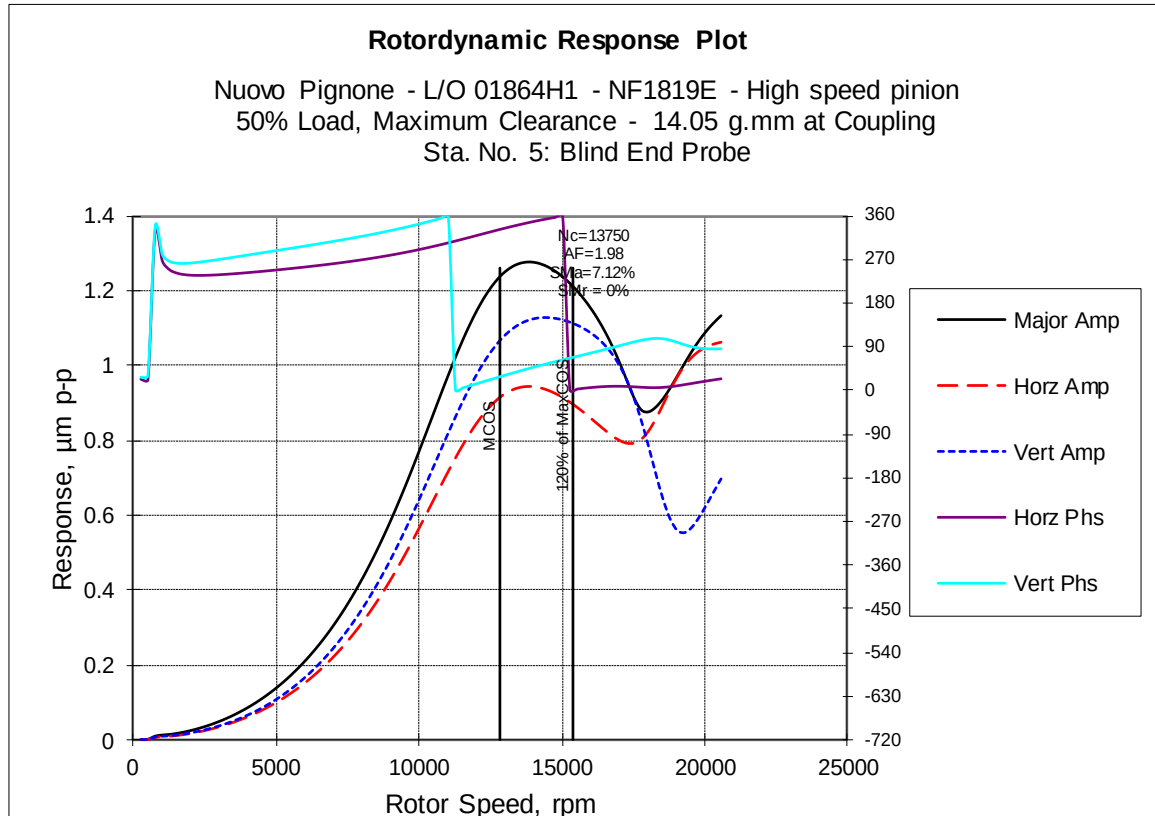
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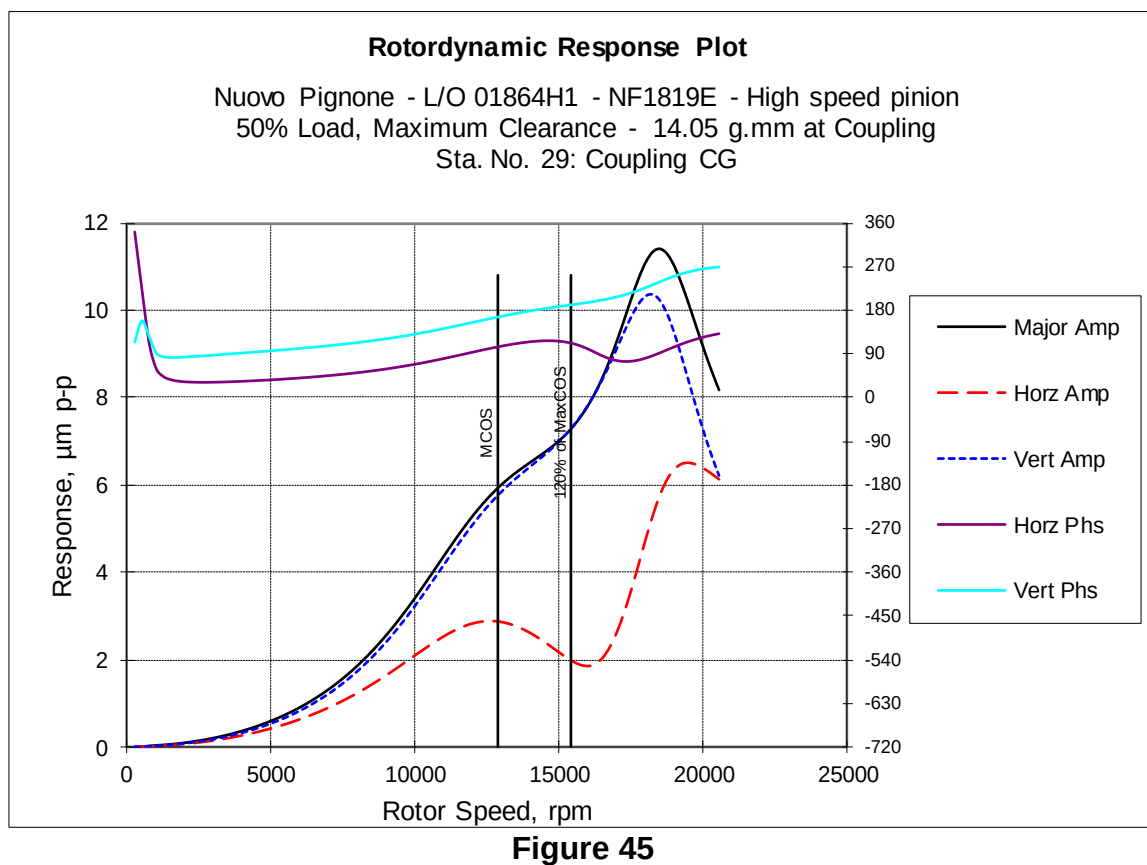
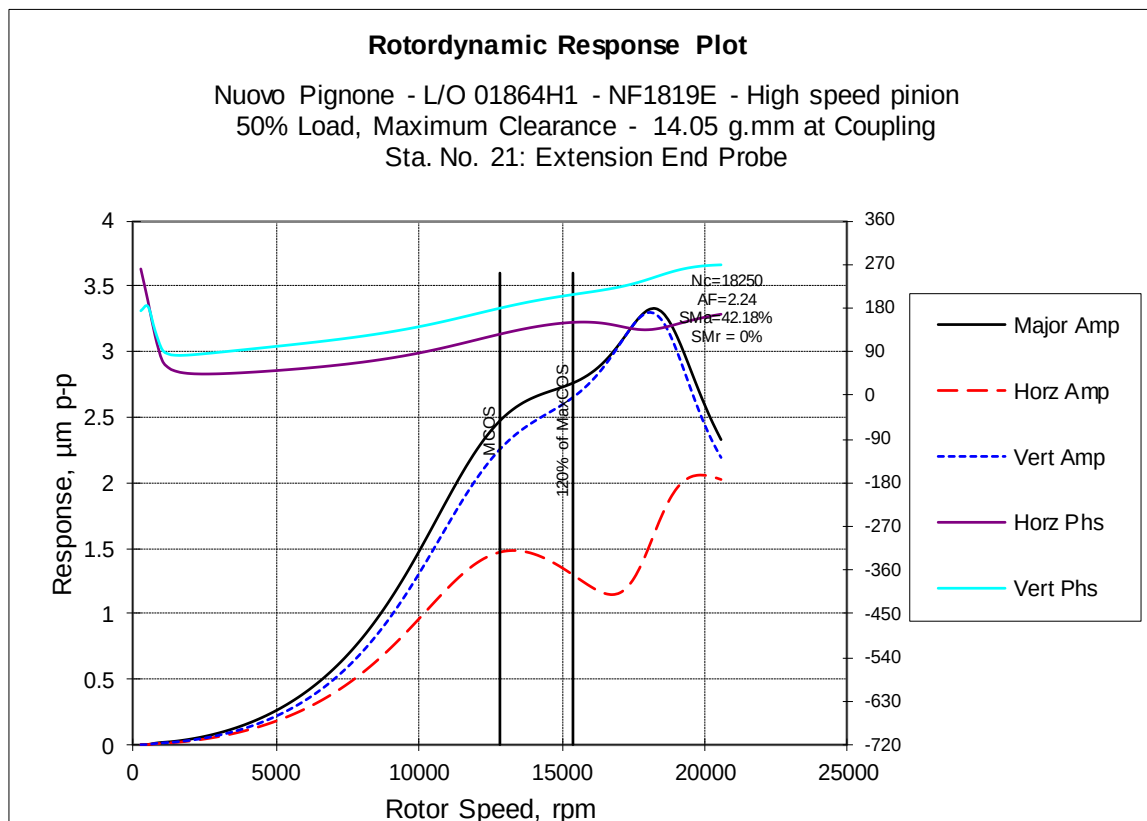
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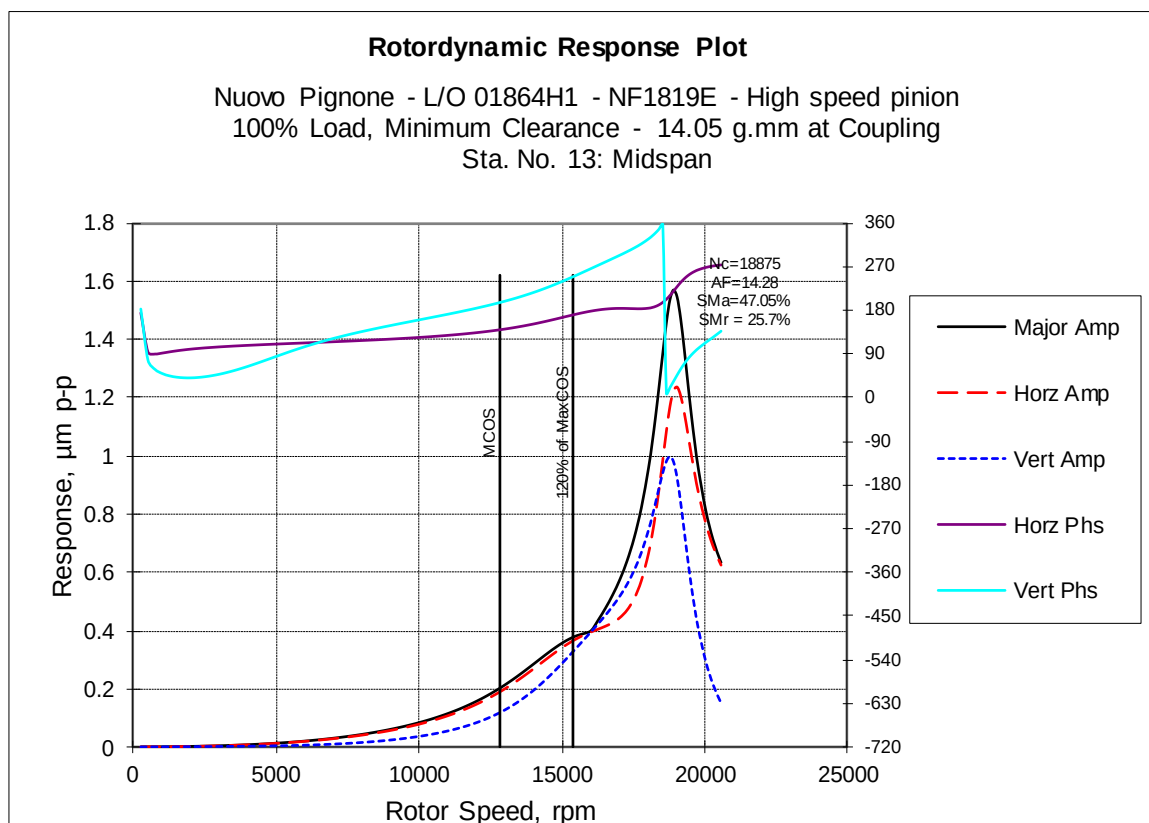
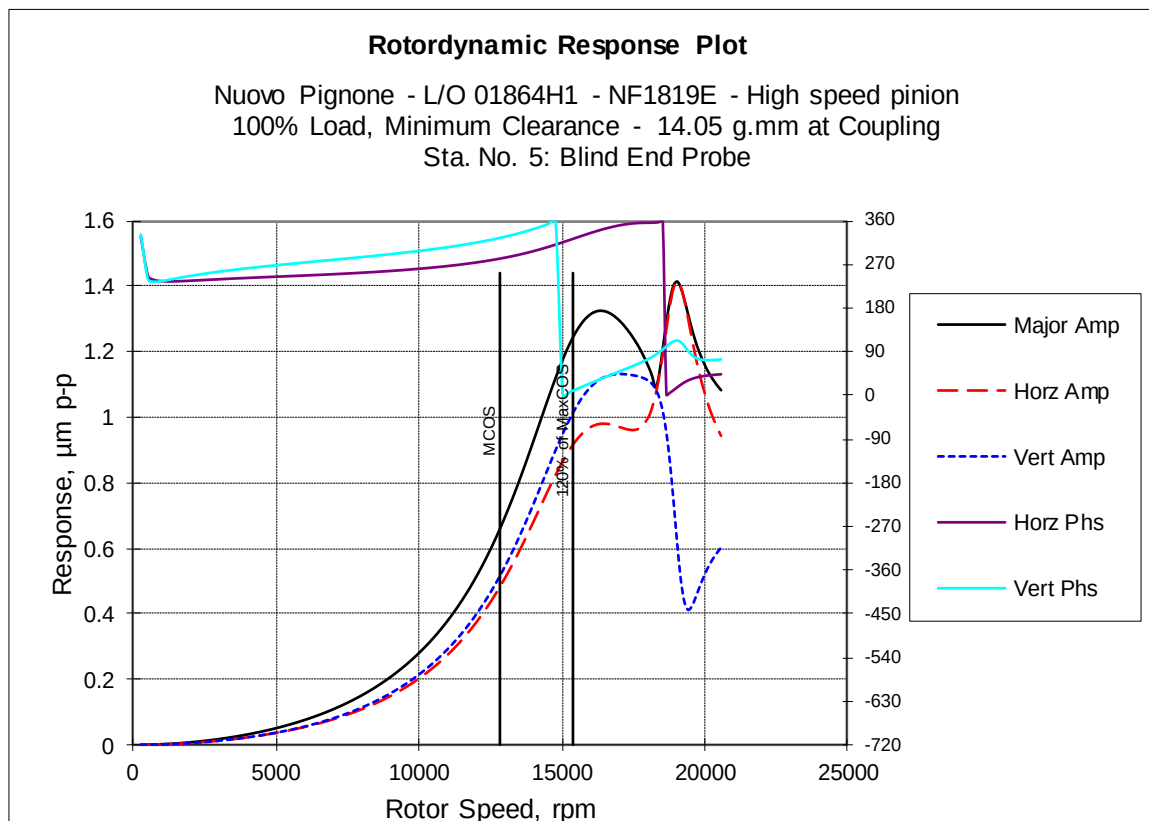
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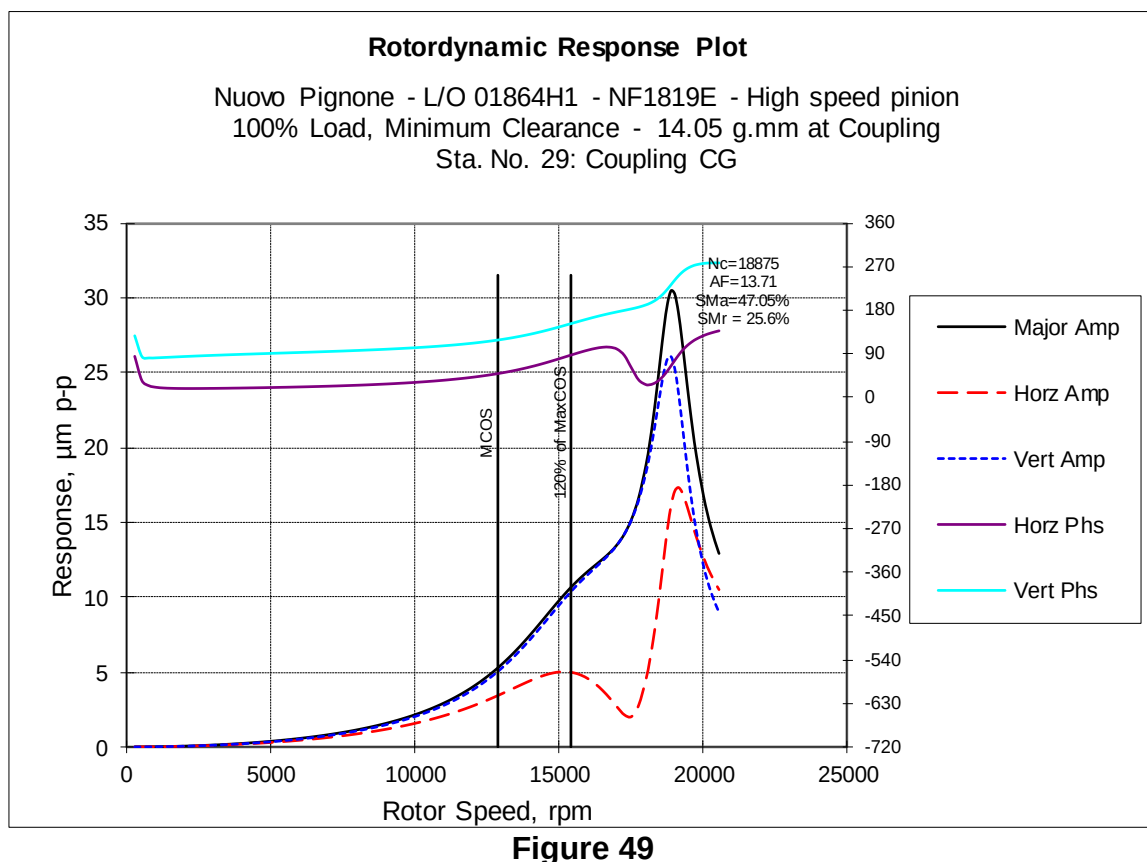
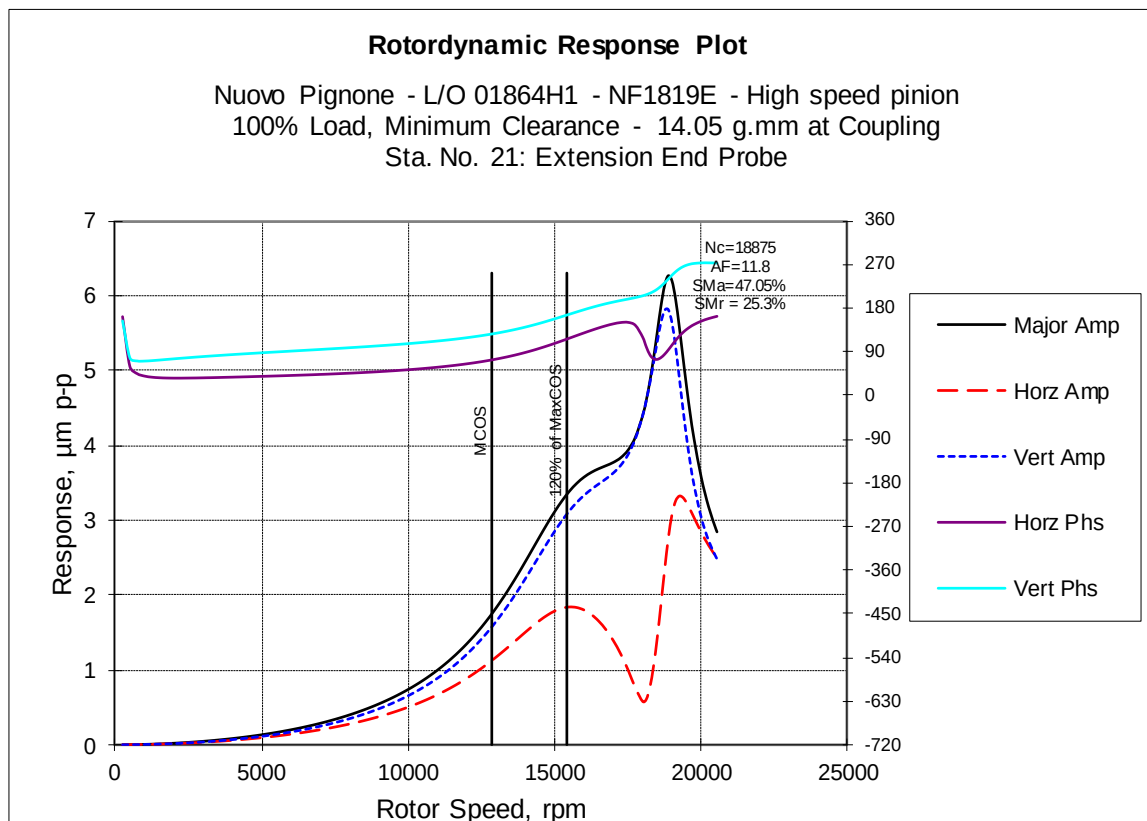
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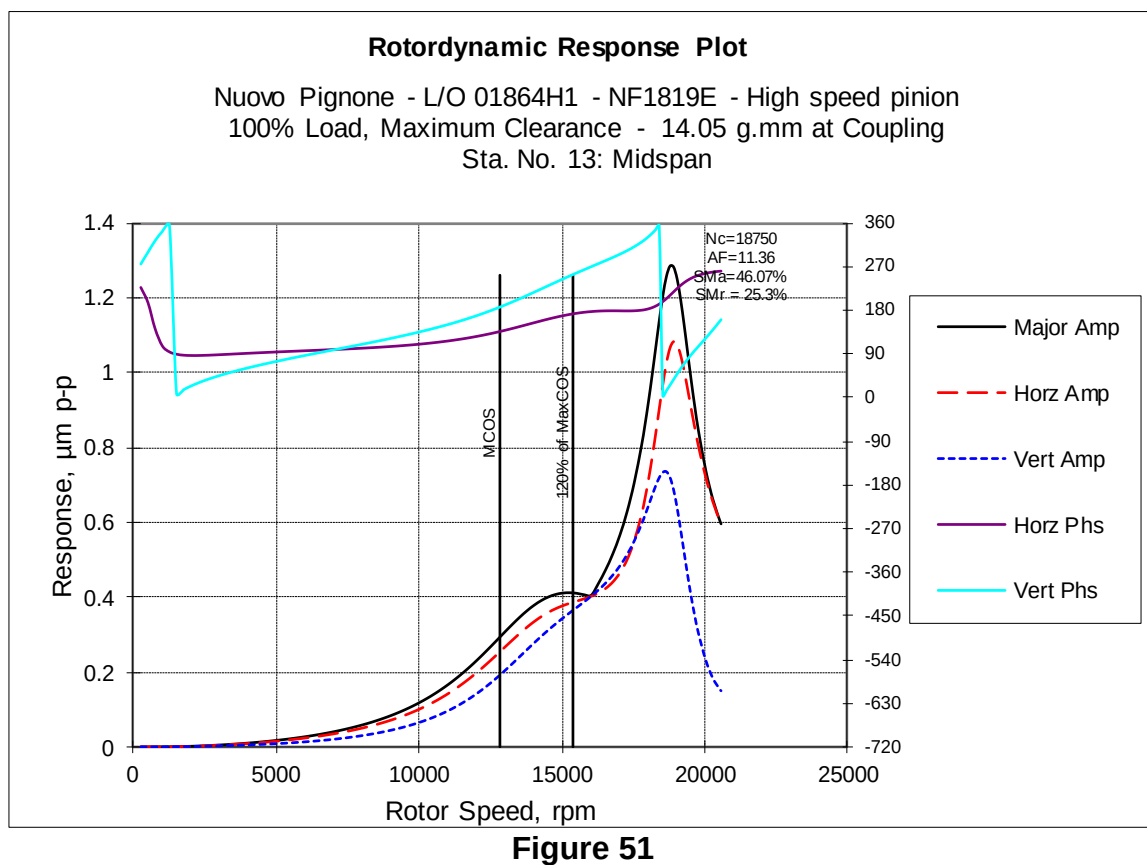
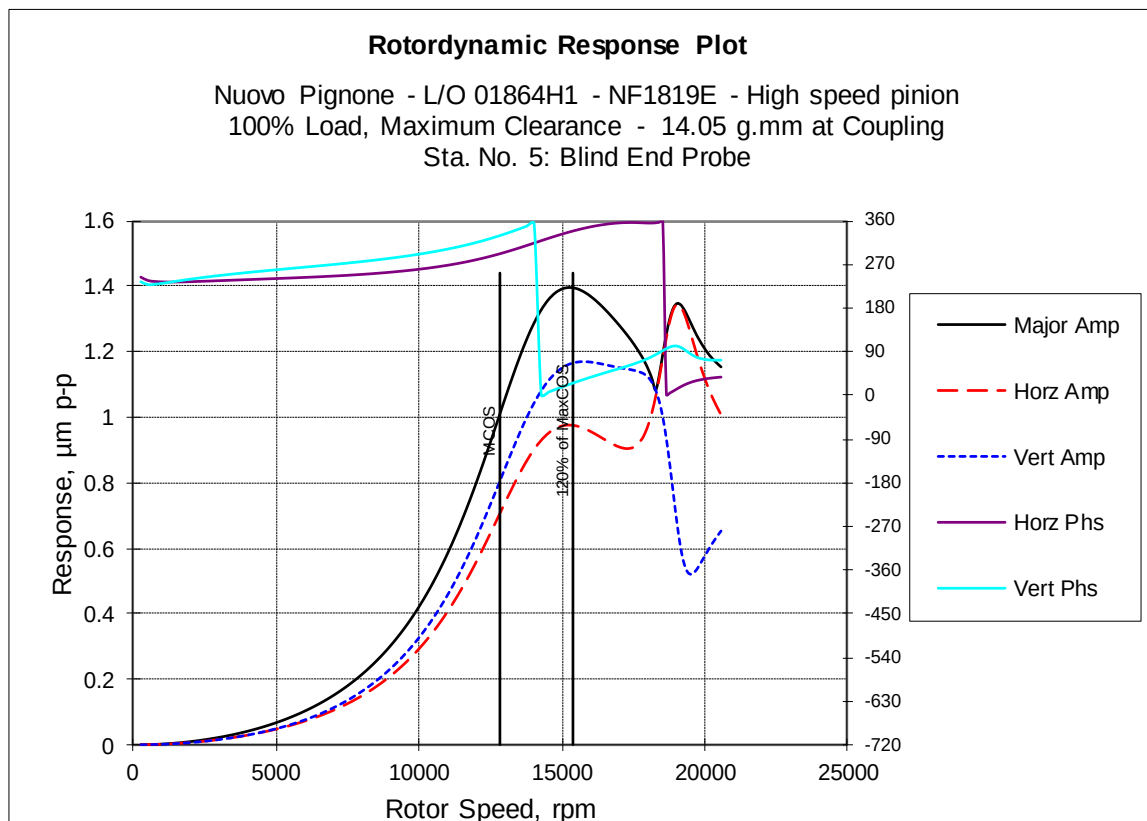
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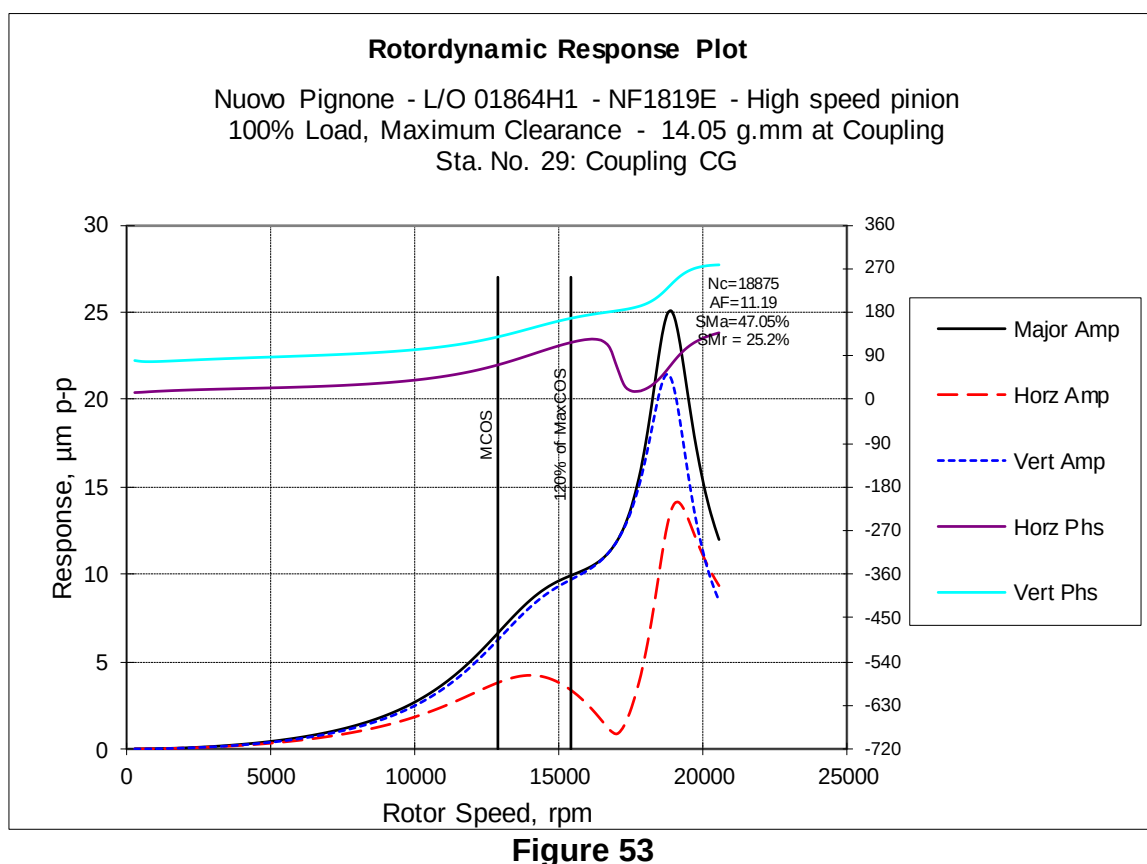
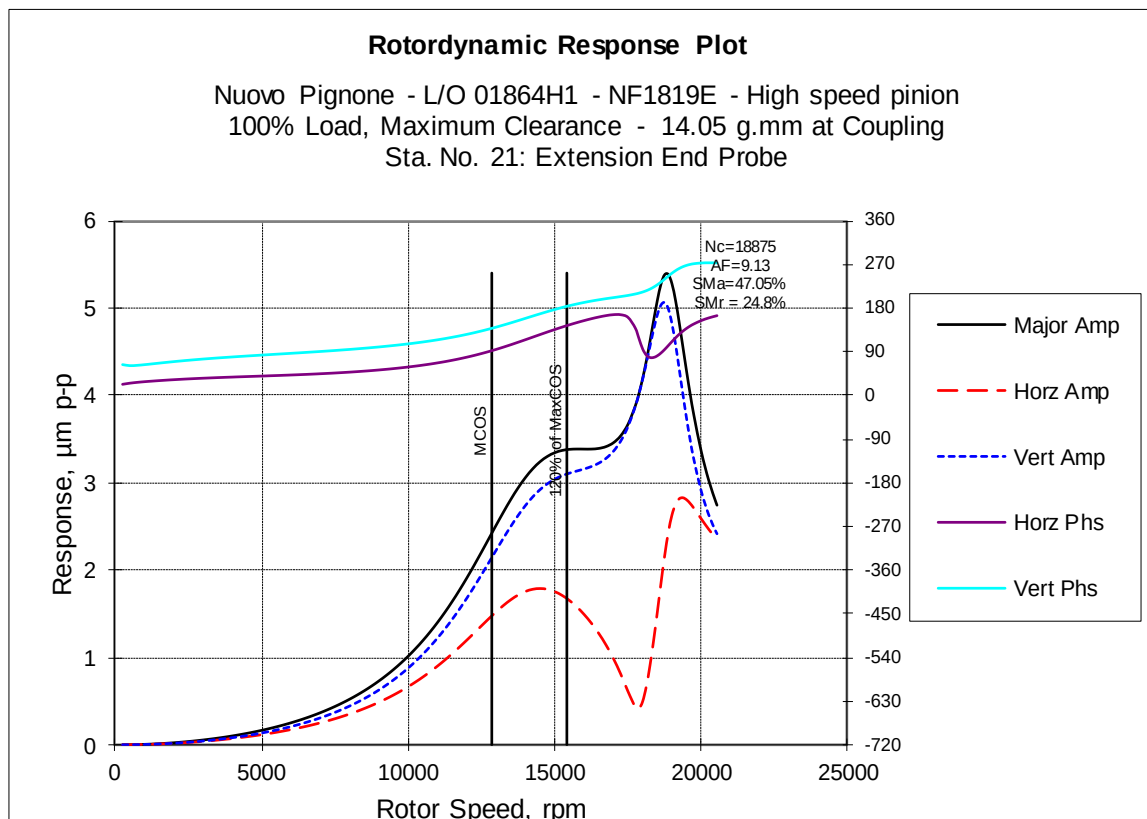
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Appendix G - Rotordynamics Coefficients

Rotordynamic Coefficients for Blind End Bearing, 10% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	7.73E+07	3.75E+07	-1.99E+08	2.18E+08	2.11E+06	-1.65E+06	-1.72E+06	9.49E+06
820	7.88E+07	4.45E+07	-2.24E+08	2.28E+08	1.28E+06	-6.44E+05	-6.34E+05	5.15E+06
1231	8.27E+07	5.35E+07	-2.59E+08	2.57E+08	1.01E+06	-3.65E+05	-3.55E+05	3.90E+06
1641	8.75E+07	6.28E+07	-2.96E+08	2.90E+08	8.80E+05	-2.48E+05	-2.39E+05	3.30E+06
2051	9.28E+07	7.16E+07	-3.32E+08	3.23E+08	7.98E+05	-1.85E+05	-1.76E+05	2.94E+06
2461	9.80E+07	7.99E+07	-3.67E+08	3.56E+08	7.40E+05	-1.46E+05	-1.38E+05	2.69E+06
2871	1.03E+08	8.75E+07	-4.01E+08	3.89E+08	6.95E+05	-1.21E+05	-1.13E+05	2.50E+06
3282	1.08E+08	9.45E+07	-4.33E+08	4.20E+08	6.58E+05	-1.03E+05	-9.48E+04	2.35E+06

Rotordynamic Coefficients for Blind End Bearing, 10% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	5.35E+07	2.13E+07	-1.45E+08	1.84E+08	1.27E+06	-1.31E+06	-1.46E+06	7.26E+06
820	5.87E+07	2.54E+07	-1.50E+08	1.56E+08	8.03E+05	-6.22E+05	-5.90E+05	3.48E+06
1231	5.76E+07	2.60E+07	-1.60E+08	1.59E+08	5.80E+05	-3.34E+05	-3.03E+05	2.39E+06
1641	5.73E+07	2.85E+07	-1.75E+08	1.68E+08	4.83E+05	-2.12E+05	-1.86E+05	1.91E+06
2051	5.75E+07	3.17E+07	-1.91E+08	1.81E+08	4.29E+05	-1.48E+05	-1.25E+05	1.65E+06
2461	5.79E+07	3.52E+07	-2.08E+08	1.94E+08	3.95E+05	-1.07E+05	-8.87E+04	1.48E+06
2871	5.86E+07	3.87E+07	-2.25E+08	2.08E+08	3.70E+05	-8.11E+04	-6.51E+04	1.35E+06
3282	5.93E+07	4.20E+07	-2.42E+08	2.23E+08	3.50E+05	-6.26E+04	-4.86E+04	1.26E+06

Rotordynamic Coefficients for Blind End Bearing, 50% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	9.05E+07	1.56E+08	-3.62E+08	7.93E+08	2.32E+06	-7.42E+04	-1.66E+06	2.62E+07
820	1.17E+08	1.42E+08	-4.13E+08	6.15E+08	1.71E+06	-6.02E+05	-1.13E+06	1.25E+07
1231	1.38E+08	1.47E+08	-4.54E+08	5.62E+08	1.47E+06	-6.46E+05	-8.63E+05	8.52E+06
1641	1.56E+08	1.58E+08	-4.92E+08	5.45E+08	1.33E+06	-6.04E+05	-6.98E+05	6.62E+06
2051	1.69E+08	1.66E+08	-5.25E+08	5.48E+08	1.22E+06	-5.48E+05	-5.78E+05	5.49E+06
2461	1.79E+08	1.75E+08	-5.56E+08	5.59E+08	1.12E+06	-4.80E+05	-4.82E+05	4.73E+06
2871	1.85E+08	1.79E+08	-5.83E+08	5.76E+08	1.02E+06	-4.06E+05	-4.00E+05	4.17E+06
3282	1.89E+08	1.82E+08	-6.09E+08	5.96E+08	9.39E+05	-3.43E+05	-3.36E+05	3.75E+06



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Rotordynamic Coefficients for Blind End Bearing, 50% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	6.40E+07	1.46E+08	-2.68E+08	7.73E+08	1.36E+06	6.74E+05	-7.66E+05	2.19E+07
820	7.77E+07	1.15E+08	-2.92E+08	5.55E+08	9.49E+05	-1.26E+05	-6.96E+05	9.73E+06
1231	8.95E+07	1.07E+08	-3.09E+08	4.72E+08	7.94E+05	-2.78E+05	-5.62E+05	6.27E+06
1641	9.75E+07	1.04E+08	-3.23E+08	4.33E+08	6.95E+05	-3.17E+05	-4.50E+05	4.68E+06
2051	1.06E+08	1.04E+08	-3.38E+08	4.09E+08	6.40E+05	-3.21E+05	-3.90E+05	3.77E+06
2461	1.15E+08	1.07E+08	-3.55E+08	3.95E+08	6.03E+05	-3.15E+05	-3.45E+05	3.18E+06
2871	1.23E+08	1.09E+08	-3.69E+08	3.87E+08	5.70E+05	-3.02E+05	-3.08E+05	2.76E+06
3282	1.28E+08	1.11E+08	-3.82E+08	3.84E+08	5.37E+05	-2.83E+05	-2.72E+05	2.45E+06

Rotordynamic Coefficients for Blind End Bearing, 100% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	1.26E+08	4.33E+08	-4.98E+08	1.90E+09	3.04E+06	4.22E+06	8.52E+05	5.13E+07
820	1.41E+08	3.27E+08	-5.77E+08	1.40E+09	1.93E+06	9.47E+05	-4.68E+05	2.30E+07
1231	1.69E+08	3.00E+08	-6.53E+08	1.20E+09	1.59E+06	2.20E+05	-6.70E+05	1.48E+07
1641	1.82E+08	2.85E+08	-6.86E+08	1.10E+09	1.37E+06	-9.41E+04	-6.04E+05	1.11E+07
2051	1.99E+08	2.81E+08	-7.25E+08	1.05E+09	1.25E+06	-2.24E+05	-5.79E+05	8.88E+06
2461	2.16E+08	2.82E+08	-7.62E+08	1.01E+09	1.18E+06	-2.99E+05	-5.56E+05	7.49E+06
2871	2.30E+08	2.85E+08	-7.95E+08	9.87E+08	1.11E+06	-3.35E+05	-5.25E+05	6.49E+06
3282	2.41E+08	2.86E+08	-8.22E+08	9.78E+08	1.04E+06	-3.51E+05	-4.77E+05	5.75E+06

Rotordynamic Coefficients for Blind End Bearing, 100% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	1.03E+08	4.39E+08	-3.62E+08	1.91E+09	2.15E+06	5.03E+06	1.91E+06	4.43E+07
820	1.08E+08	3.12E+08	-4.35E+08	1.34E+09	1.25E+06	1.47E+06	-3.34E+04	1.89E+07
1231	1.11E+08	2.62E+08	-4.47E+08	1.12E+09	9.33E+05	5.91E+05	-2.04E+05	1.18E+07
1641	1.20E+08	2.38E+08	-4.70E+08	9.89E+08	7.90E+05	2.32E+05	-2.78E+05	8.60E+06
2051	1.30E+08	2.25E+08	-4.94E+08	9.05E+08	7.14E+05	5.22E+04	-3.22E+05	6.76E+06
2461	1.34E+08	2.15E+08	-5.03E+08	8.52E+08	6.39E+05	-1.55E+04	-2.88E+05	5.51E+06
2871	1.39E+08	2.06E+08	-5.14E+08	8.16E+08	5.81E+05	-5.17E+04	-2.57E+05	4.66E+06
3282	1.52E+08	2.05E+08	-5.39E+08	7.79E+08	5.66E+05	-1.33E+05	-2.85E+05	4.13E+06



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LANGUAGE

Rotordynamic Coefficients for Extension End Bearing, 10% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	1.39E+08	3.10E+07	-3.49E+08	4.27E+08	3.21E+06	-3.58E+06	-3.87E+06	1.61E+07
820	1.41E+08	4.28E+07	-3.46E+08	3.74E+08	1.84E+06	-1.51E+06	-1.50E+06	7.71E+06
1231	1.39E+08	4.88E+07	-3.63E+08	3.80E+08	1.32E+06	-8.38E+05	-8.28E+05	5.29E+06
1641	1.40E+08	5.61E+07	-3.89E+08	3.99E+08	1.08E+06	-5.63E+05	-5.53E+05	4.21E+06
2051	1.43E+08	6.35E+07	-4.18E+08	4.22E+08	9.37E+05	-4.18E+05	-4.08E+05	3.59E+06
2461	1.47E+08	7.05E+07	-4.47E+08	4.48E+08	8.44E+05	-3.31E+05	-3.21E+05	3.19E+06
2871	1.50E+08	7.75E+07	-4.77E+08	4.75E+08	7.78E+05	-2.71E+05	-2.62E+05	2.90E+06
3282	1.54E+08	8.38E+07	-5.07E+08	5.03E+08	7.27E+05	-2.30E+05	-2.21E+05	2.68E+06

Rotordynamic Coefficients for Extension End Bearing, 10% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	9.94E+07	1.31E+07	-2.78E+08	4.01E+08	2.00E+06	-2.73E+06	-3.22E+06	1.34E+07
820	1.09E+08	2.47E+07	-2.61E+08	3.00E+08	1.27E+06	-1.40E+06	-1.41E+06	5.94E+06
1231	1.08E+08	2.57E+07	-2.56E+08	2.74E+08	8.78E+05	-8.16E+05	-7.82E+05	3.76E+06
1641	1.05E+08	2.52E+07	-2.58E+08	2.68E+08	6.66E+05	-5.30E+05	-4.96E+05	2.78E+06
2051	1.04E+08	2.58E+07	-2.65E+08	2.68E+08	5.50E+05	-3.84E+05	-3.49E+05	2.24E+06
2461	1.03E+08	2.68E+07	-2.75E+08	2.72E+08	4.77E+05	-2.96E+05	-2.62E+05	1.91E+06
2871	1.03E+08	2.80E+07	-2.86E+08	2.79E+08	4.27E+05	-2.39E+05	-2.06E+05	1.69E+06
3282	1.03E+08	2.94E+07	-2.98E+08	2.87E+08	3.90E+05	-1.99E+05	-1.66E+05	1.53E+06

Rotordynamic Coefficients for Extension End Bearing, 50% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	1.46E+08	1.57E+08	-5.53E+08	1.12E+09	2.87E+06	-1.79E+06	-3.79E+06	3.50E+07
820	1.69E+08	1.41E+08	-5.56E+08	8.48E+08	2.09E+06	-1.50E+06	-2.16E+06	1.61E+07
1231	1.92E+08	1.47E+08	-5.84E+08	7.51E+08	1.75E+06	-1.23E+06	-1.51E+06	1.06E+07
1641	2.11E+08	1.57E+08	-6.13E+08	7.09E+08	1.54E+06	-1.04E+06	-1.15E+06	7.99E+06
2051	2.23E+08	1.65E+08	-6.38E+08	6.96E+08	1.39E+06	-8.81E+05	-9.23E+05	6.49E+06
2461	2.33E+08	1.74E+08	-6.63E+08	6.94E+08	1.26E+06	-7.50E+05	-7.55E+05	5.51E+06
2871	2.38E+08	1.78E+08	-6.86E+08	7.03E+08	1.15E+06	-6.30E+05	-6.24E+05	4.80E+06
3282	2.41E+08	1.80E+08	-7.08E+08	7.17E+08	1.04E+06	-5.29E+05	-5.22E+05	4.27E+06



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Rotordynamic Coefficients for Extension End Bearing, 50% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	9.41E+07	1.41E+08	-4.11E+08	1.14E+09	1.68E+06	-5.68E+05	-2.49E+06	3.00E+07
820	1.07E+08	1.09E+08	-4.02E+08	8.05E+08	1.16E+06	-8.27E+05	-1.46E+06	1.31E+07
1231	1.28E+08	1.06E+08	-4.22E+08	6.65E+08	9.82E+05	-7.17E+05	-1.09E+06	8.15E+06
1641	1.35E+08	1.02E+08	-4.25E+08	6.01E+08	8.40E+05	-6.45E+05	-8.22E+05	5.95E+06
2051	1.47E+08	1.05E+08	-4.37E+08	5.57E+08	7.65E+05	-5.83E+05	-6.81E+05	4.70E+06
2461	1.56E+08	1.07E+08	-4.48E+08	5.29E+08	7.09E+05	-5.29E+05	-5.80E+05	3.90E+06
2871	1.64E+08	1.09E+08	-4.58E+08	5.11E+08	6.59E+05	-4.82E+05	-5.00E+05	3.35E+06
3282	1.69E+08	1.10E+08	-4.67E+08	5.00E+08	6.13E+05	-4.38E+05	-4.34E+05	2.93E+06

Rotordynamic Coefficients for Extension End Bearing, 100% Load, Minimum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	1.70E+08	4.37E+08	-7.33E+08	2.38E+09	3.27E+06	2.71E+06	-1.53E+06	6.18E+07
820	1.93E+08	3.32E+08	-7.69E+08	1.72E+09	2.20E+06	1.18E+05	-1.71E+06	2.73E+07
1231	2.03E+08	2.93E+08	-7.75E+08	1.48E+09	1.72E+06	-4.18E+05	-1.24E+06	1.74E+07
1641	2.24E+08	2.81E+08	-8.14E+08	1.34E+09	1.52E+06	-5.37E+05	-1.09E+06	1.28E+07
2051	2.43E+08	2.76E+08	-8.47E+08	1.26E+09	1.37E+06	-5.75E+05	-9.43E+05	1.02E+07
2461	2.62E+08	2.80E+08	-8.82E+08	1.20E+09	1.27E+06	-5.83E+05	-8.49E+05	8.48E+06
2871	2.81E+08	2.84E+08	-9.16E+08	1.17E+09	1.21E+06	-5.92E+05	-7.92E+05	7.33E+06
3282	2.96E+08	2.88E+08	-9.44E+08	1.14E+09	1.15E+06	-5.77E+05	-7.23E+05	6.45E+06

Rotordynamic Coefficients for Extension End Bearing, 100% Load, Maximum Clearance

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
410	1.09E+08	4.42E+08	-4.46E+08	2.47E+09	2.10E+06	3.91E+06	5.42E+05	5.40E+07
820	1.24E+08	3.07E+08	-5.31E+08	1.70E+09	1.27E+06	7.96E+05	-6.00E+05	2.31E+07
1231	1.31E+08	2.58E+08	-5.44E+08	1.41E+09	9.84E+05	1.63E+05	-6.30E+05	1.41E+07
1641	1.42E+08	2.34E+08	-5.64E+08	1.24E+09	8.36E+05	-5.62E+04	-5.79E+05	1.01E+07
2051	1.57E+08	2.22E+08	-5.91E+08	1.12E+09	7.64E+05	-1.46E+05	-5.69E+05	7.81E+06
2461	1.73E+08	2.17E+08	-6.18E+08	1.04E+09	7.28E+05	-2.61E+05	-5.75E+05	6.50E+06
2871	1.78E+08	2.09E+08	-6.23E+08	9.90E+08	6.63E+05	-2.59E+05	-4.98E+05	5.45E+06
3282	1.85E+08	2.05E+08	-6.35E+08	9.48E+08	6.19E+05	-2.72E+05	-4.54E+05	4.73E+06

Rotordynamic Coefficients for Gear Mesh

Speed (rpm)	Kxx (N/m)	Kxy (N/m)	Kyx (N/m)	Kyy (N/m)	Cxx (N.s/m)	Cxy (N.s/m)	Cyx (N.s/m)	Cyy (N.s/m)
1	1.62E+08	-4.44E+08	-4.44E+08	1.22E+09	8.00E+03	0.00E+00	0.00E+00	2.20E+04
50000	1.62E+08	-4.44E+08	-4.44E+08	1.22E+09	8.00E+03	0.00E+00	0.00E+00	2.20E+04



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Appendix H - Rotor-Bearing Model

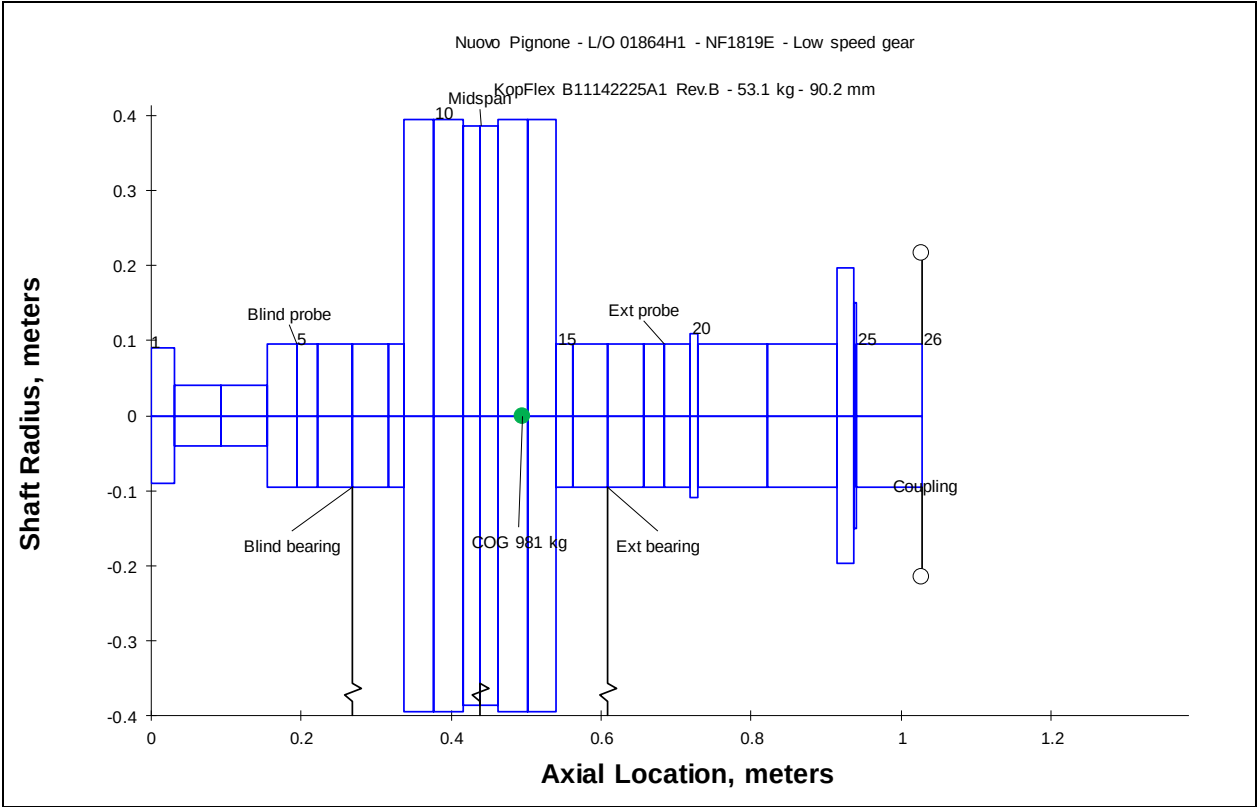


Figure 54

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Appendix I - Undamped Critical Speed Map

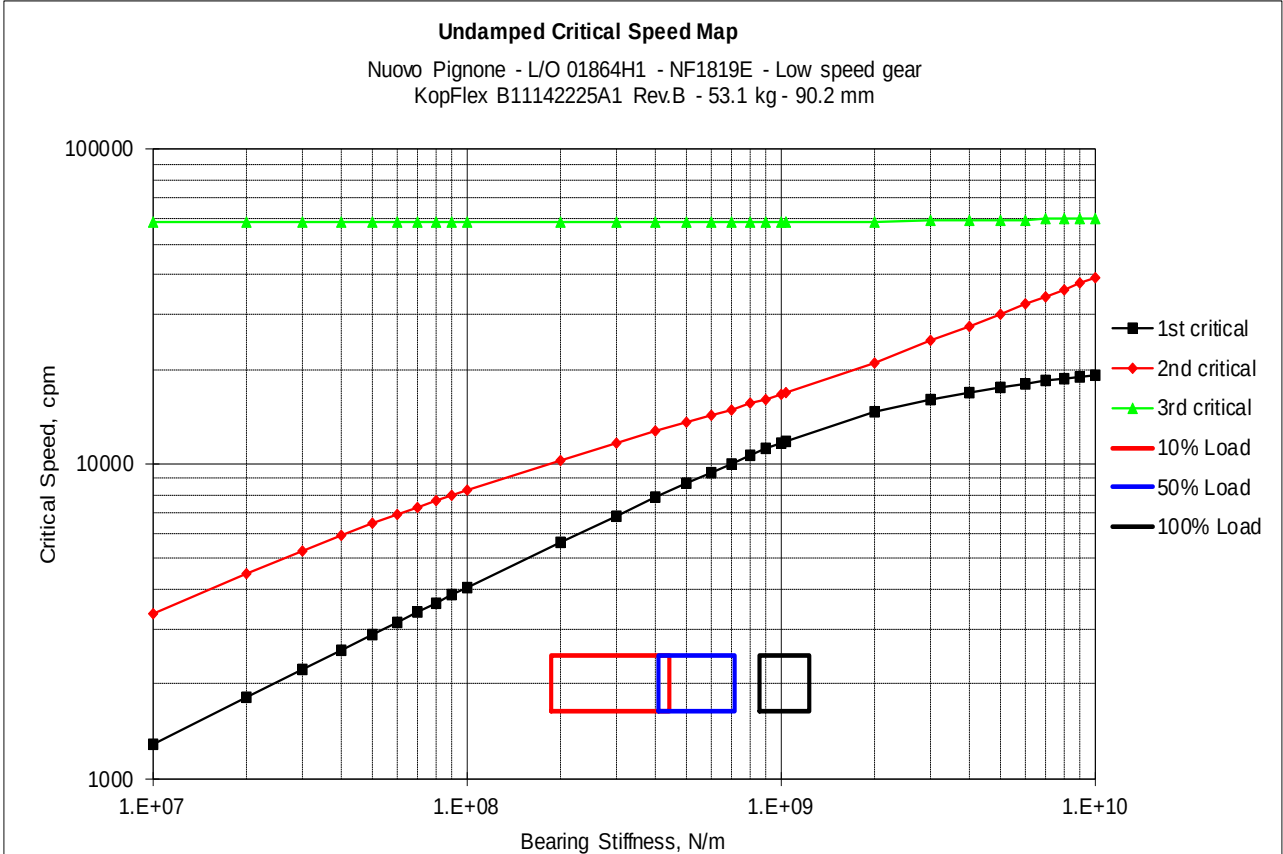
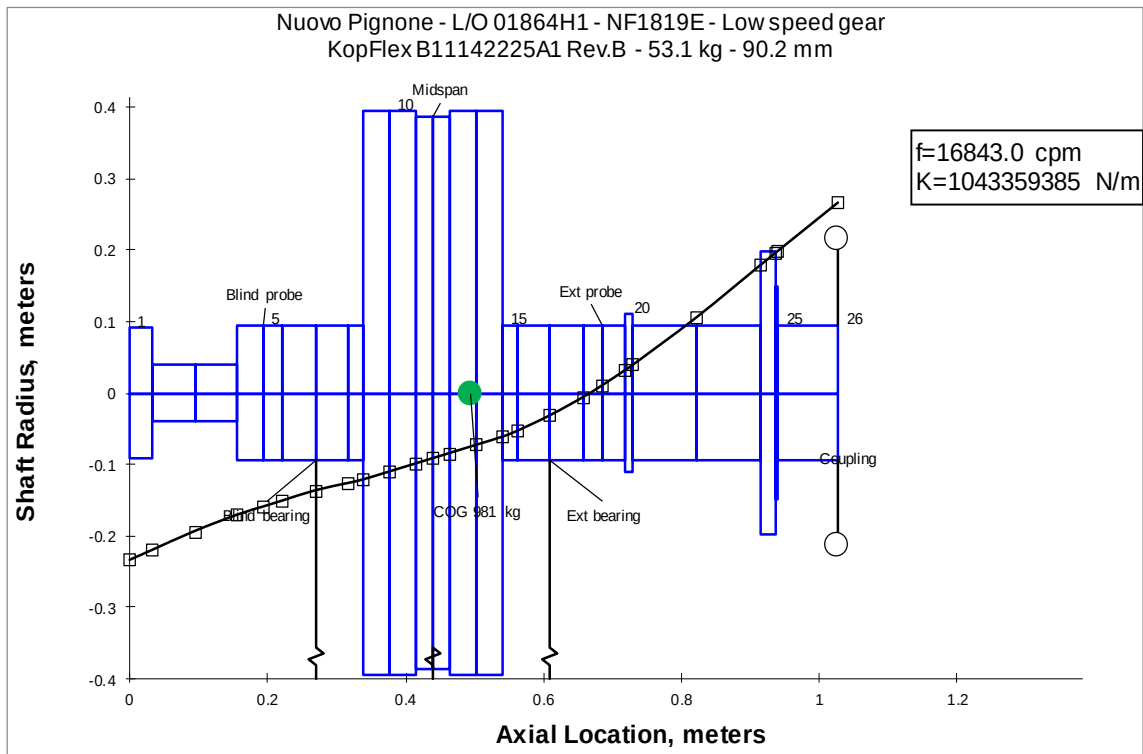
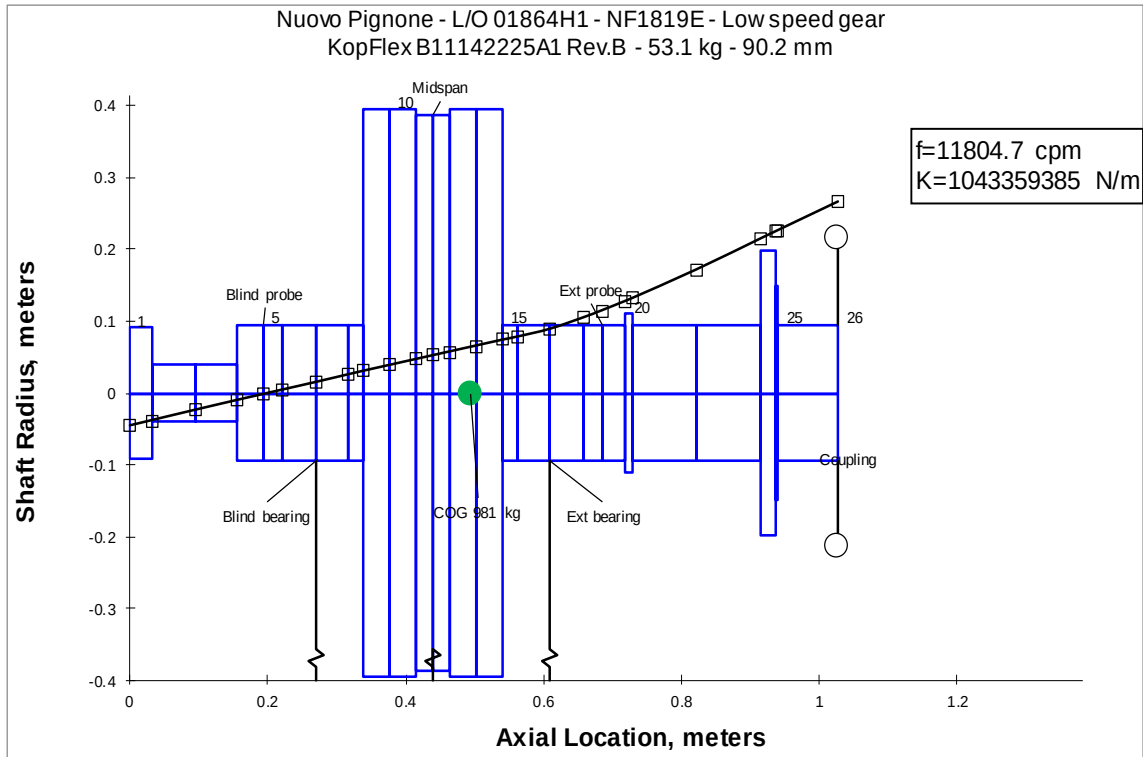


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Appendix J - Undamped Mode Shape Plots



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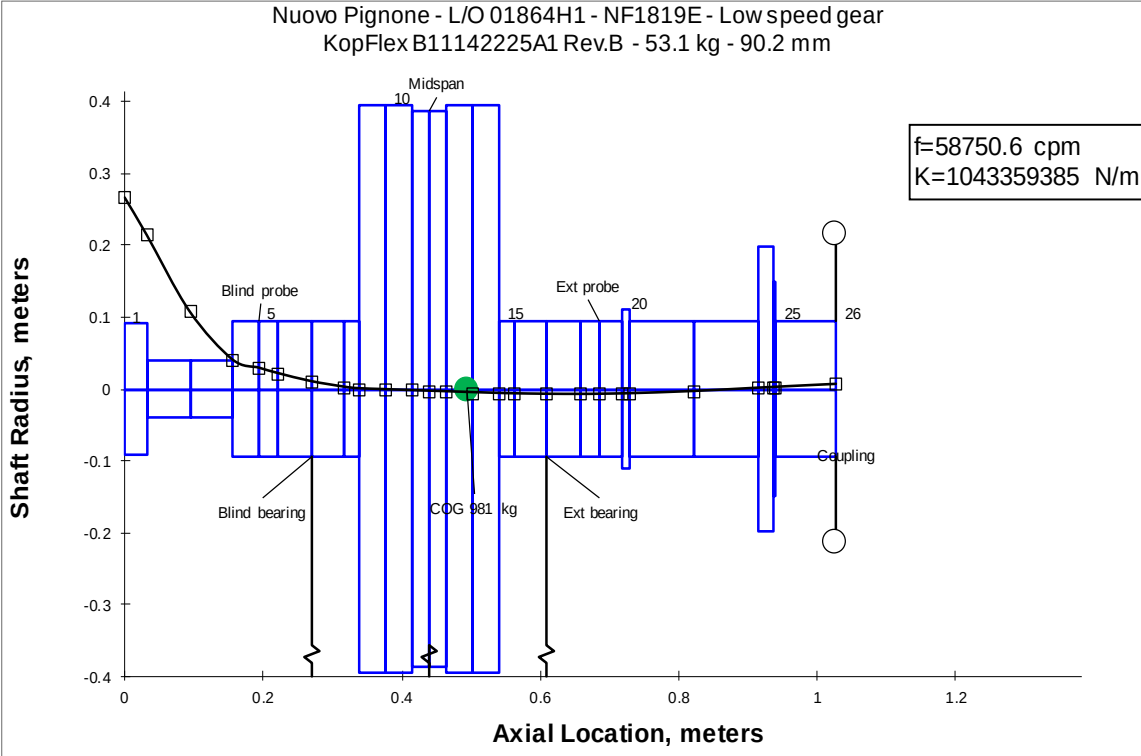
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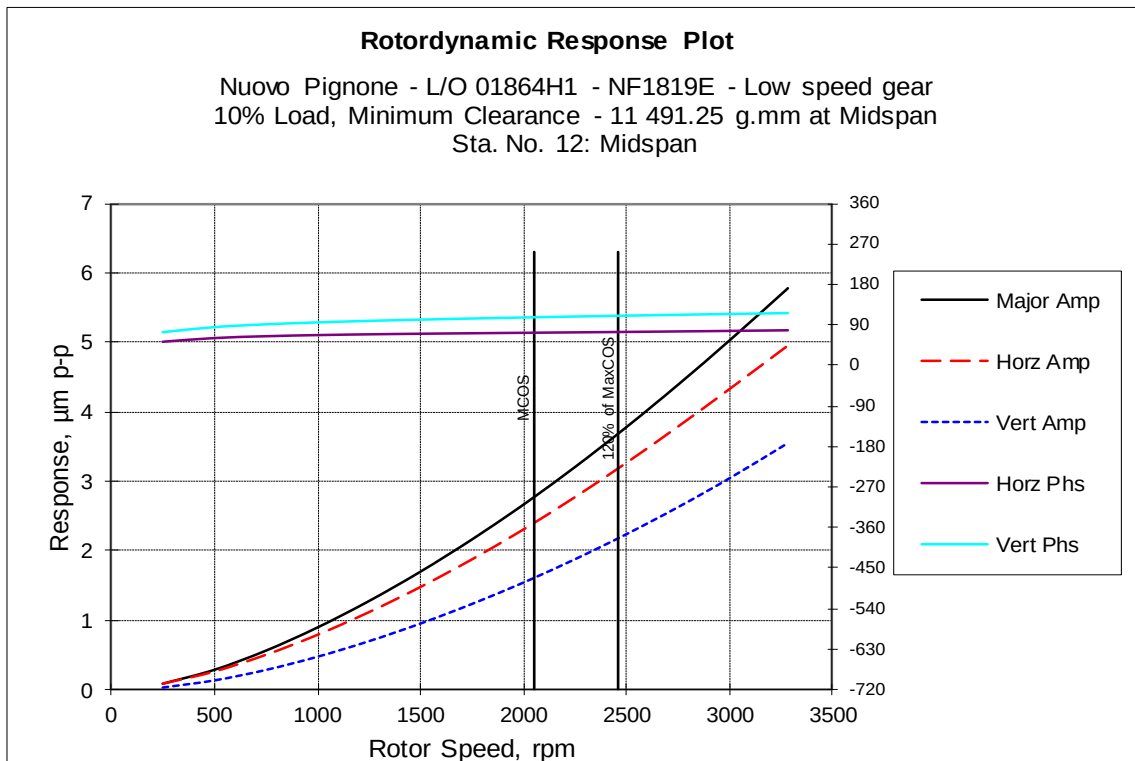
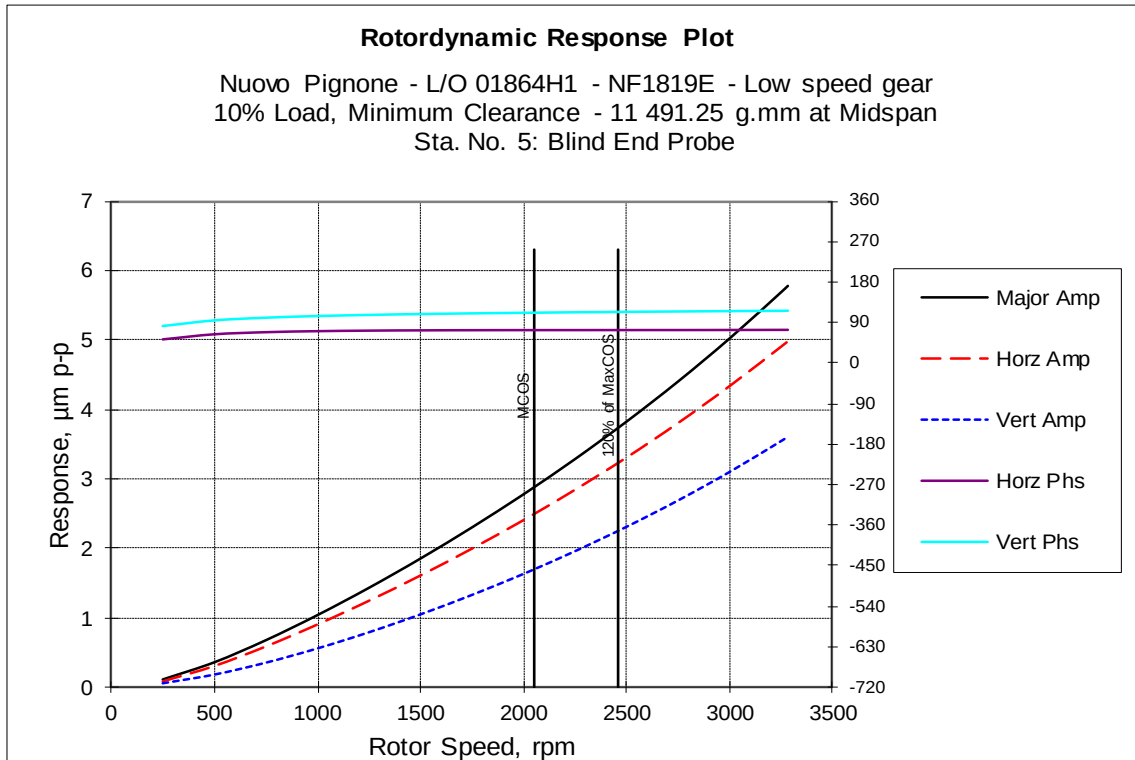
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Appendix K - Midspan Unbalance Response Plots



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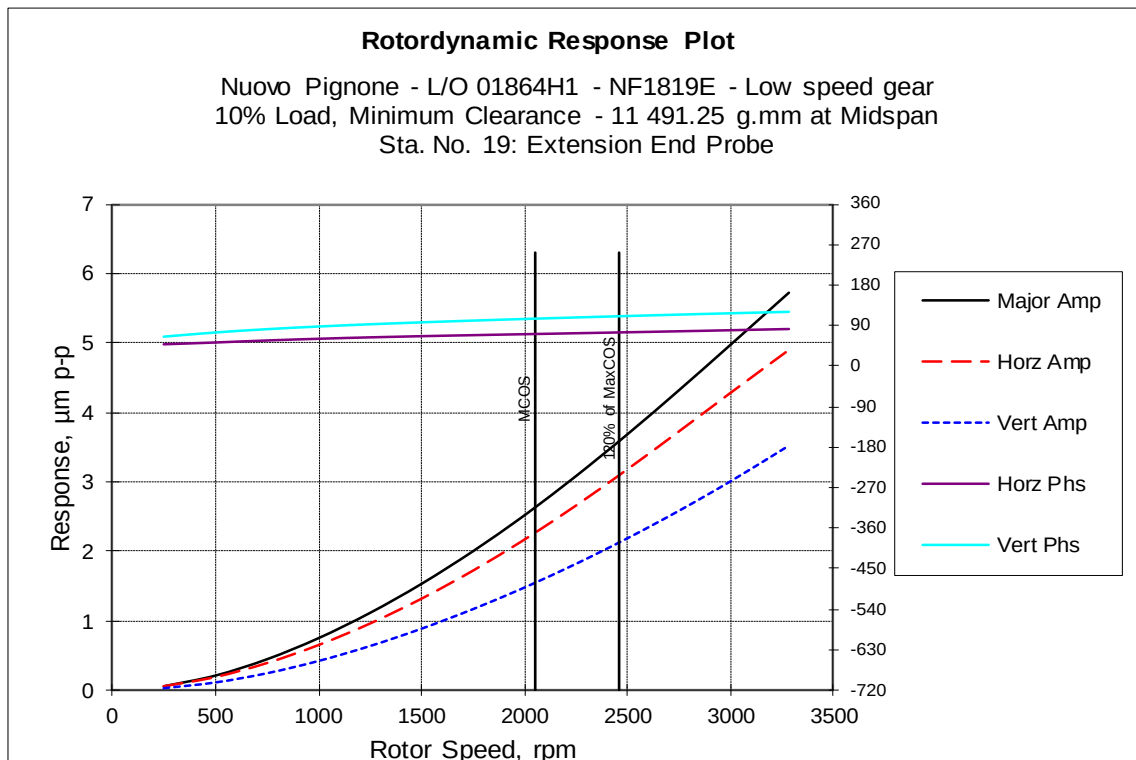


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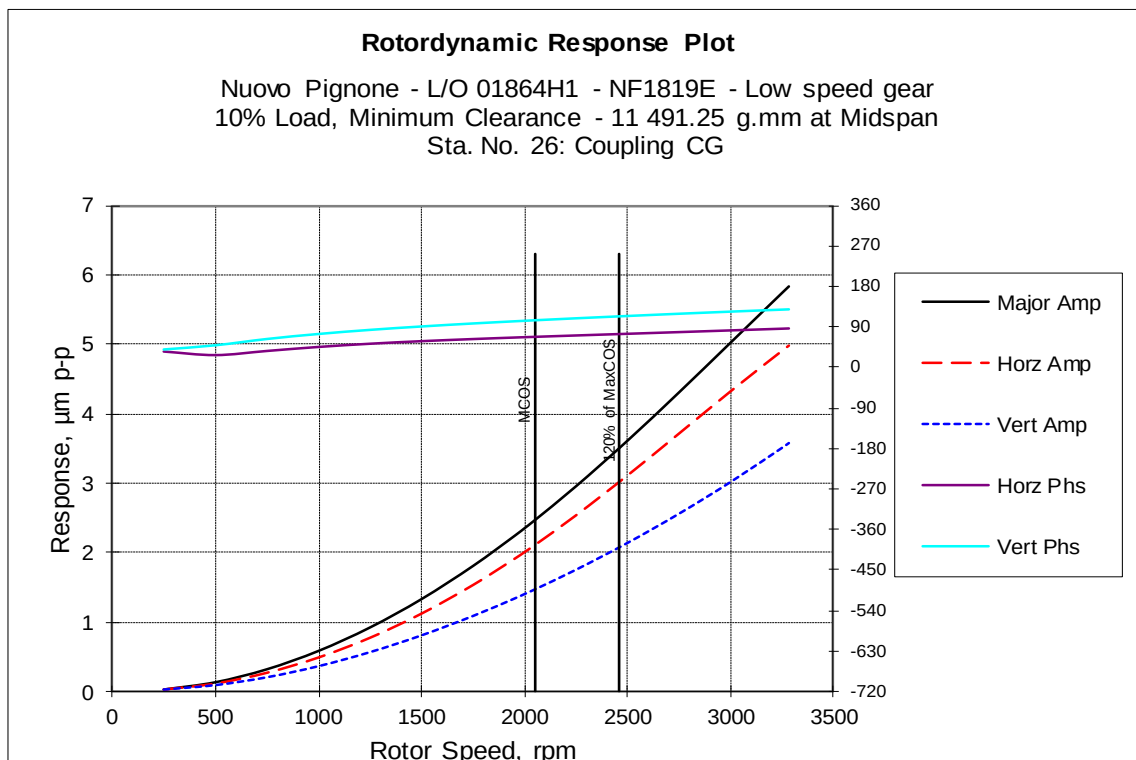
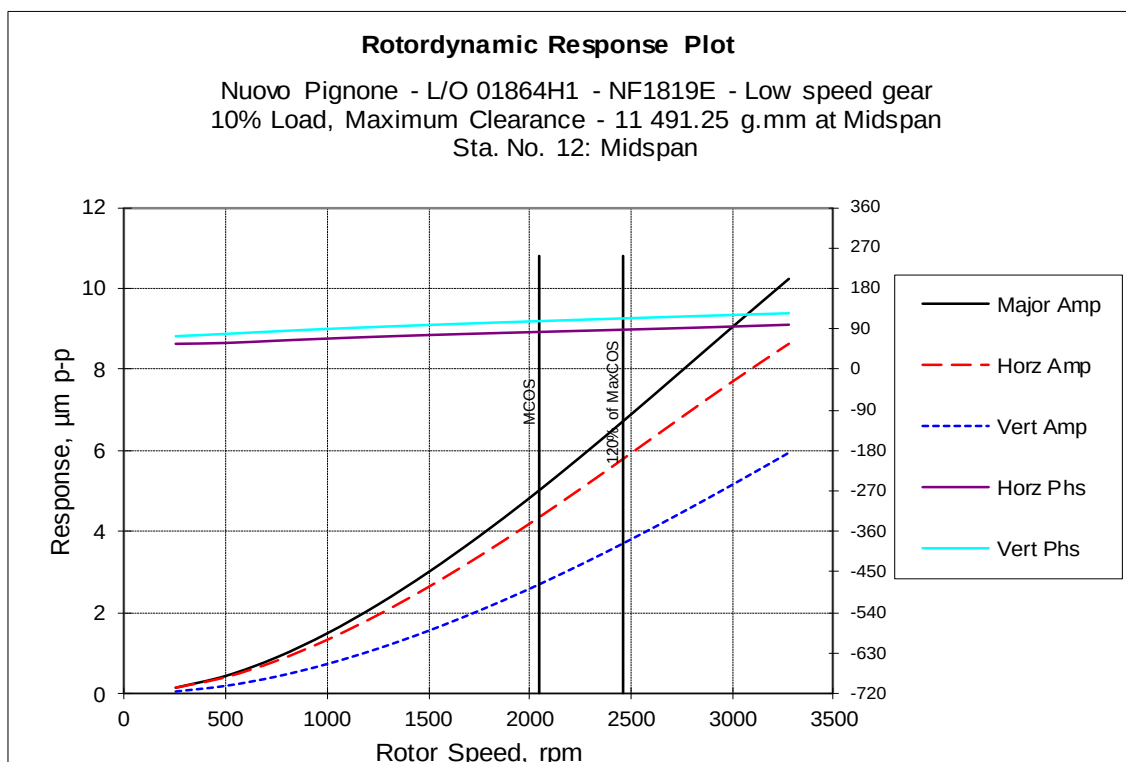
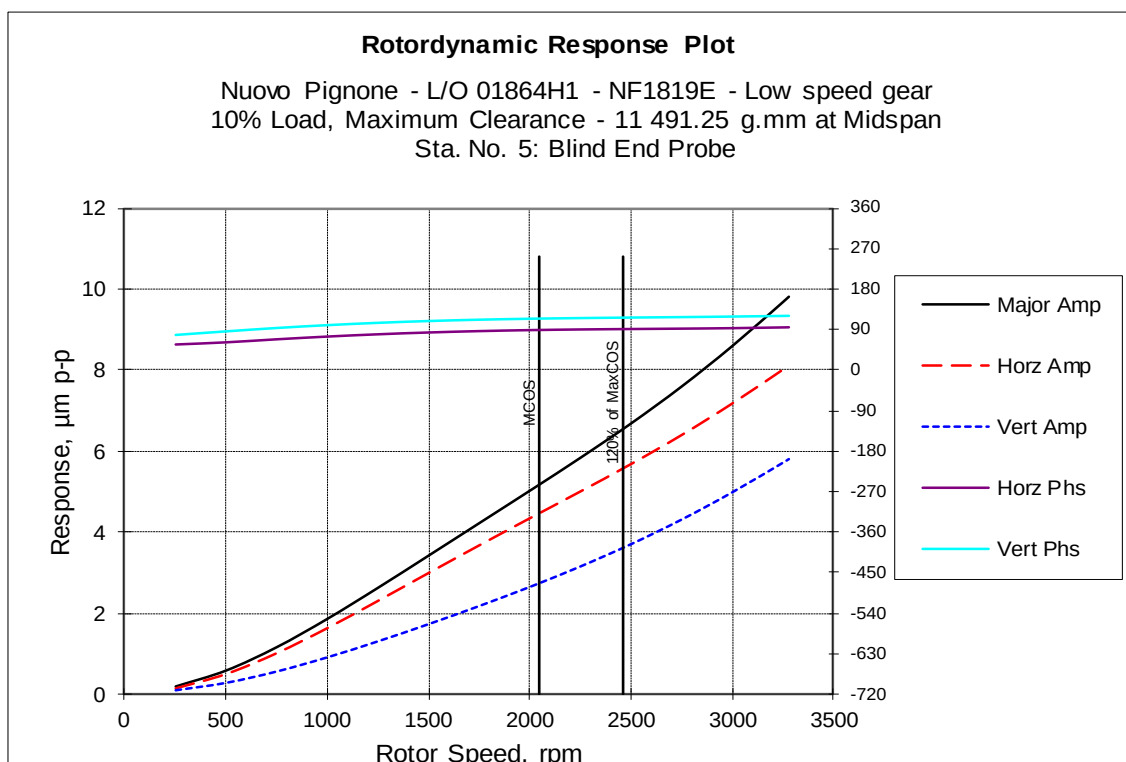


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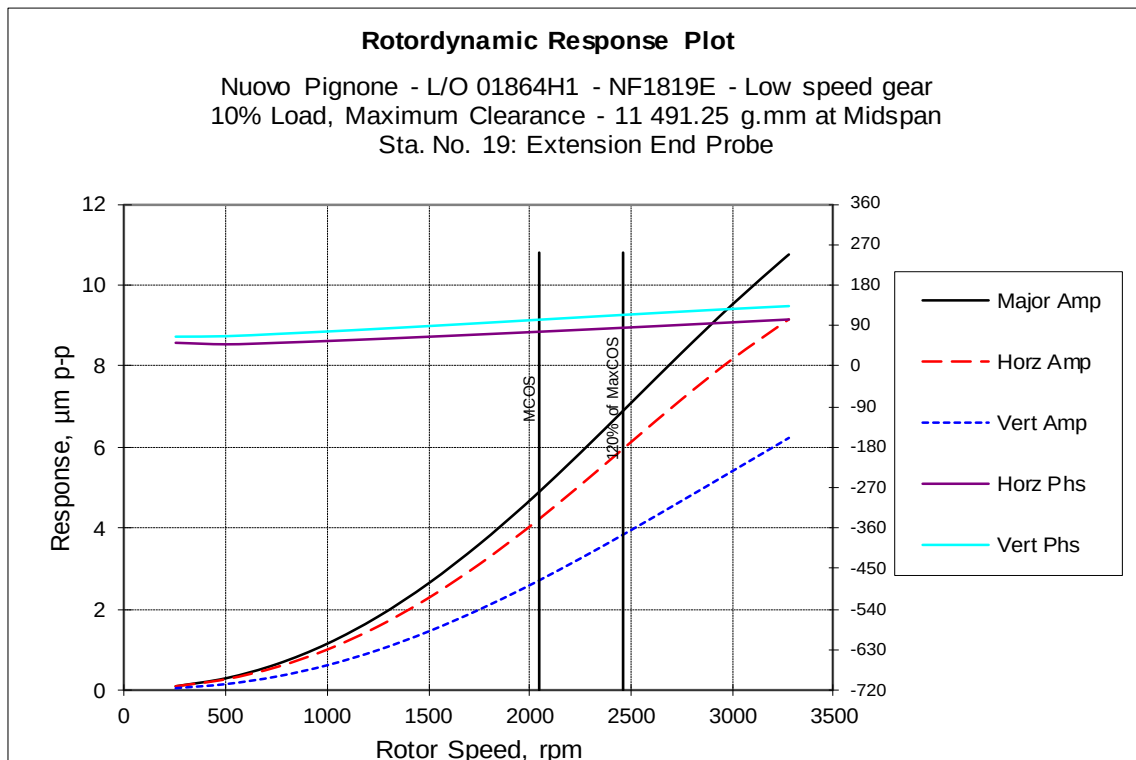


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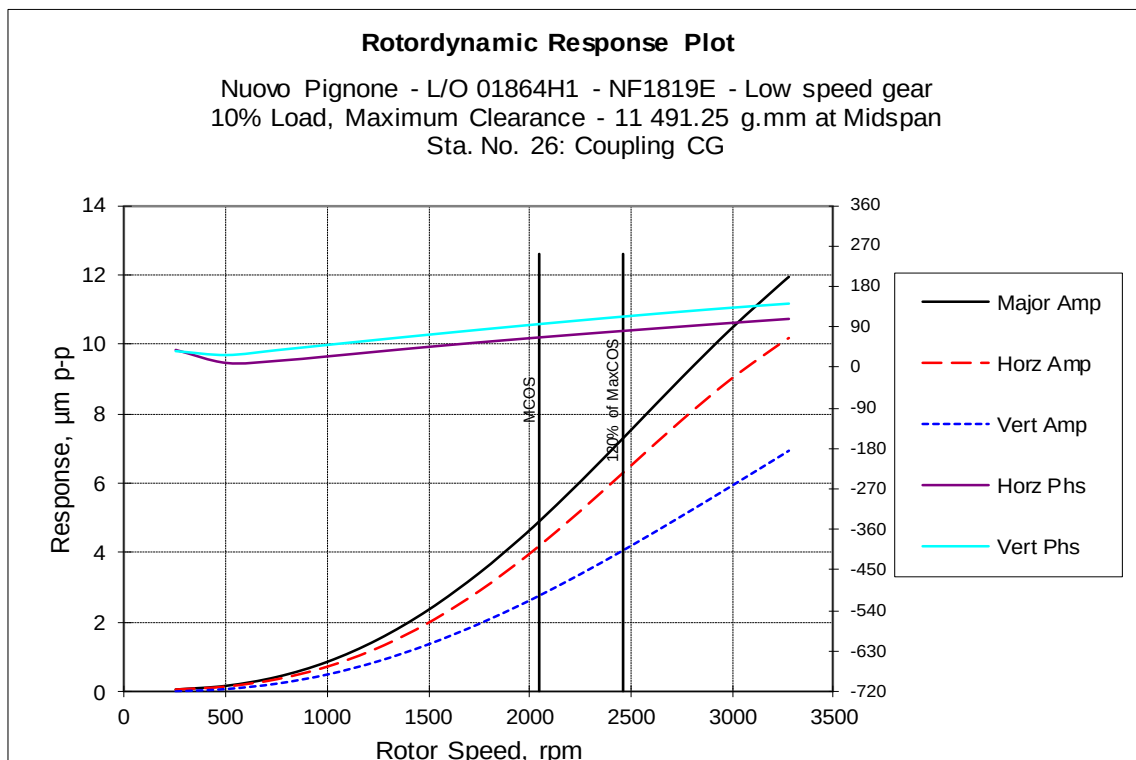


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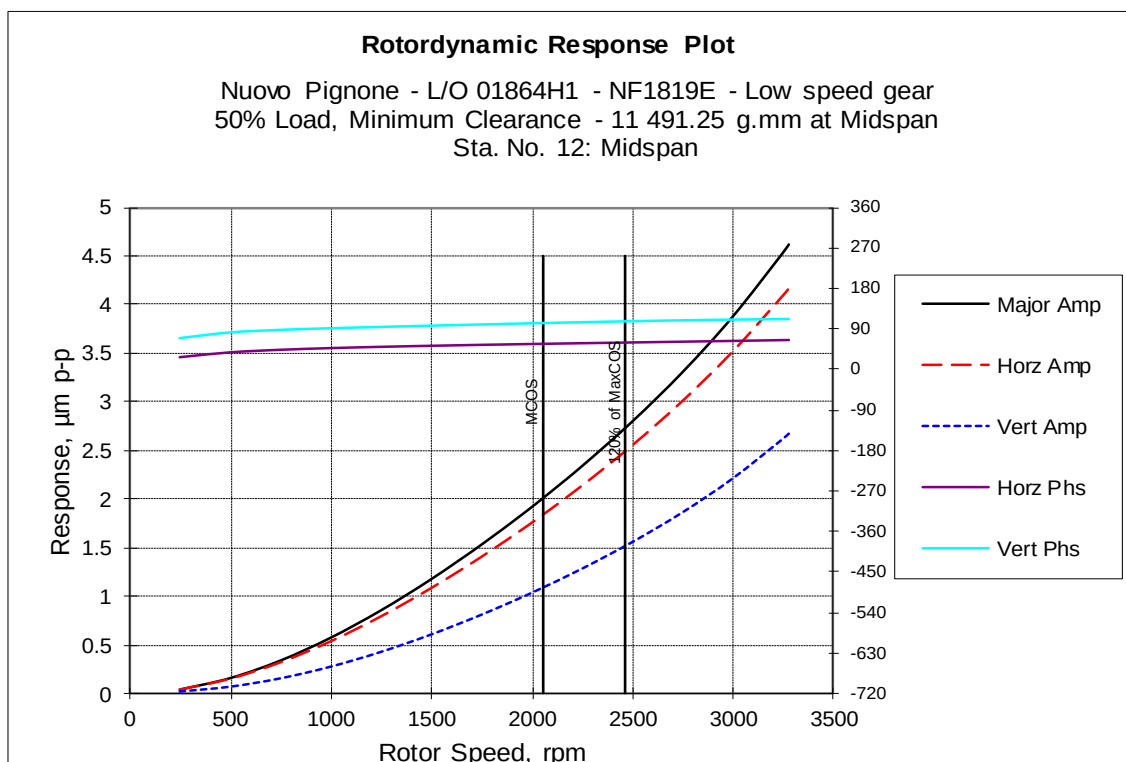
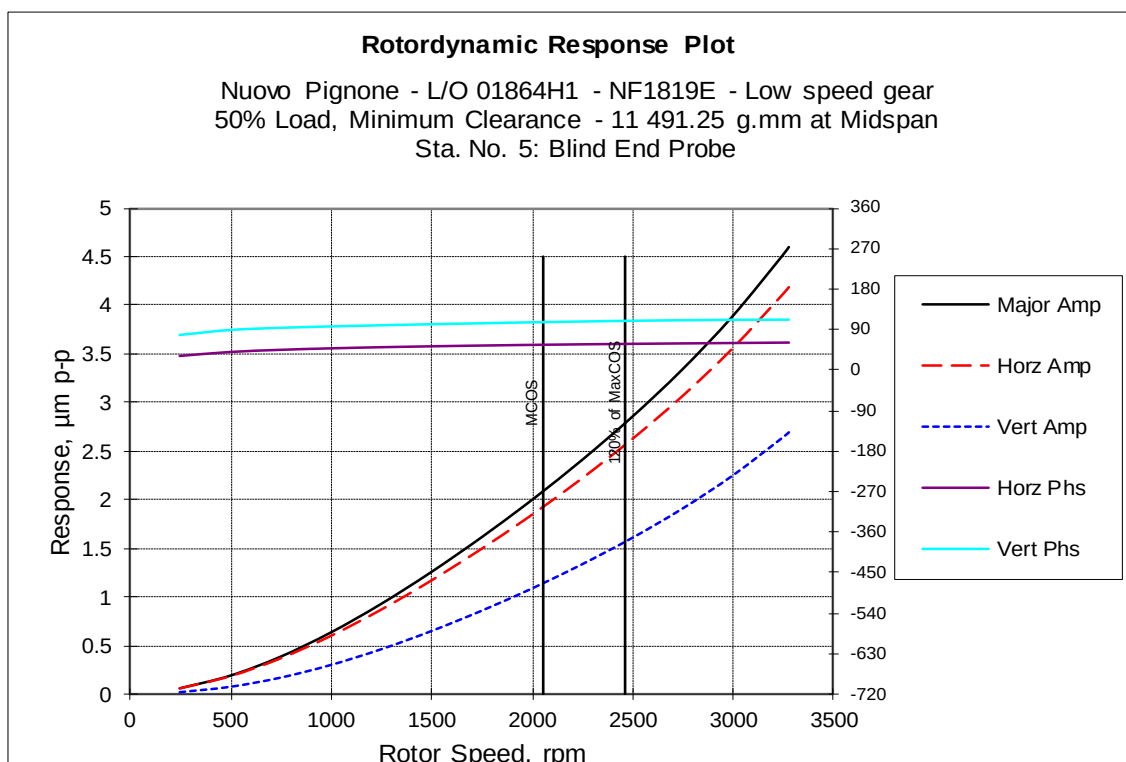
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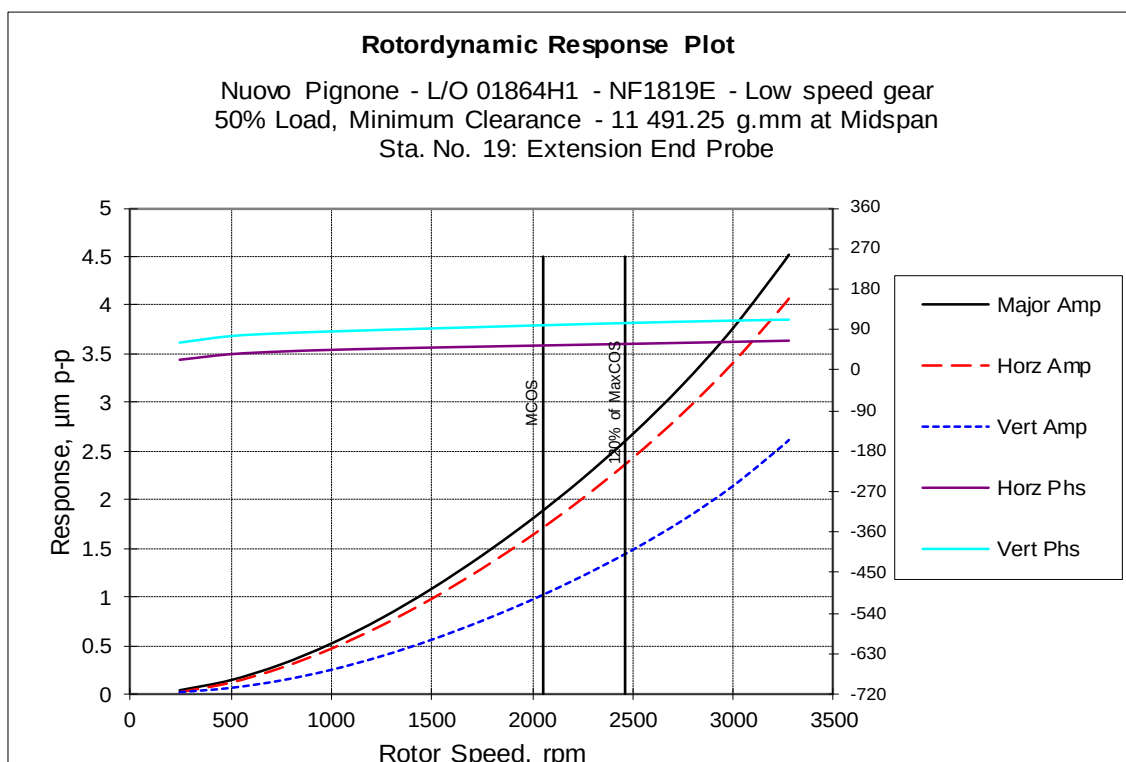


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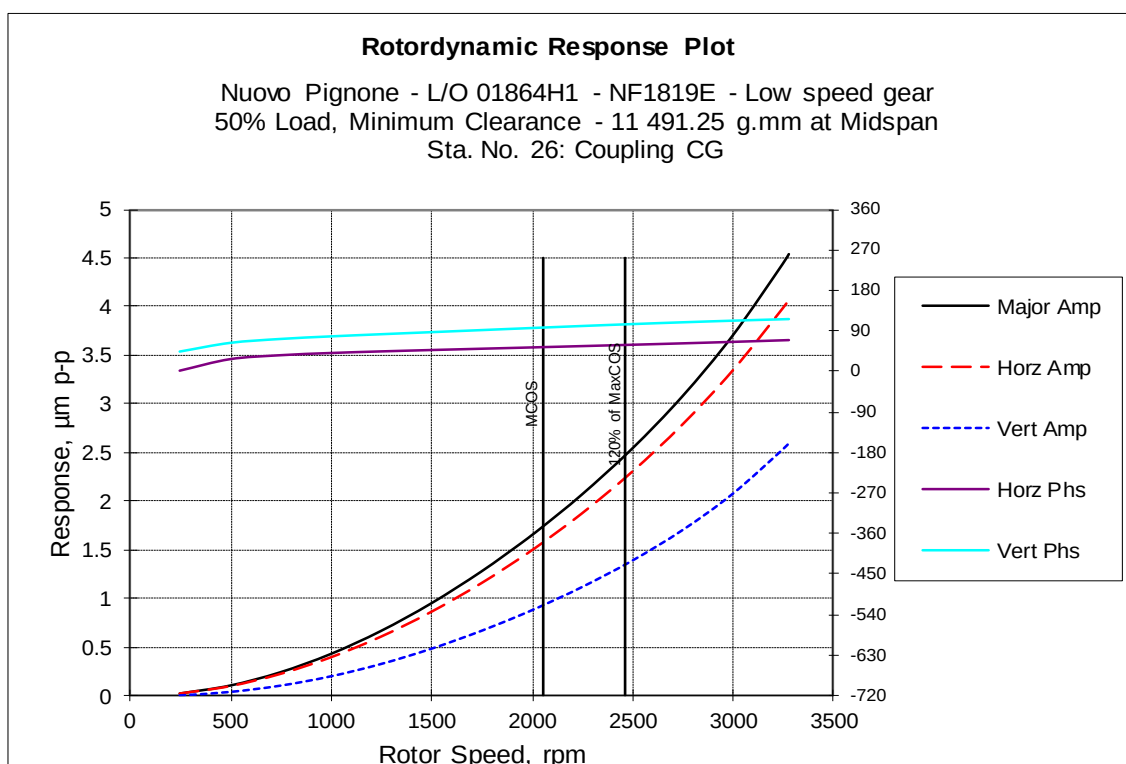


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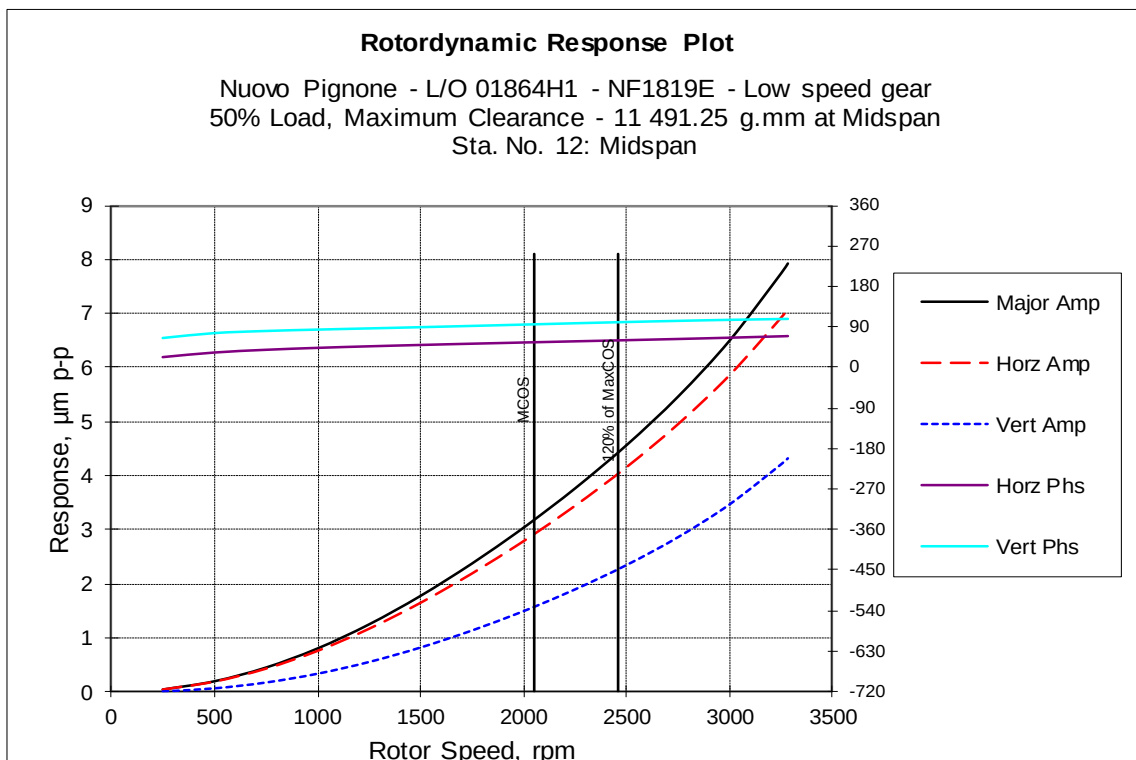
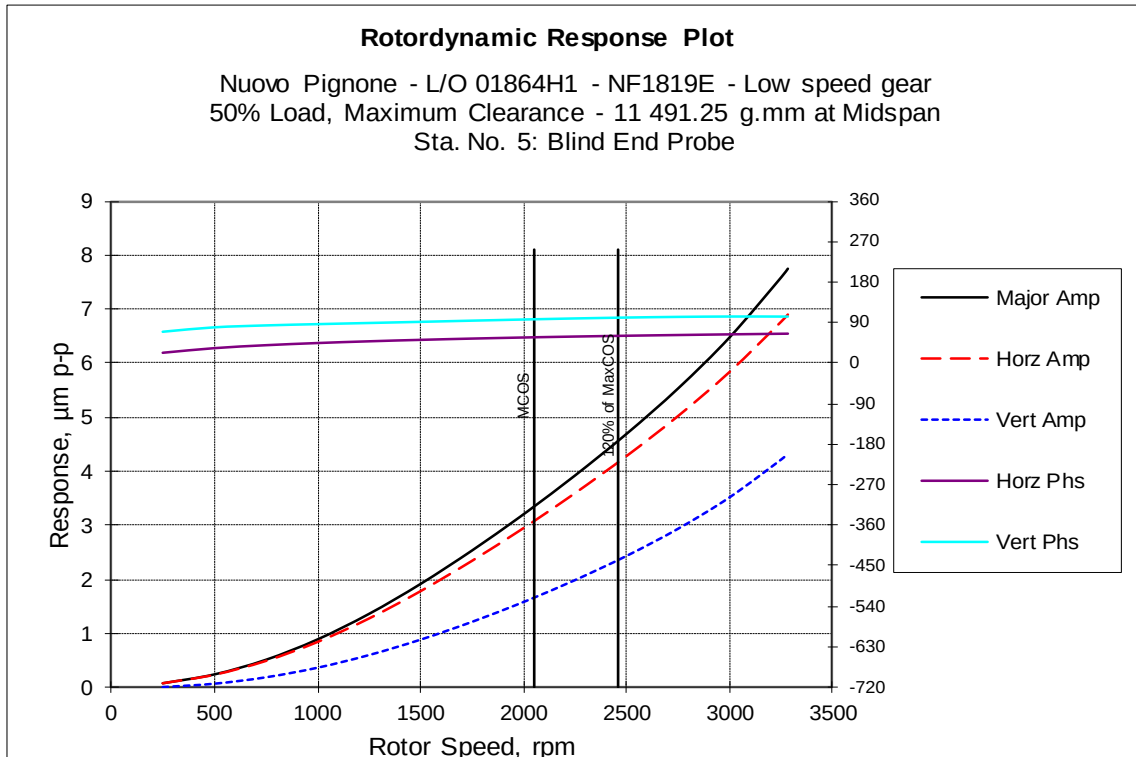
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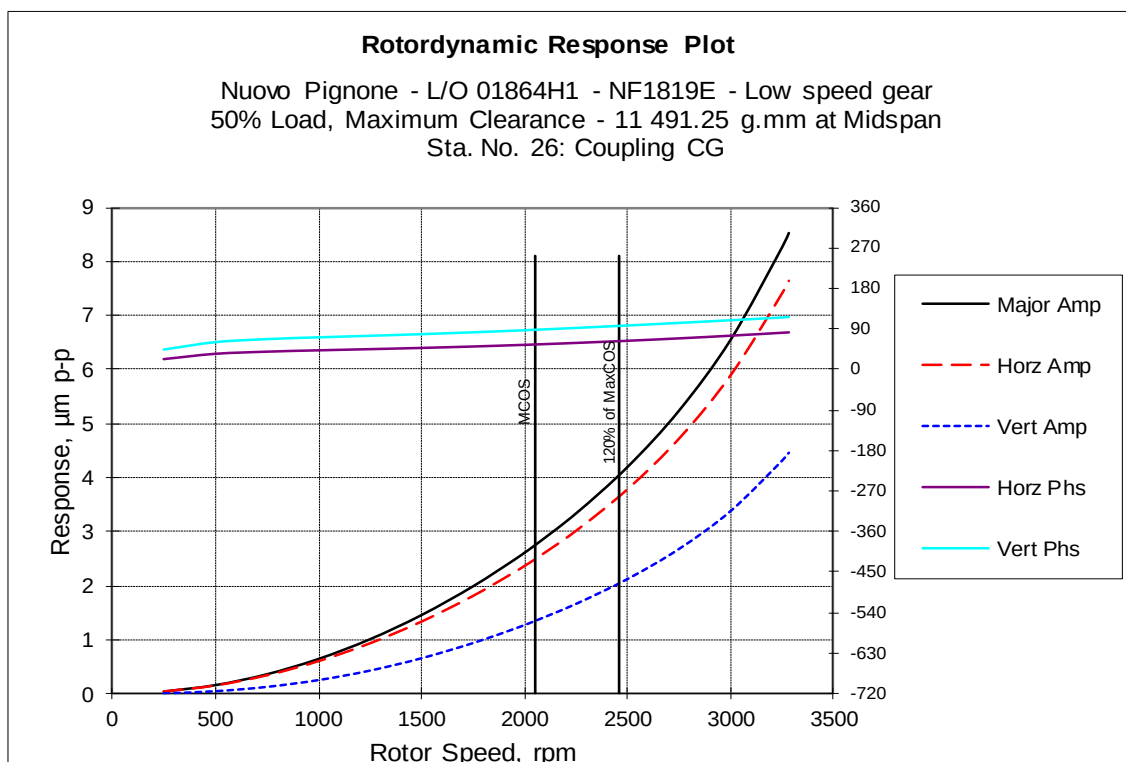
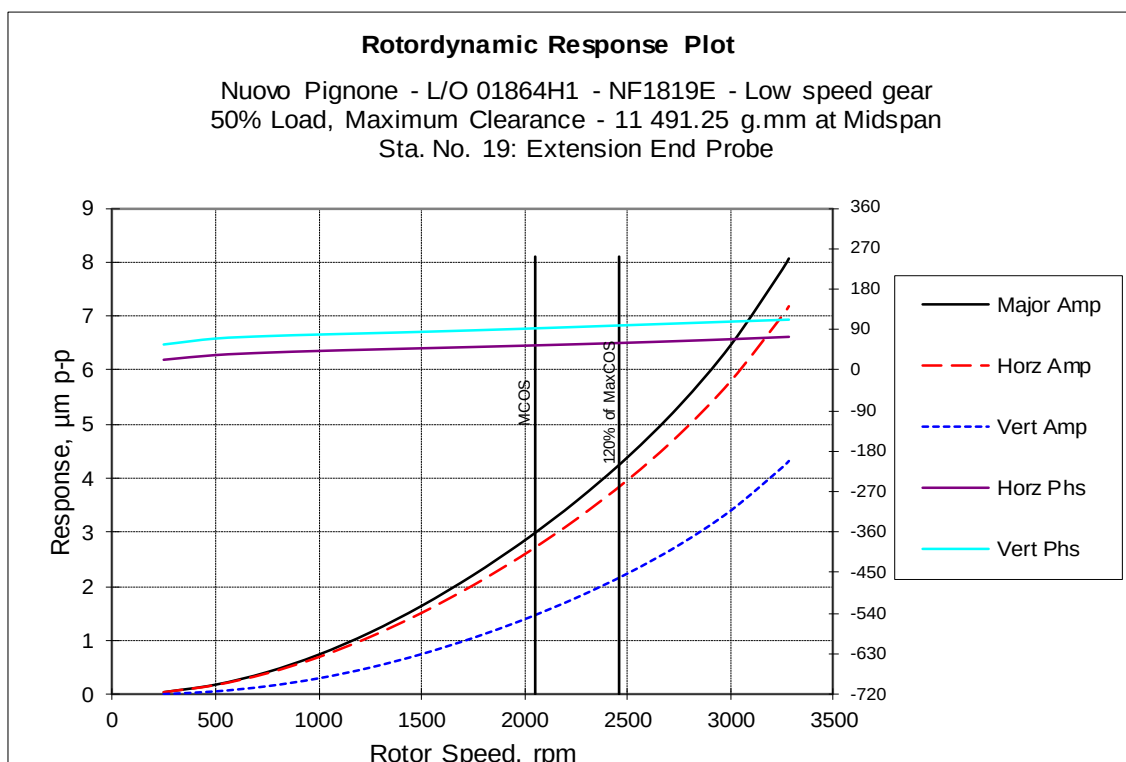
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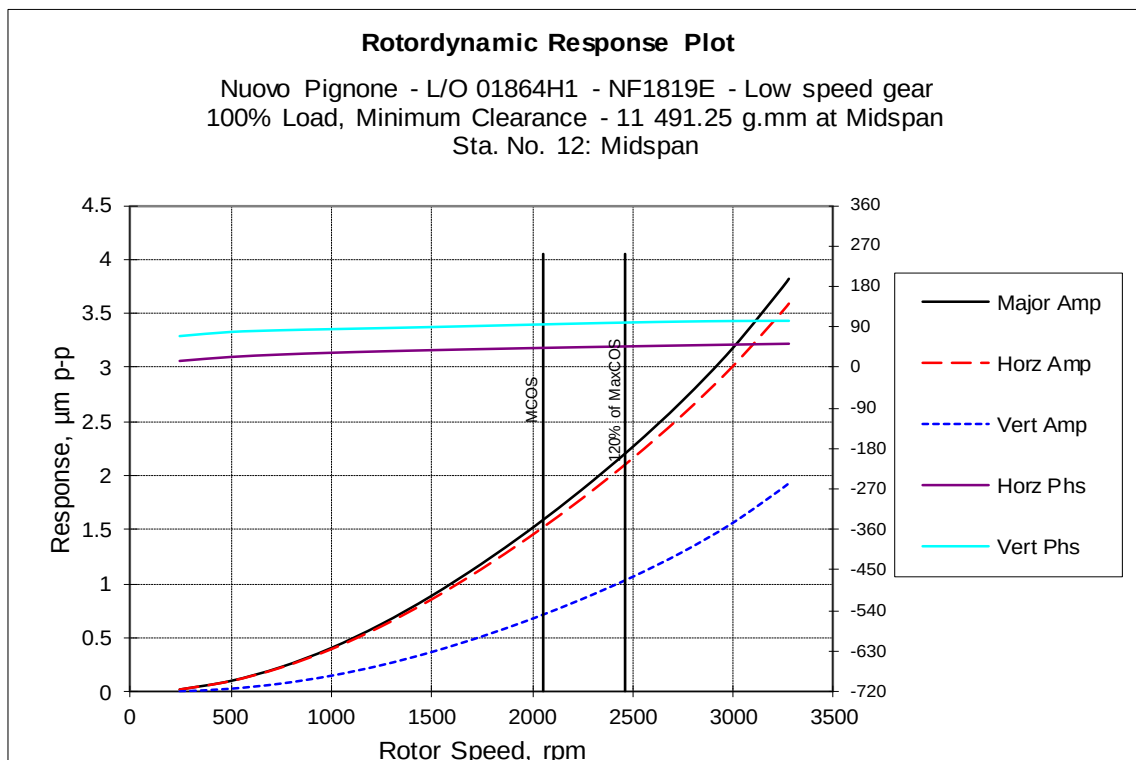
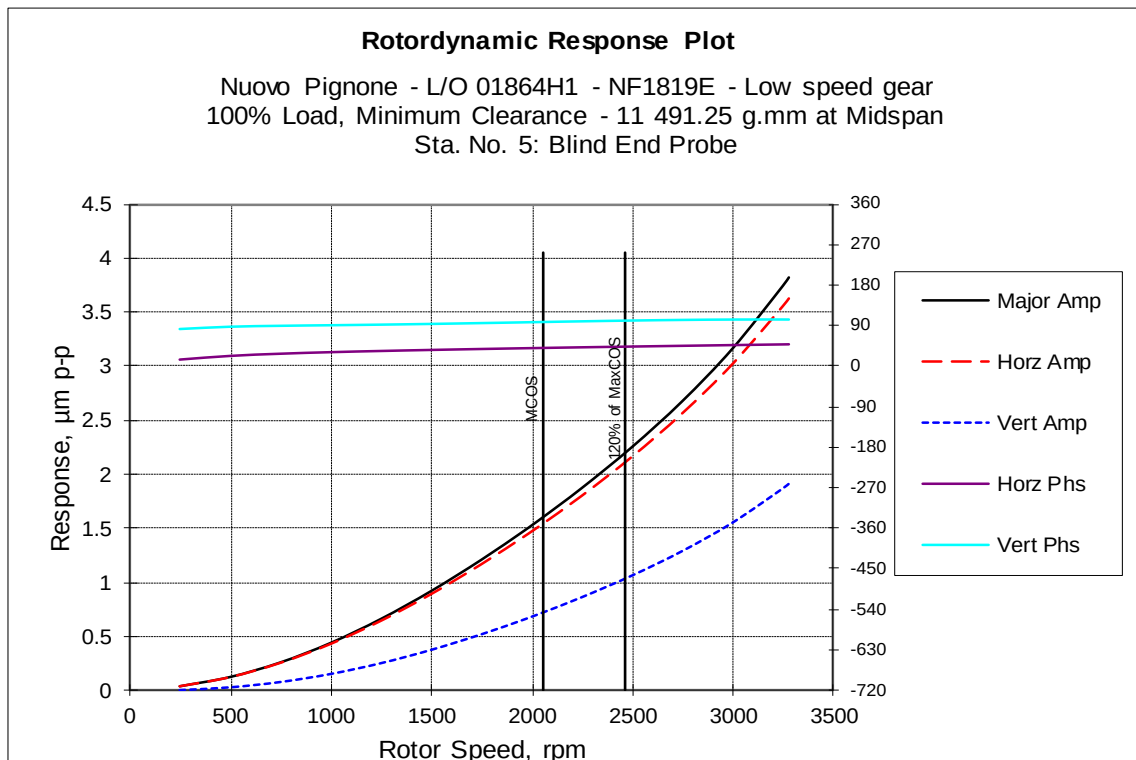
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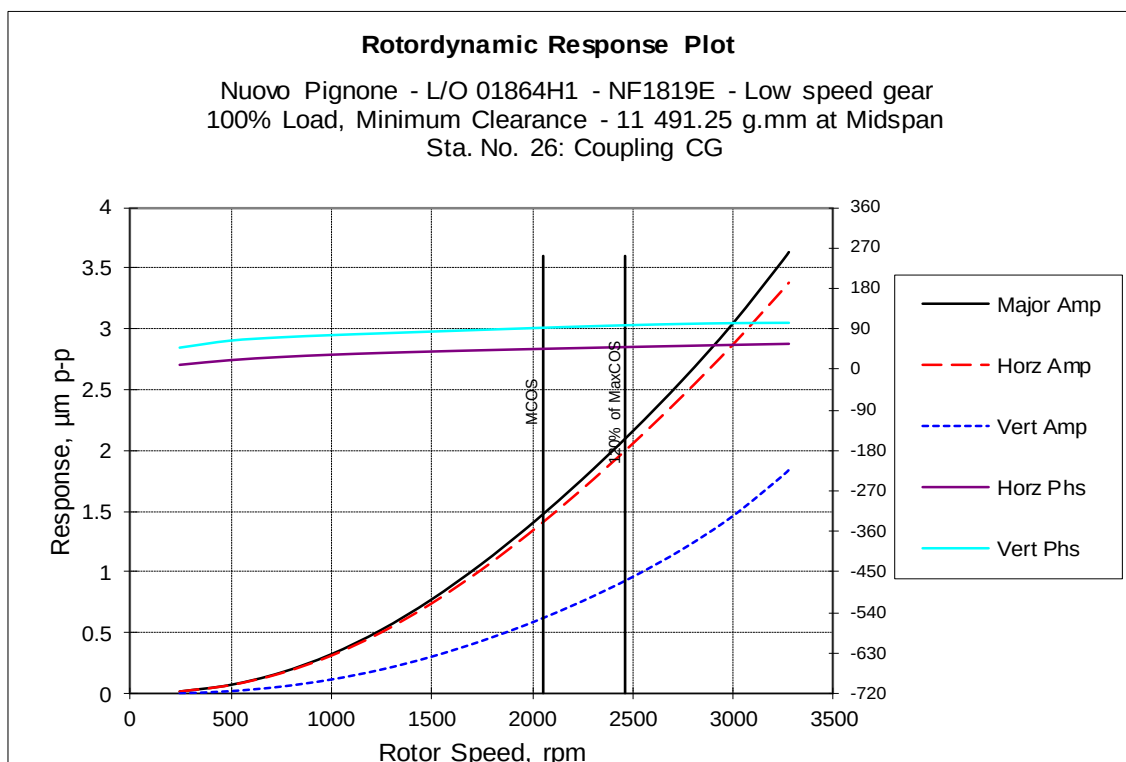
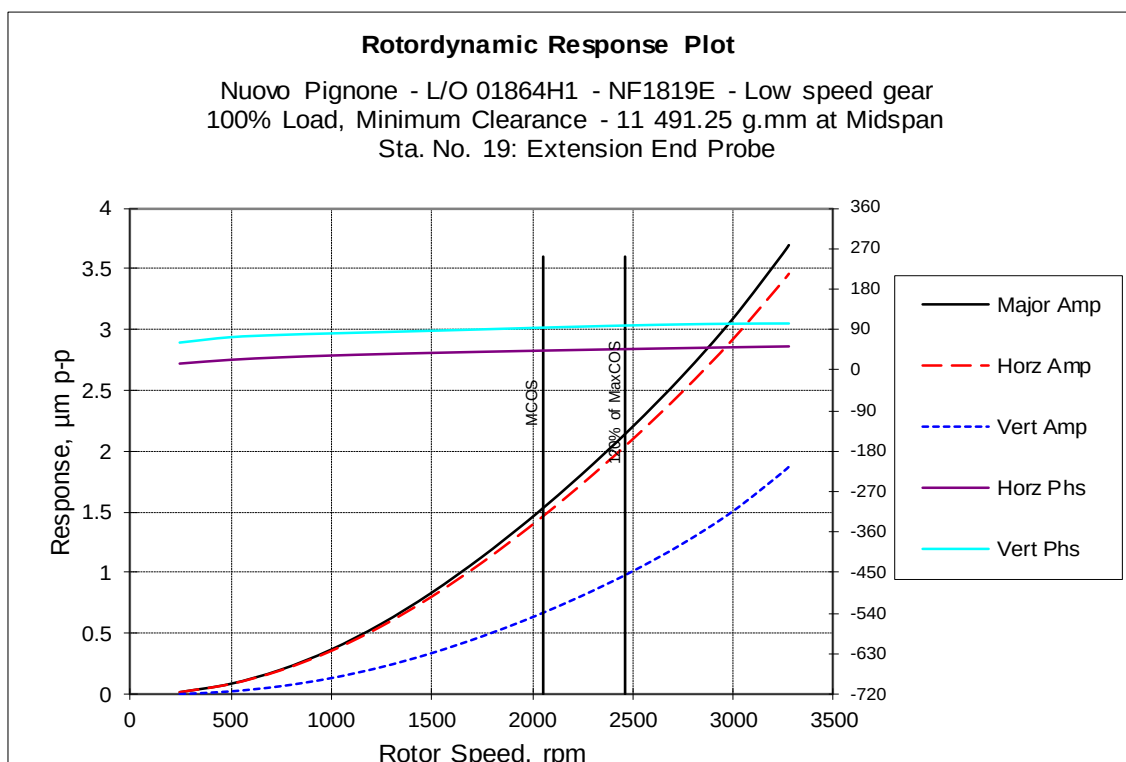
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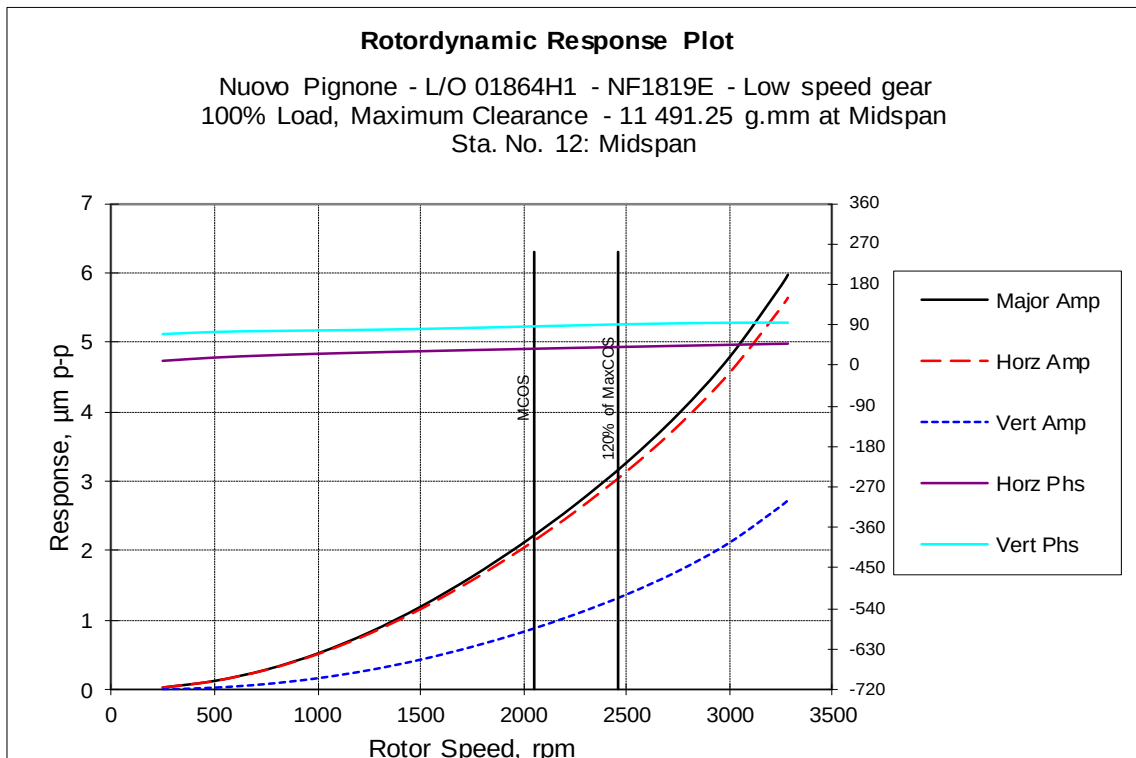
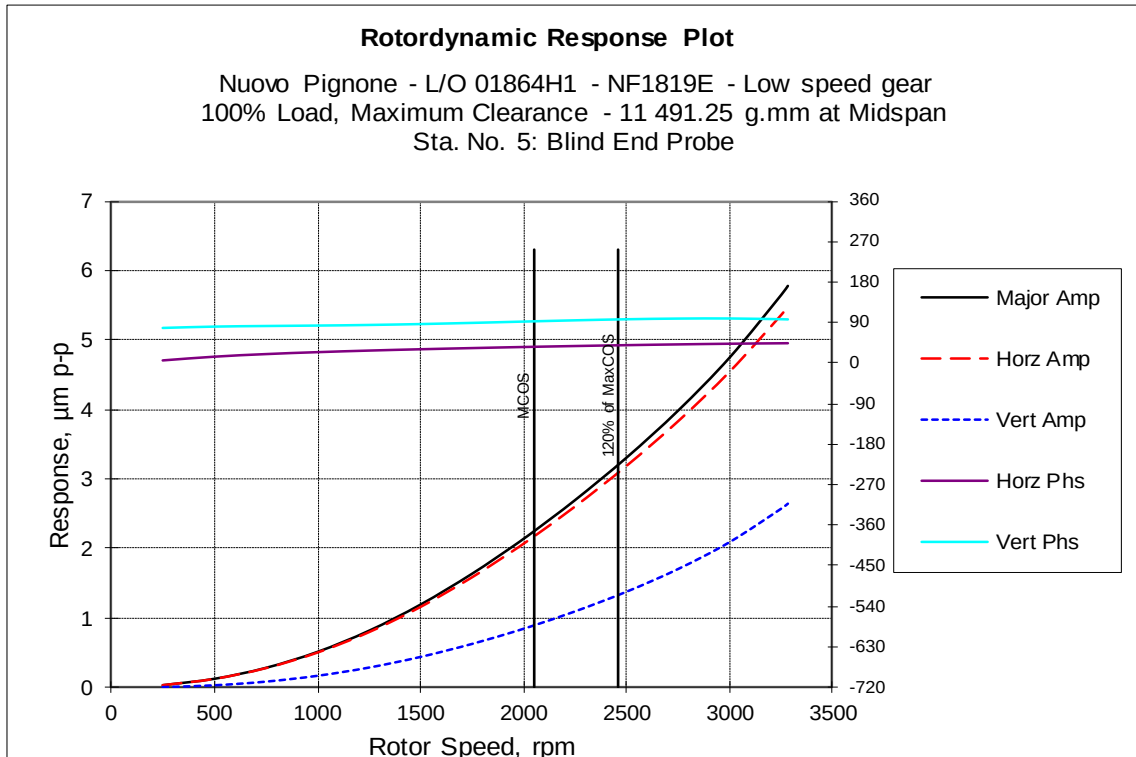
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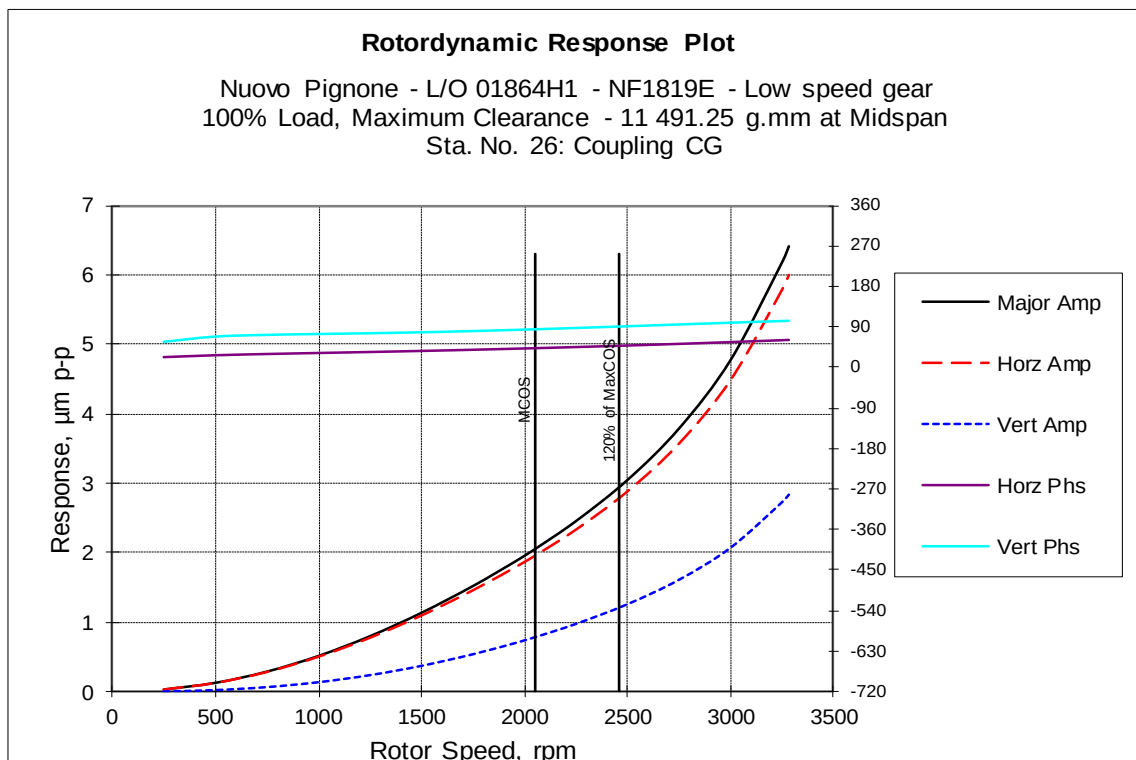
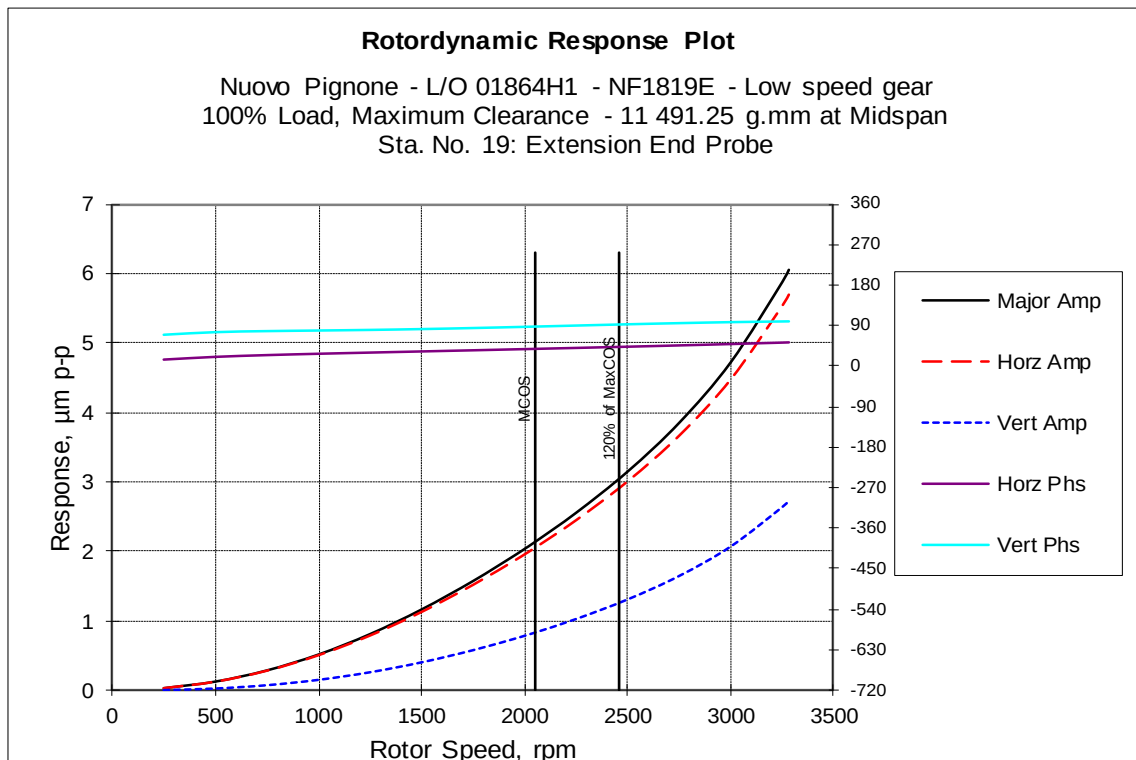
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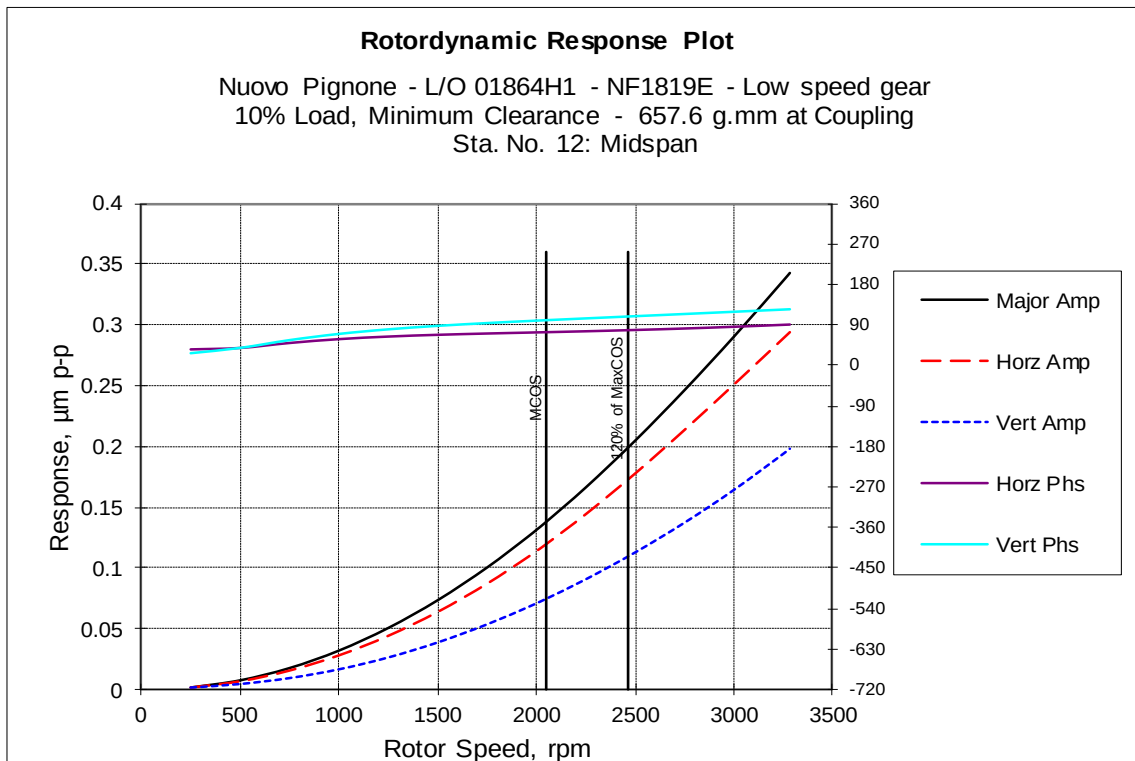
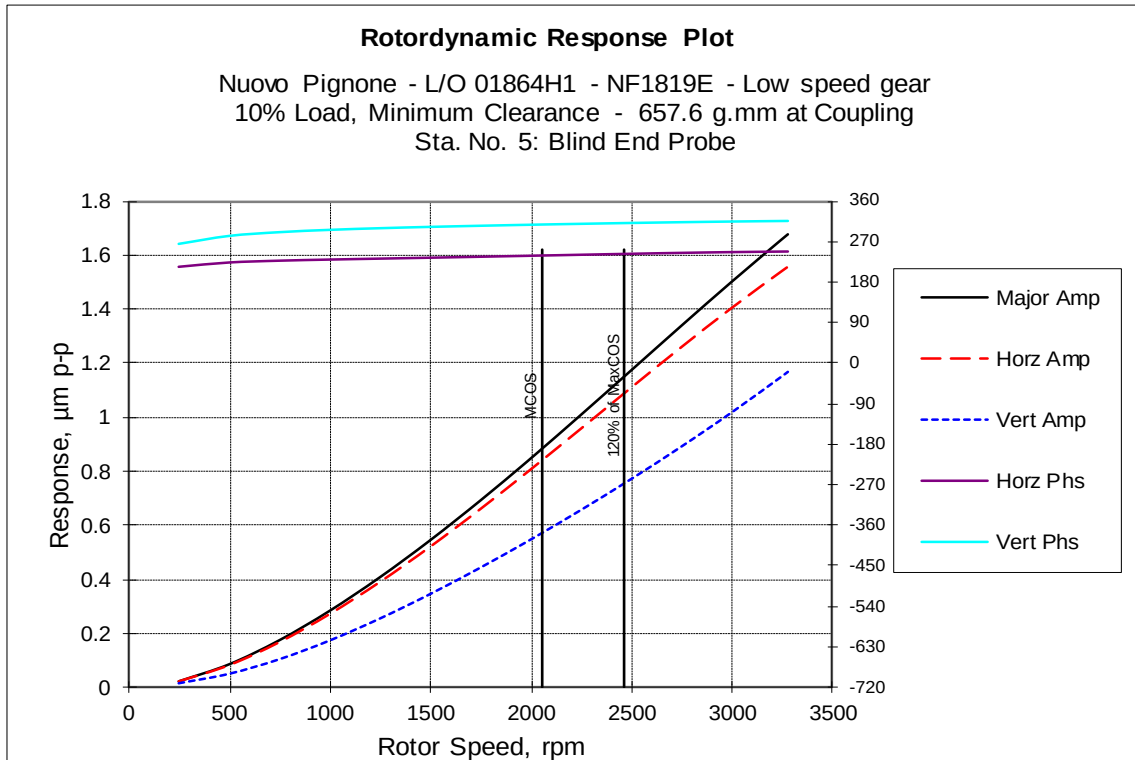


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Appendix L - Coupling Unbalance Response Plots



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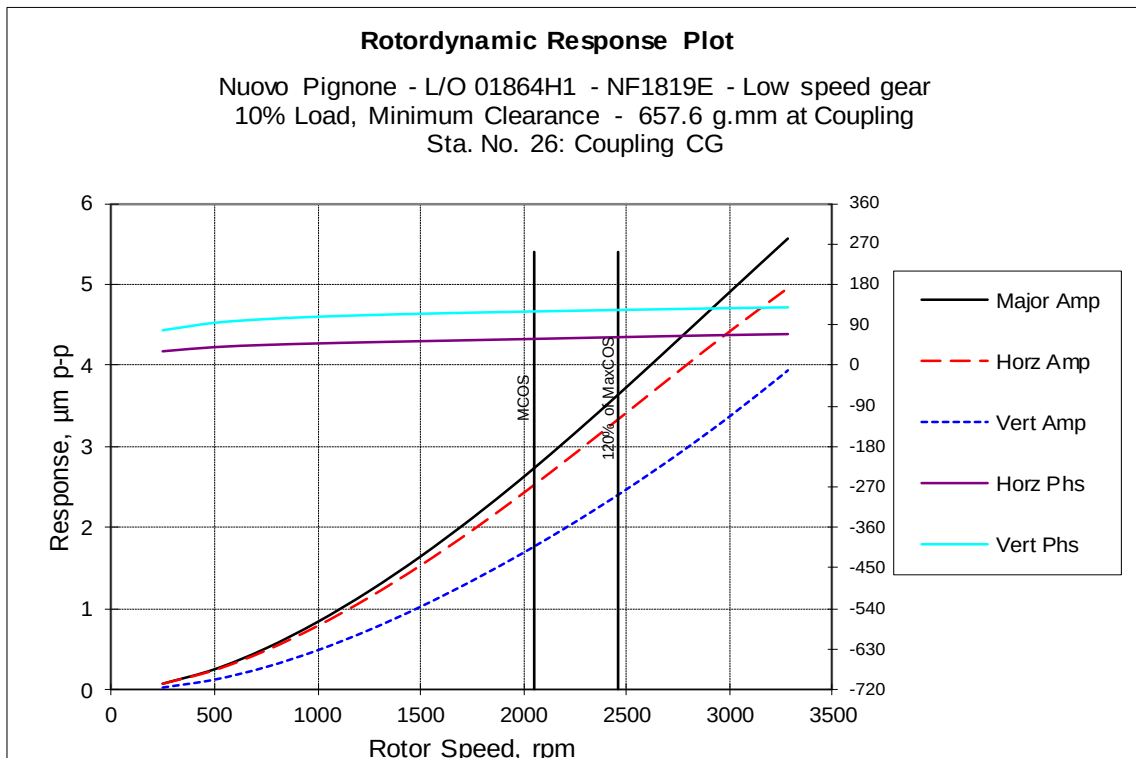
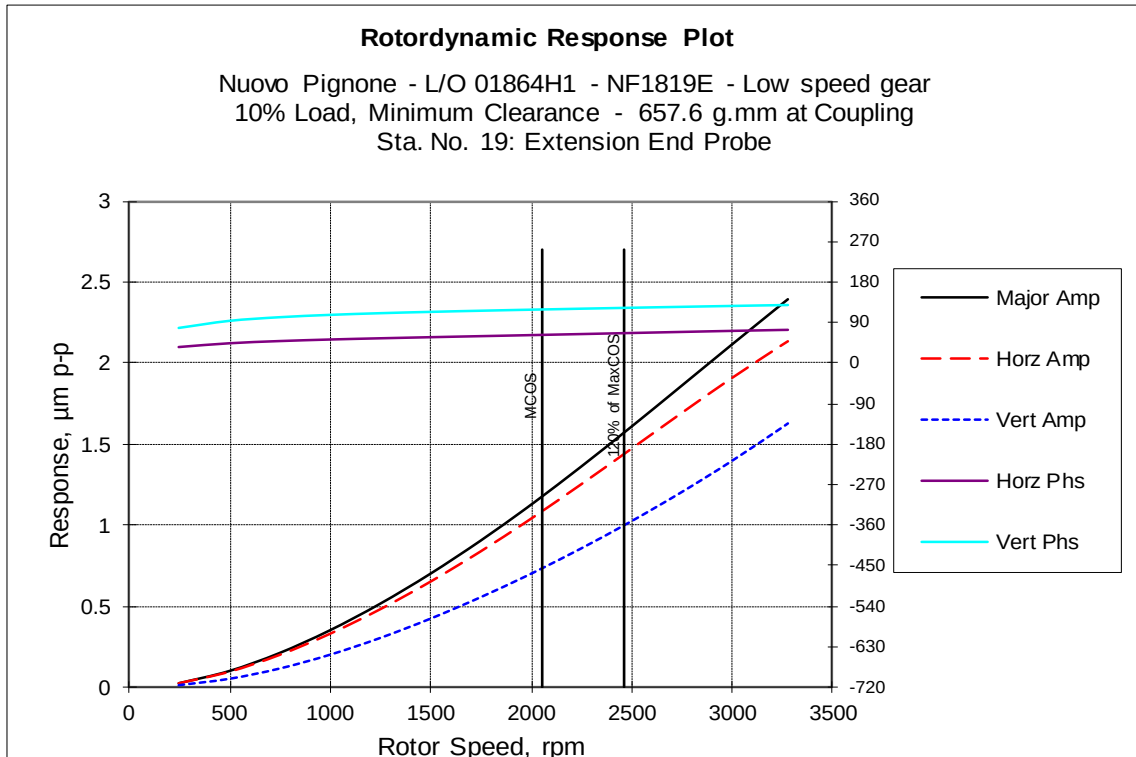
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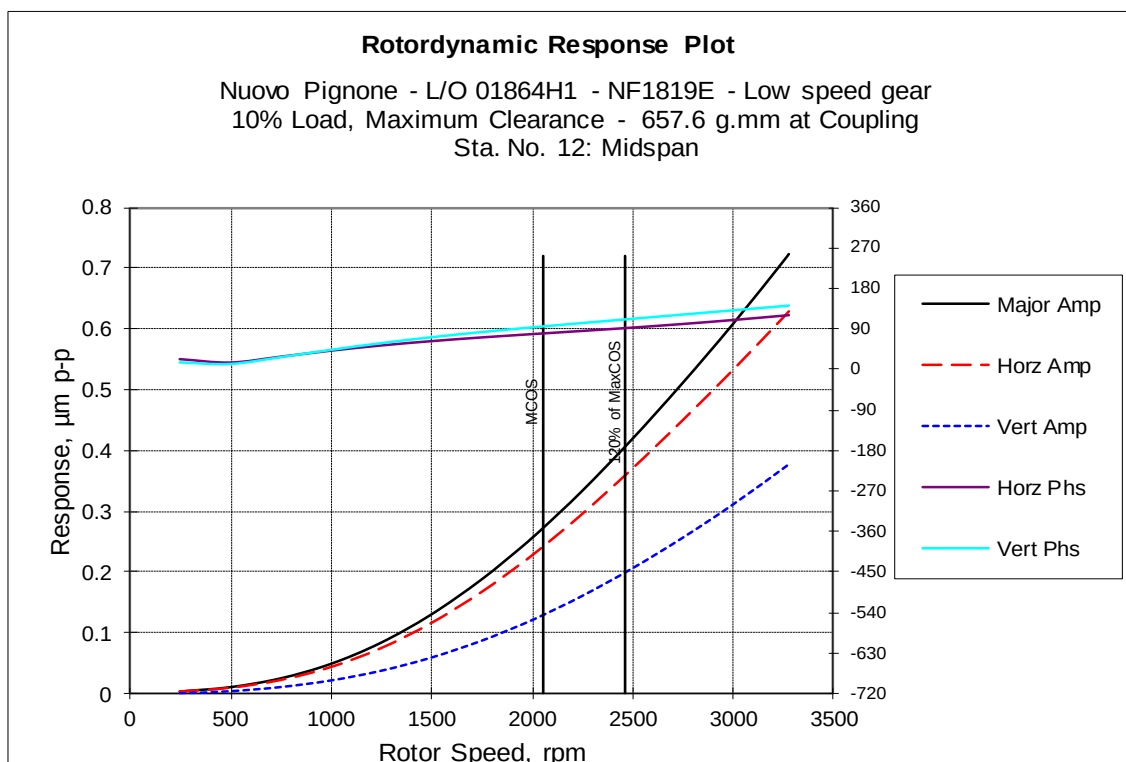
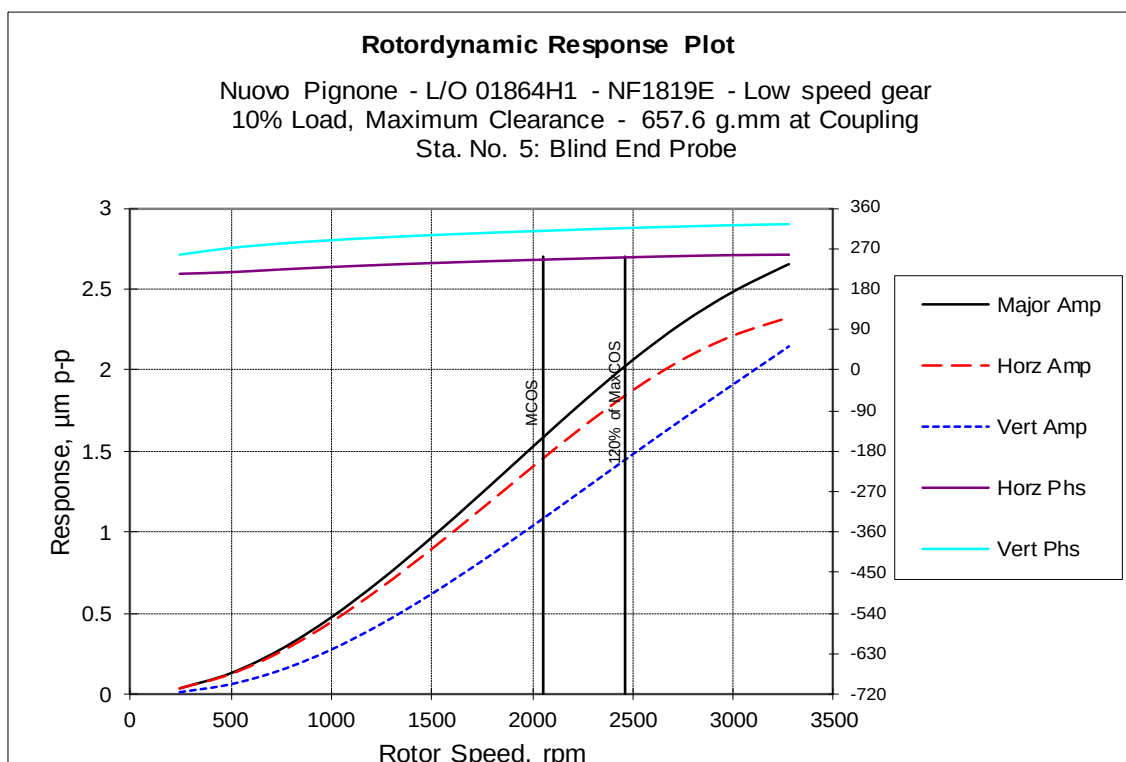
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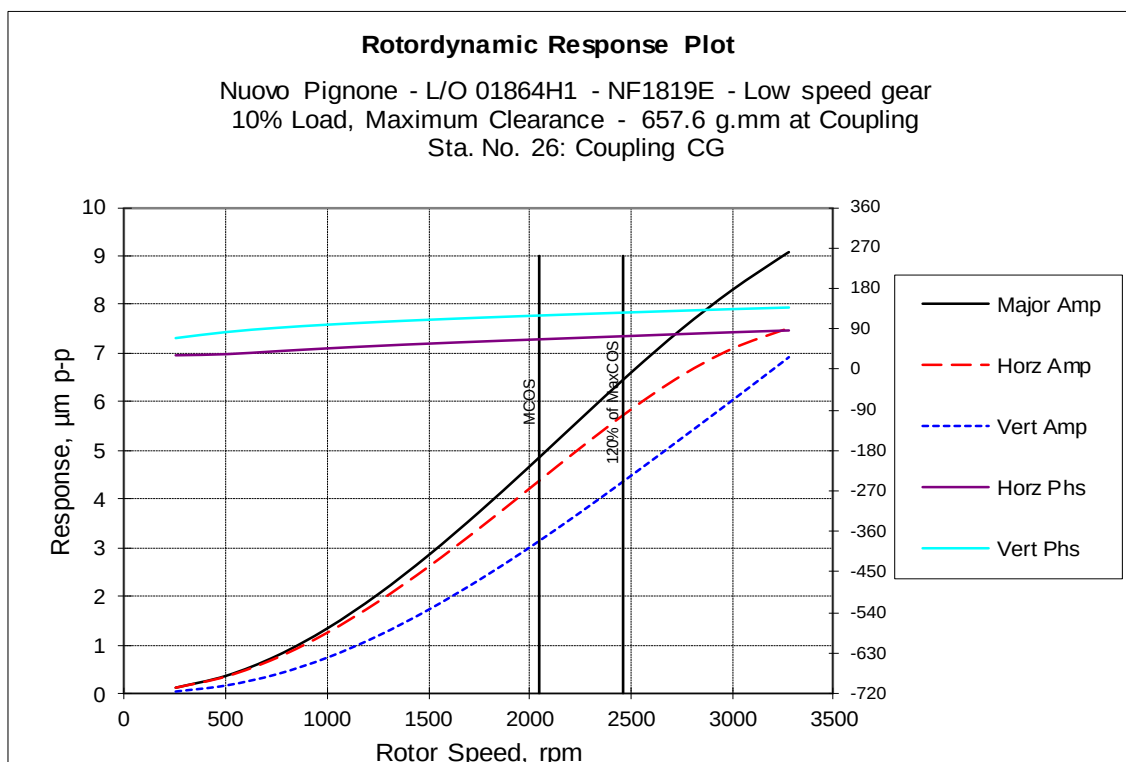
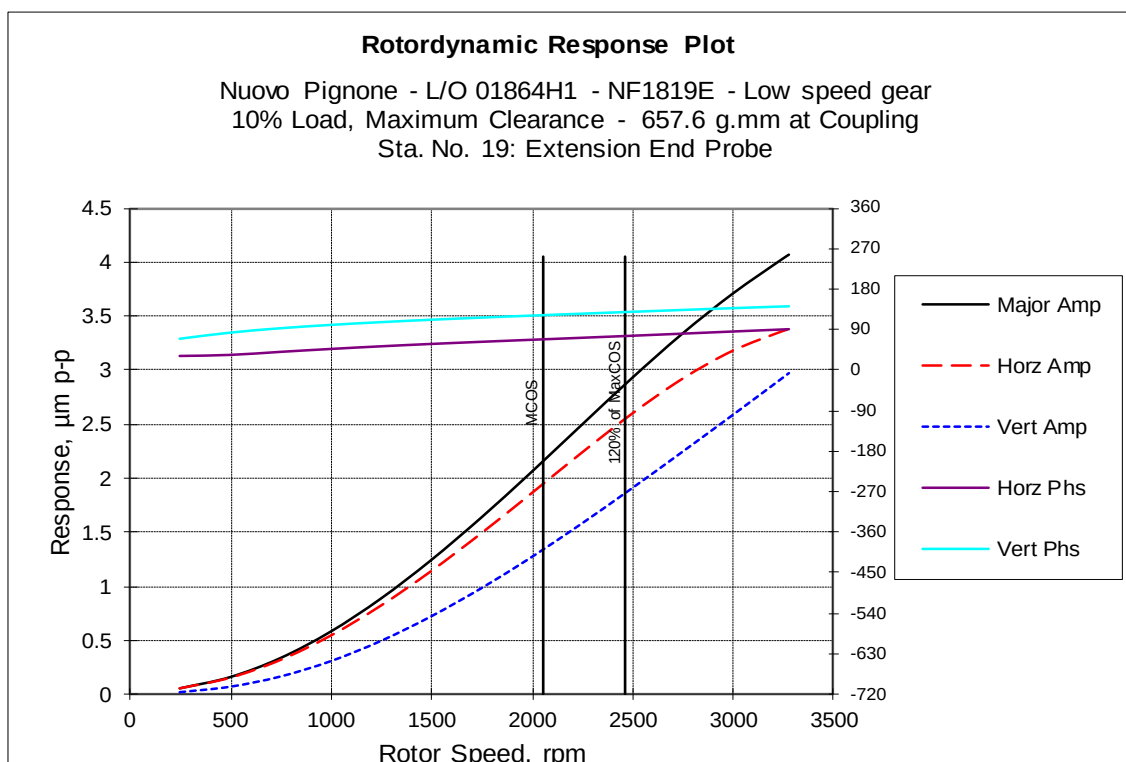
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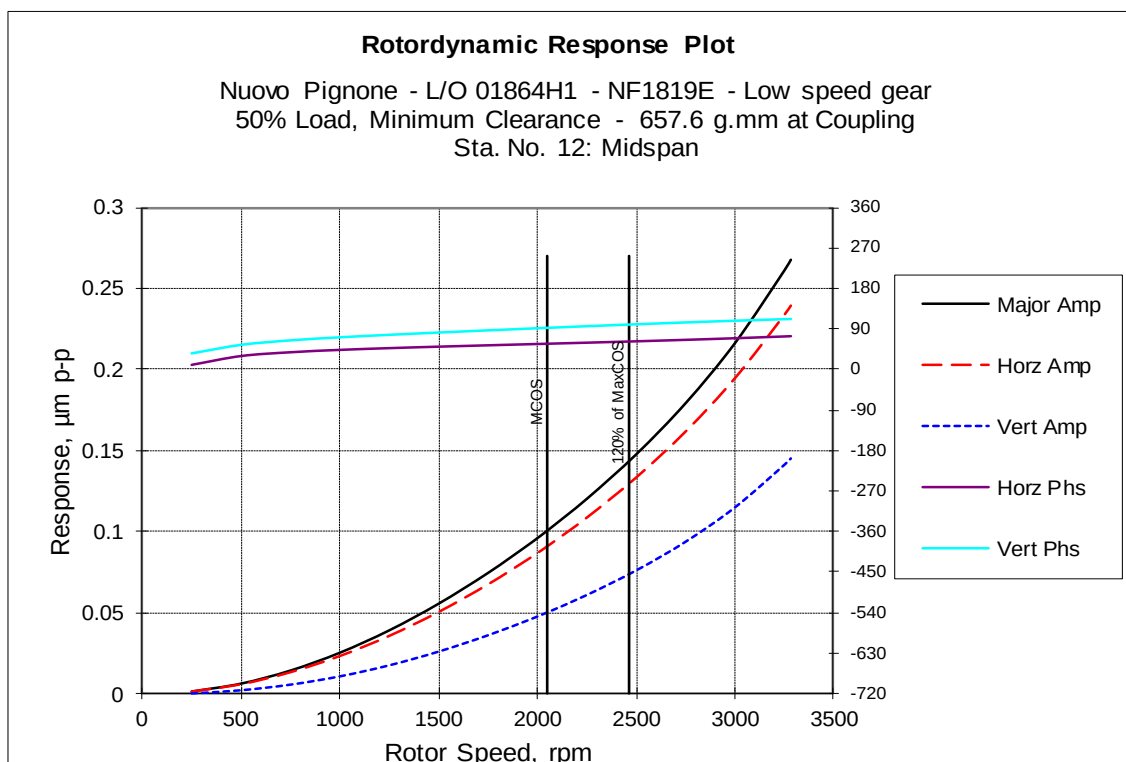
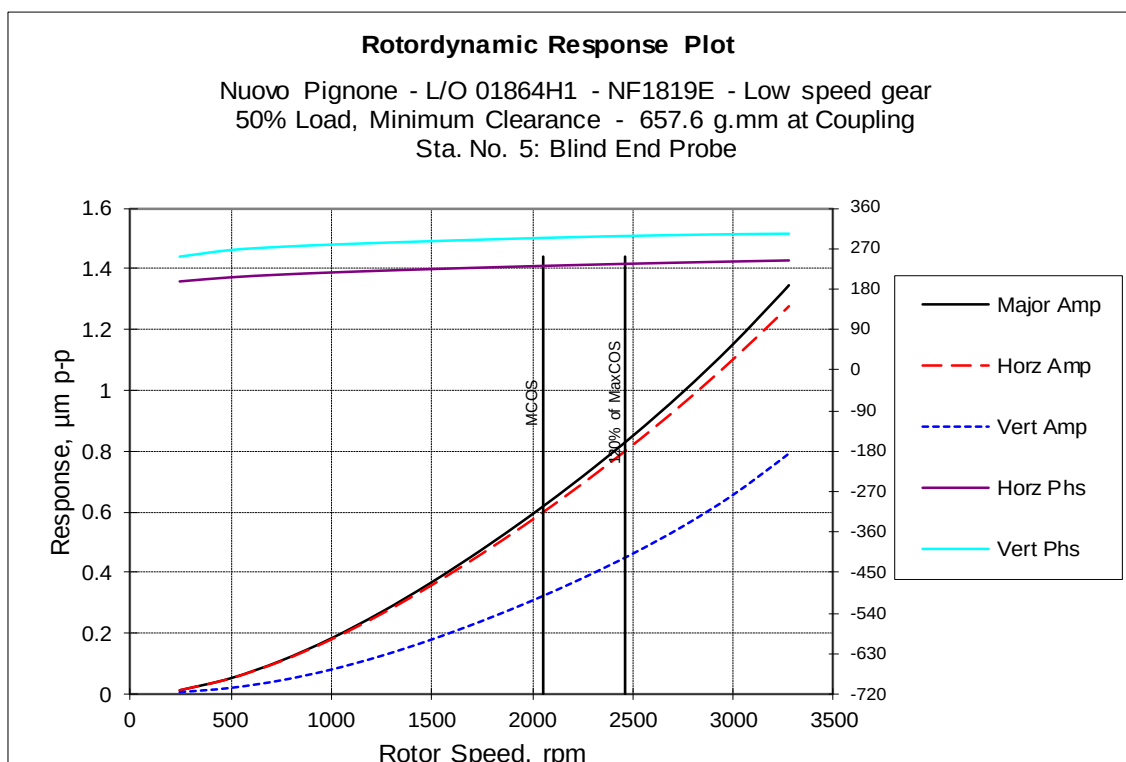
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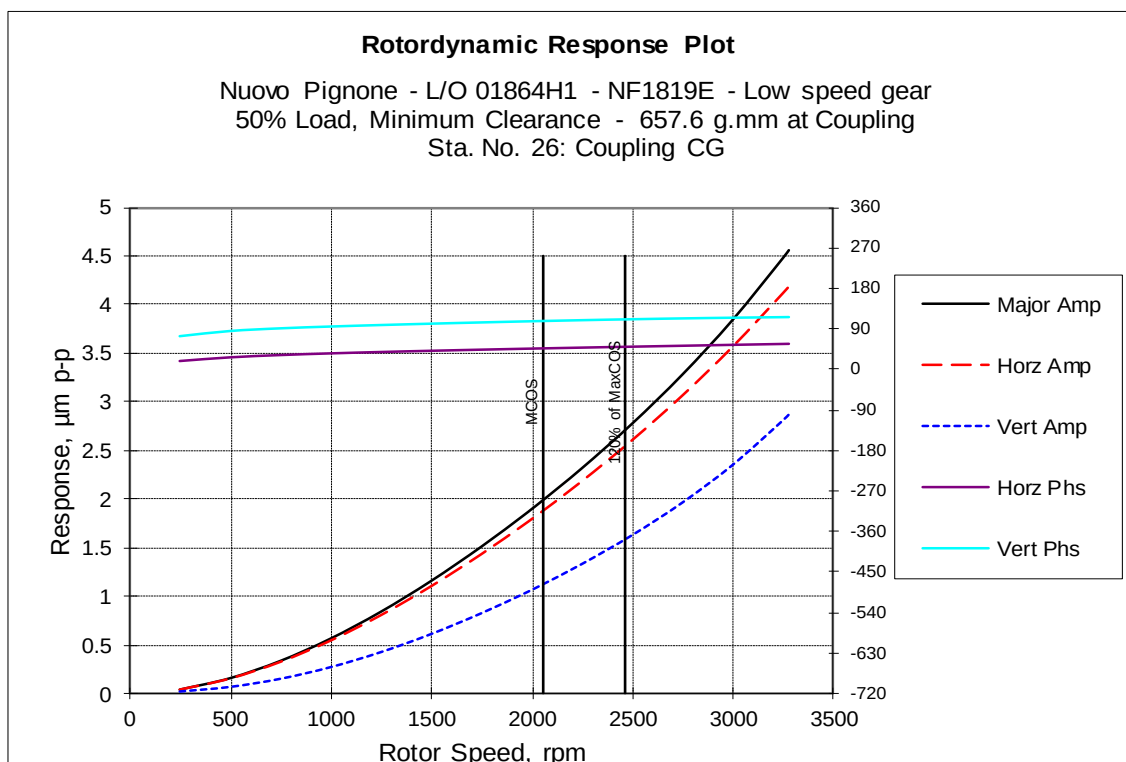
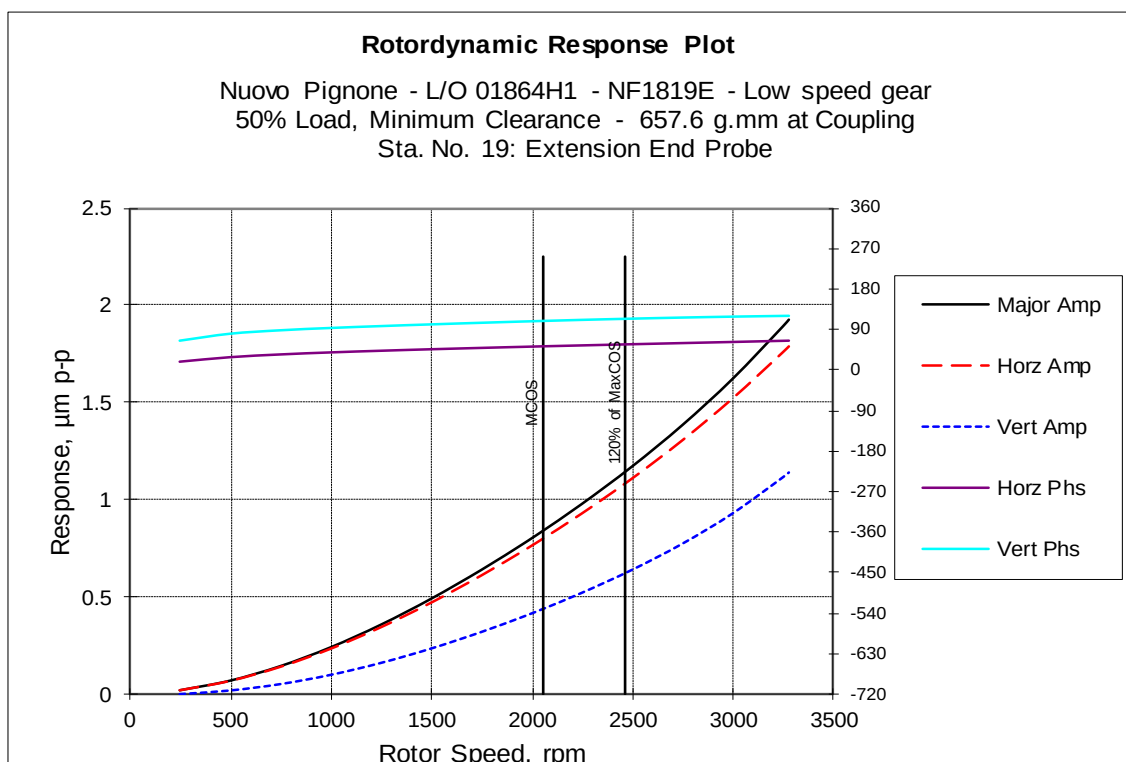
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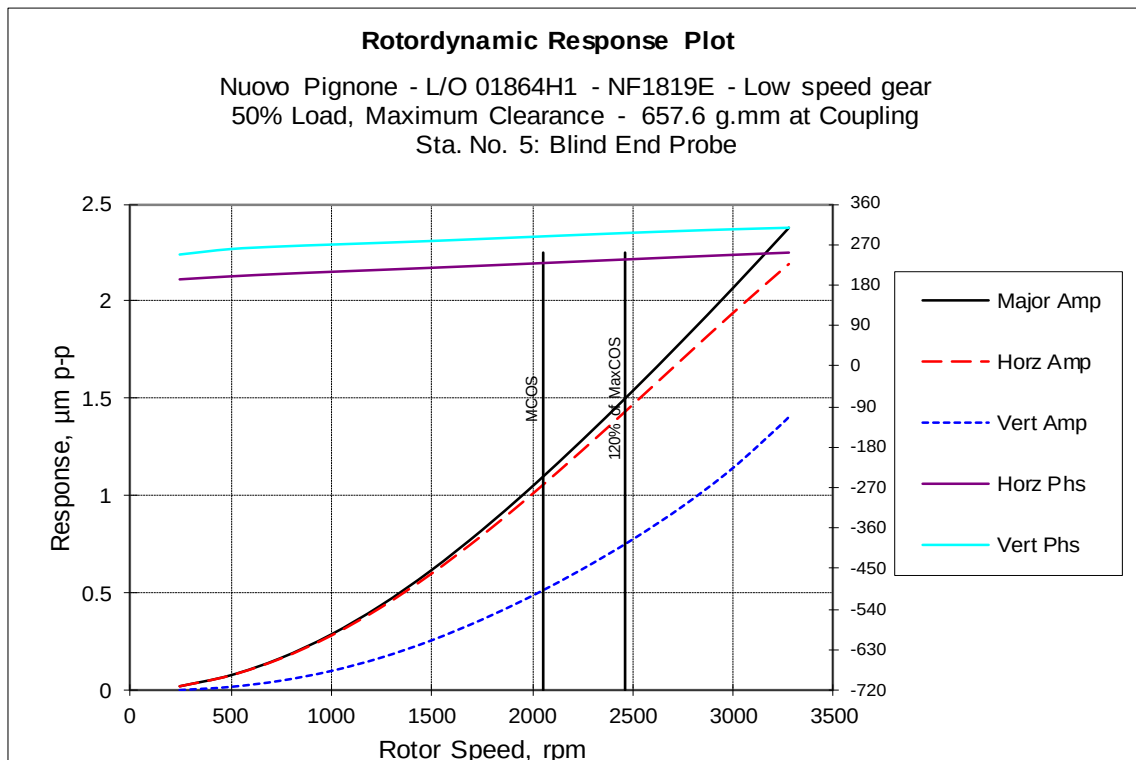


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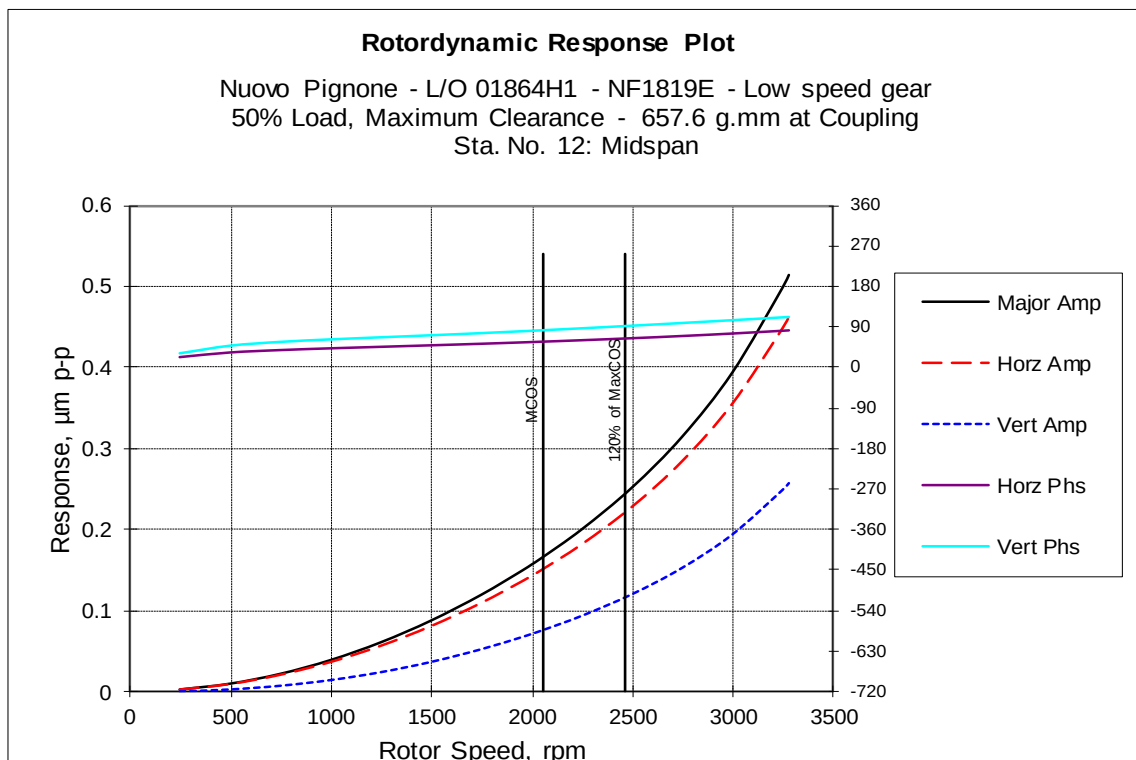
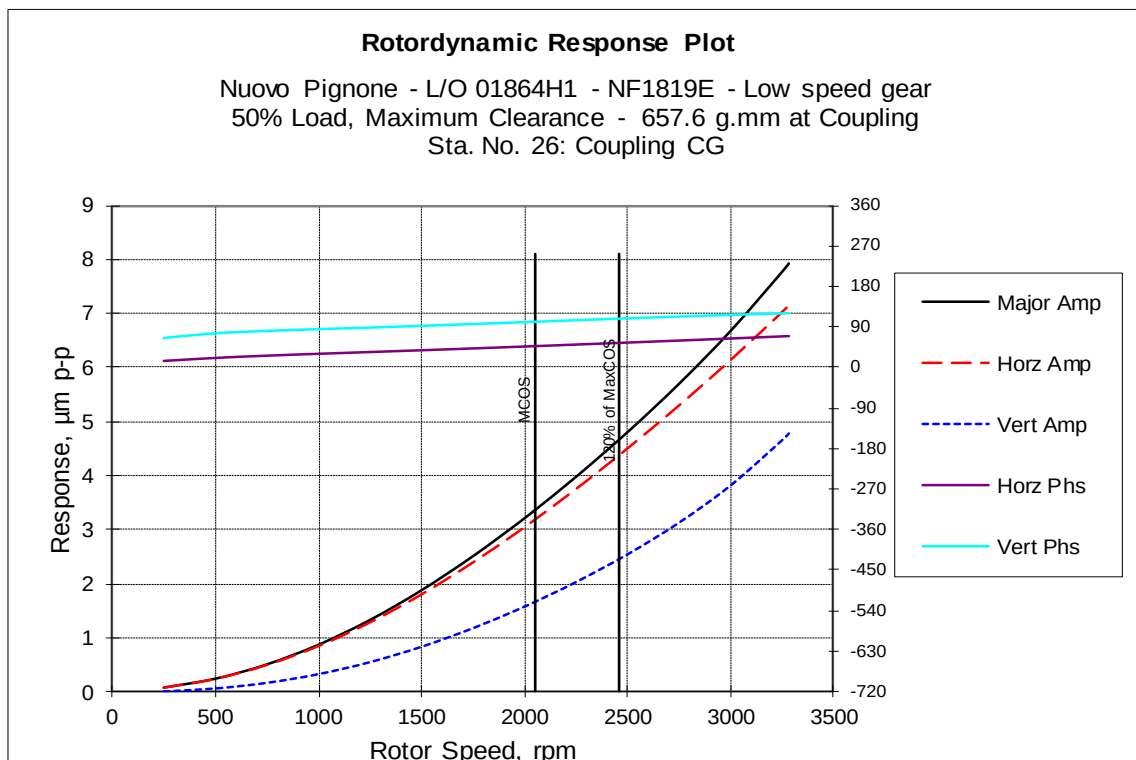
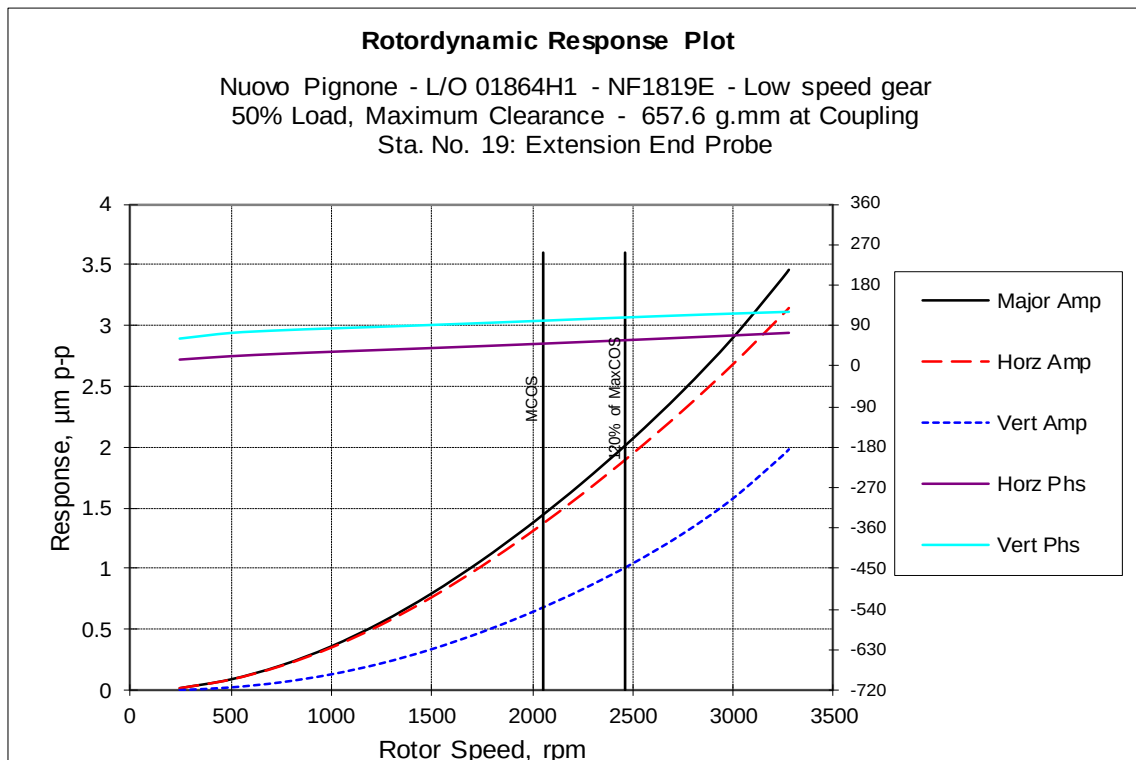
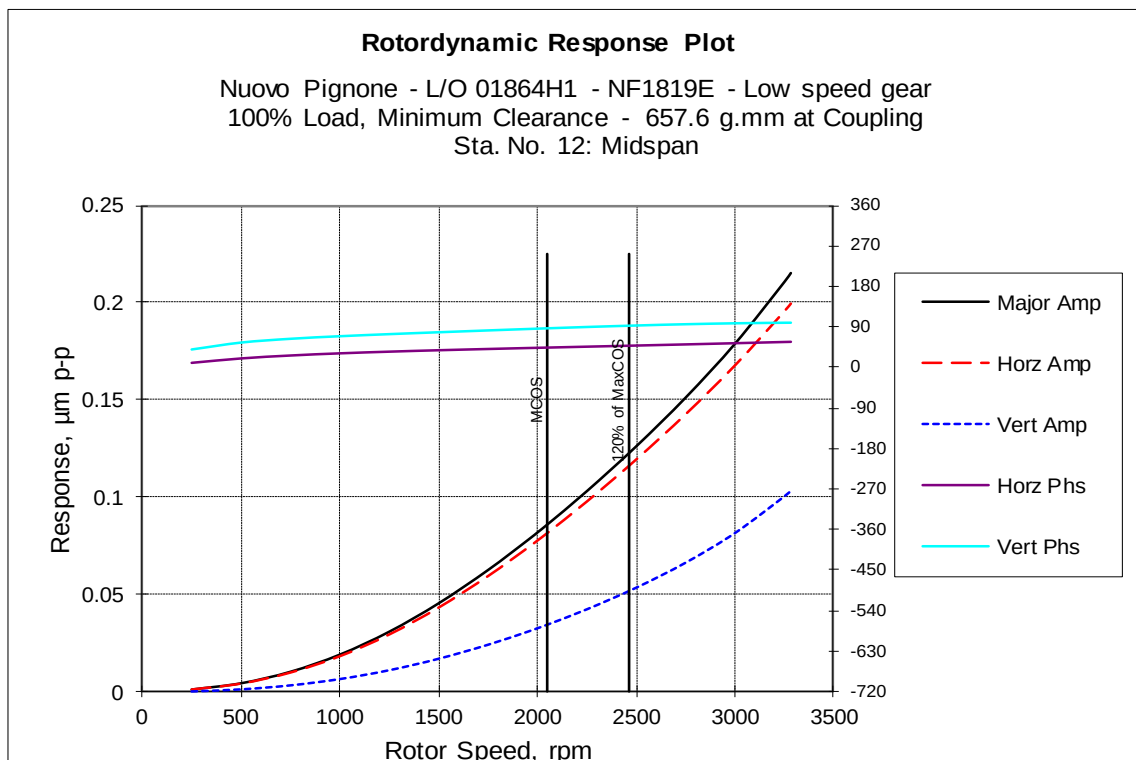
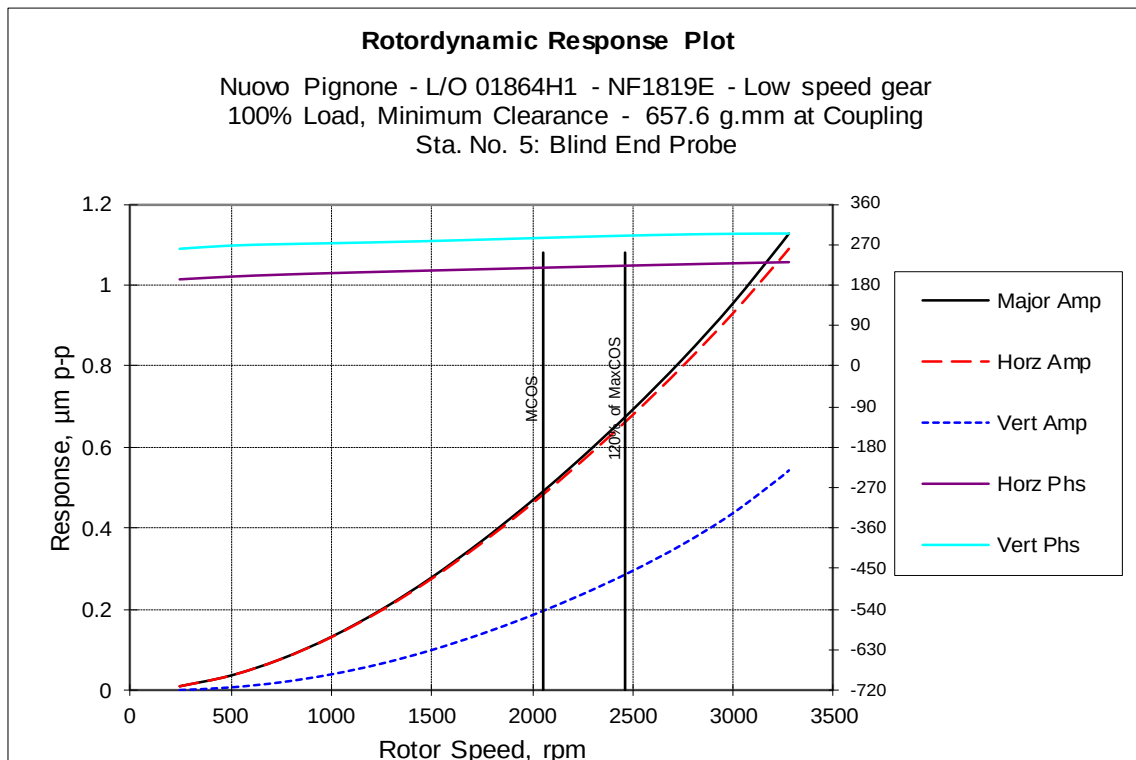


Figure 96

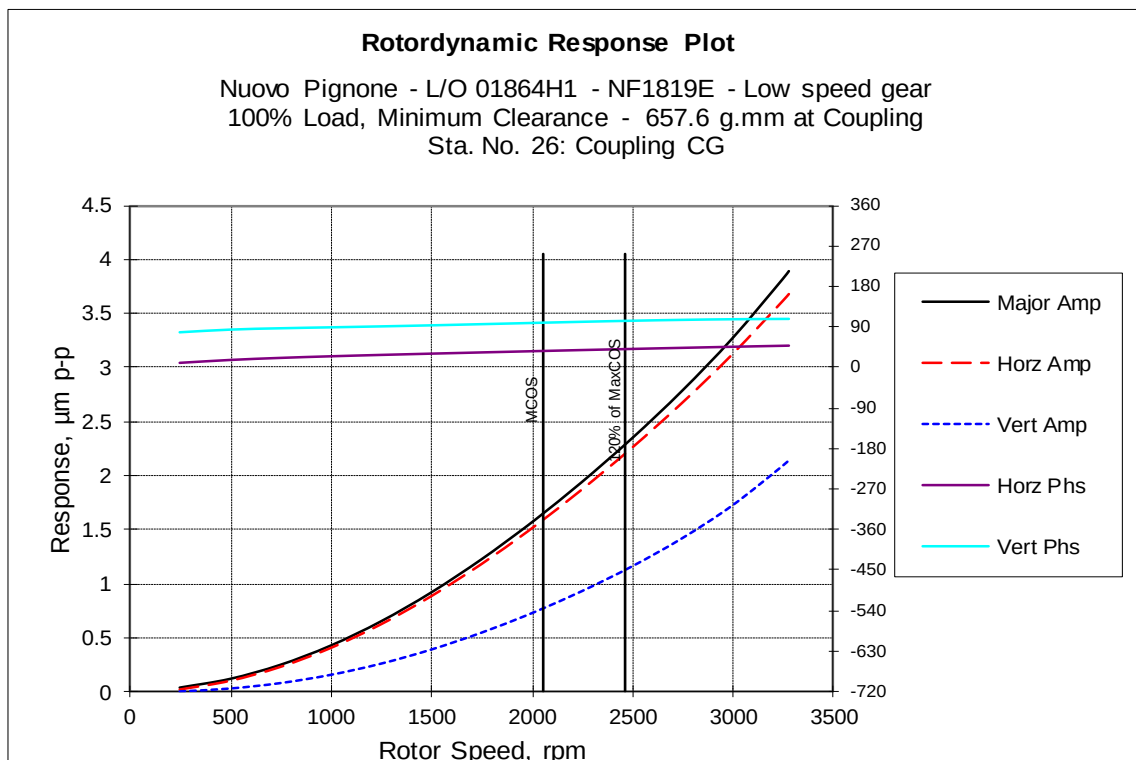
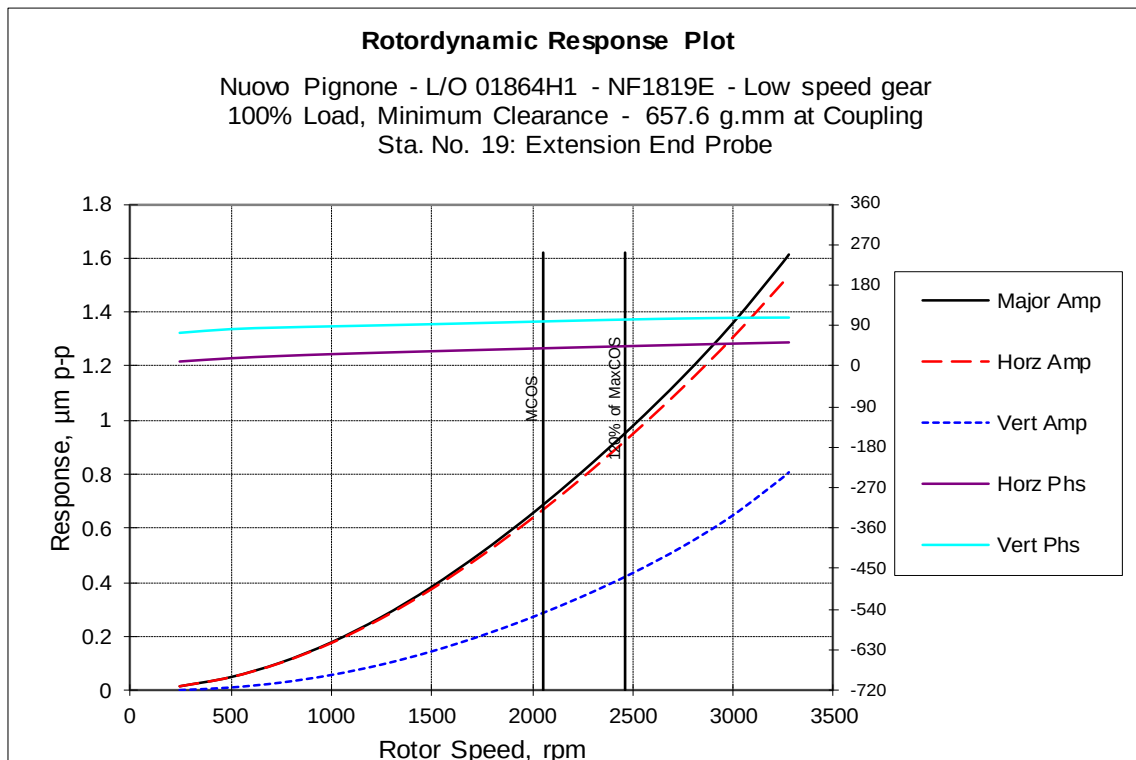
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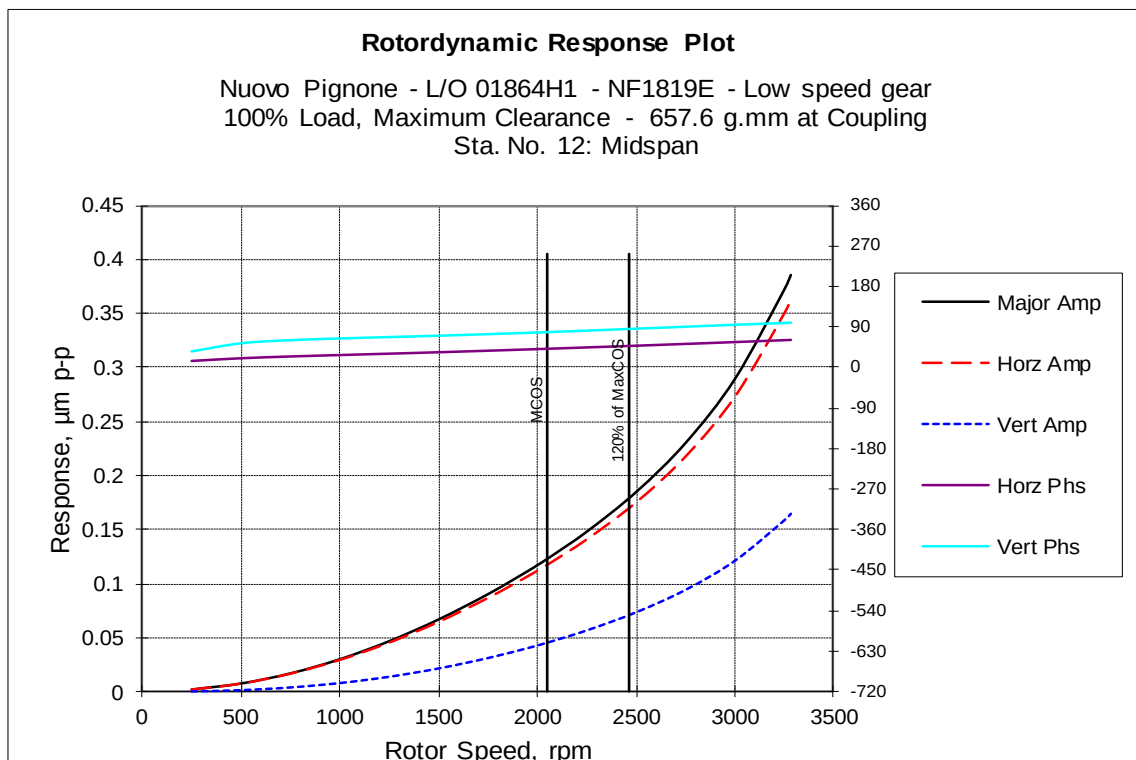
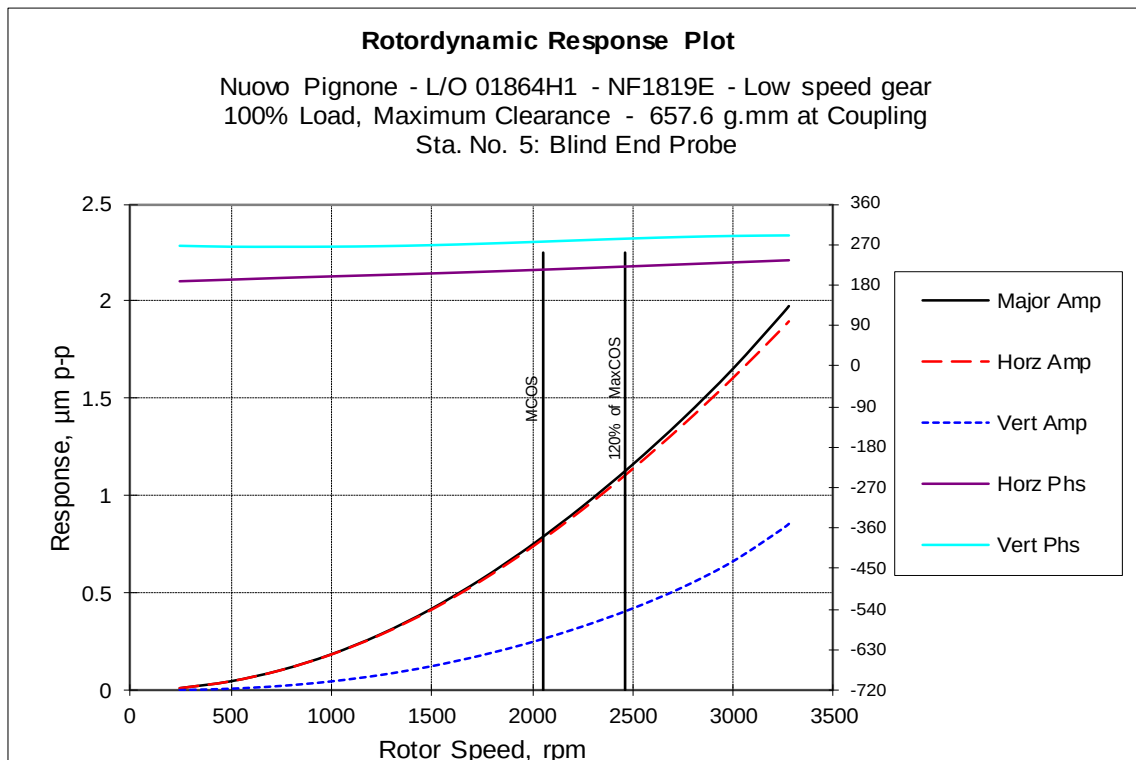
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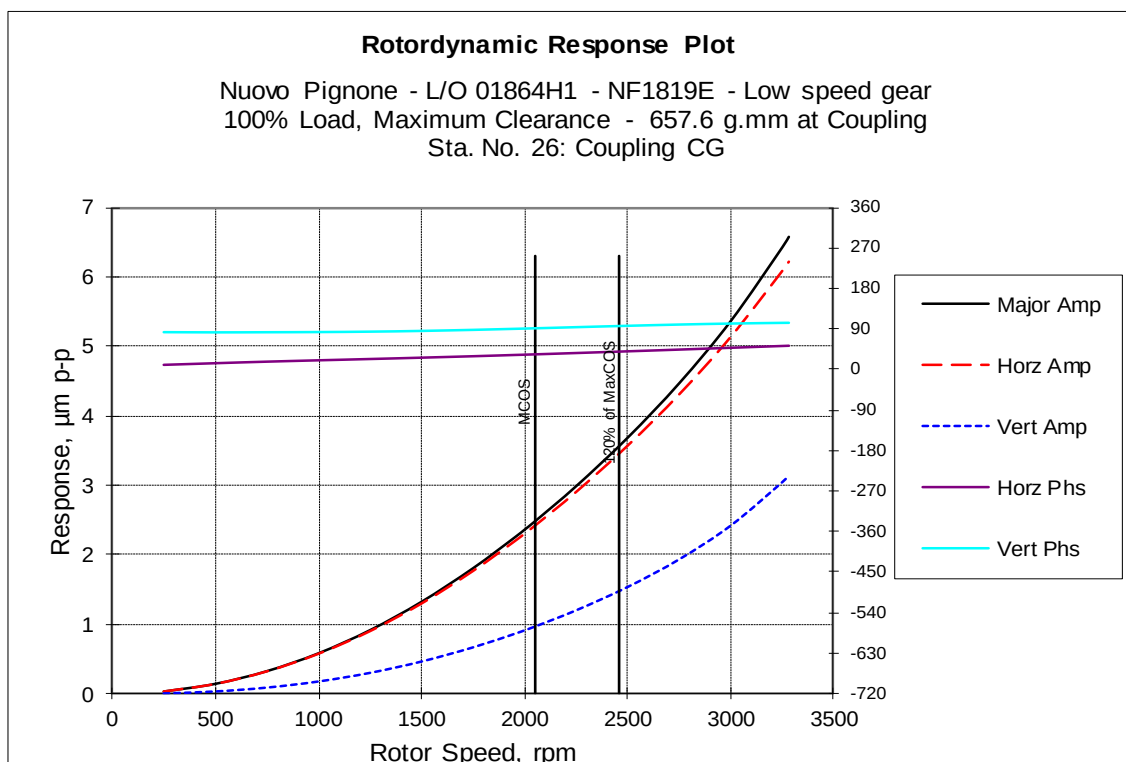
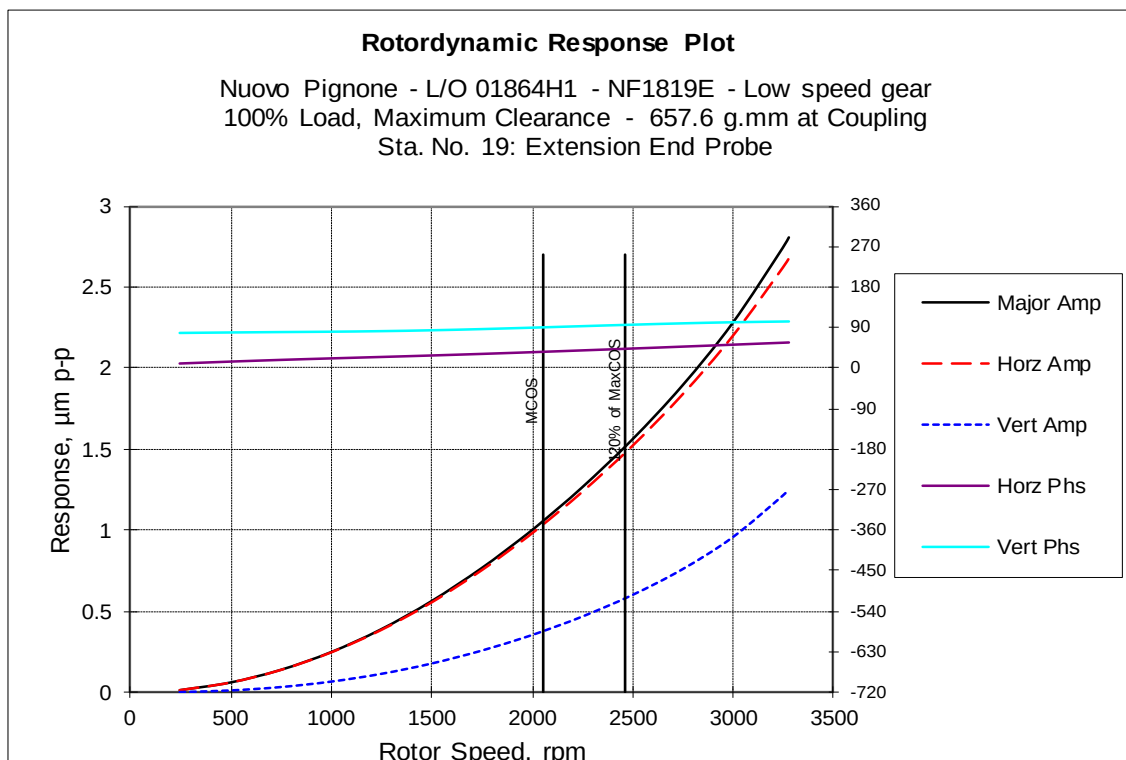
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